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 $^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10**

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$J^\pi=0^+$  for  $^{34}\text{S}$  ground state.

- 1976Me12:** 0.7-2.1-MeV proton beams were produced from the 1.1-MV Cockcroft-Walton generator and a 3-MV Van de Graaff accelerator. Targets were ZnS (46%  $^{34}\text{S}$ ) and CdS (37%  $^{34}\text{S}$ ) on tantalum backings.  $\gamma$  rays were detected using a 10-cm by 10-cm NaI detector and a 40-cm<sup>3</sup> Ge(Li) detector. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $\gamma(\theta)$ . Deduced level energies for 33 bound states and 47 resonances, J,  $\pi$ , resonance strengths,  $\gamma$  branching ratios, mixing ratios, and lifetimes for 25 bound states using DSAM.
- 1976Sp08:** 1.21, 1.95-2.91-MeV protons were produced from the Utrecht University 3-MV Van de Graaff accelerator. Targets were Ag<sub>2</sub>S (95%  $^{34}\text{S}$ ) evaporated onto a tantalum backing.  $\gamma$  rays were detected using a 10 cm by 10 cm diameter NaI scintillator and a 120- or 60-cm<sup>3</sup> Ge(Li) detector. Measured reaction yields,  $E_\gamma$ . Deduced Q value, 84 resonance energies, resonance strengths, widths, J,  $\pi$ , and  $\gamma$ -branching ratios.
- 1976Sp09:** similar setup as **1976Sp08** with 2.33-2.79-MeV protons. Measured reaction yields,  $E_\gamma$ ,  $\gamma(\theta)$ . Deduced levels, J,  $\pi$ , and  $\gamma$ -branching ratios.
- 1972Hu10:** 0.7-2.1-MeV protons were produced from the 4-MV Van de Graaff at the Centre d'Etudes Nucleaires de Bordeaux-Gradignan. Targets were 100- $\mu\text{g}/\text{cm}^2$  Ag<sub>2</sub>S (90%  $^{34}\text{S}$ ).  $\gamma$  rays were detected using a 12.7-cm by 12.7-cm NaI(Tl) detector and a 60-cm<sup>3</sup> Ge(Li) detector. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ . Deduced 59 resonances, 32 bound states, resonance strengths, and  $\gamma$  branching ratios.
- 1972Hu11:** similar setup as **1972Hu10**. Deduced lifetimes using DSAM.
- 1973Fa07:** 1-2-MeV protons were produced from the Helsinki 2-MV Van de Graaff accelerator. Targets were  $^{34}\text{S}$  on tantalum backings.  $\gamma$  rays were detected using a 55-cm<sup>3</sup> Ge(Li) detector. Measured reaction yields,  $E_\gamma$ ,  $I_\gamma$ . Deduced 28 resonances, resonance strengths,  $\gamma$ -branching ratios, and 9 lifetimes using DSAM.
- 2001Vo24:** 1.183-2.820-MeV protons were produced from the 4-MeV electrostatic Van de Graaff accelerator (NSC KhPEI). Enriched  $^{34}\text{S}$  target (100%). A 63 cm<sup>3</sup> Ge(Li) detector and a 15-cm-diam by 10-cm-long NaI(Tl) detector. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ . Deduced 12 resonances and resonance strengths.
- 2002Vo17:** 2.187-MeV protons were produced from the ESU-4 accelerator (NSC KhPEI). Enriched  $^{34}\text{S}$  target. A 63 cm<sup>3</sup> Ge(Li) detector. Measured  $E_\gamma$ ,  $I_\gamma$ . Deduced levels, resonance strengths, widths.
- 1981Bi05:** 1.4-2.8-MeV protons were produced from the Groningen 5-MV Van de Graaff accelerator. Targets were 4 and 1- $\mu\text{g}/\text{cm}^2$  Sb<sub>2</sub>S<sub>3</sub> (90% in  $^{34}\text{S}$ ) evaporated onto 10- $\mu\text{g}/\text{cm}^2$  carbon backings. Scattered protons were detected using three silicon surface barrier detectors.  $\gamma$  rays were detected using a 10-cm by 10-cm NaI(Tl) detector and a Ge(Li) detector. Measured  $\sigma(E_p, \theta)$ ,  $E_\gamma$ ,  $\gamma(\theta)$  for two  $\gamma$  rays. Deduced levels, J,  $\pi$ , widths, and resonance strengths.
- 1971Wi13:** protons were produced from the Ontario Cancer Institute. Targets were CdS.  $\gamma$  rays were detected using NaI(Tl) and Ge(Li) detectors. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ ,  $\gamma$ -linear polarization. Deduced 1.020-, 1.214-, 1.267-, and 1.510-MeV resonances,  $J^\pi$ , branchings, mixing ratios, and half-lives using DSAM.
- 1971Pr11:** 1.211-MeV protons were produced from the 2-MeV ARL Van de Graaff at the Aerospace Research Laboratories (ARL). Targets were Ag<sub>2</sub>S (85.6%  $^{34}\text{S}$ ).  $\gamma$  rays were detected using an 80-cm<sup>3</sup> Ge(Li) detector. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ ,  $p\gamma$ -delay. Deduced resonances, levels, J,  $\pi$ , branchings.
- 1963Ha32:** 0.7-1.2-MeV protons were produced from the Laboratorium Technische Fysika. Target was  $^{34}\text{S}$ .  $\gamma$  rays were detected using NaI(Tl) scintillation spectrometers. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ . Deduced levels, branchings.
- 1967Da12:** 0.8-2.5-MeV protons were produced from the 2.5 MeV Van de Graaff generator of the U.A.R. Atomic Energy Establishment. Targets were  $^{34}\text{S}$  on tantalum backings.  $\gamma$  rays were detected using two 15.2-cm-diam by 12.7-cm-thick NaI(Tl) crystals. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ . Deduced resonances, levels, branchings.
- 1960An06:** 0.7-1.9-MeV protons were produced from the electrostatic generator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Targets were isotopic  $^{34}\text{S}$  (4.2%).  $\gamma$  rays were detected using a 2-cm-thick by 3-cm-diam CsI(Tl) crystal. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $\gamma(\theta)$ . Deduced levels, J, branchings.
- 1966Wa09:** 1.214 MeV proton beam. Measured  $E_\gamma$ ,  $\gamma$ -polarization. Deduced level energy and J,  $\pi$  for the level of 3163 and 7540 level.
- 1966En04:** 0.3-2.1-MeV protons were produced from the Utrecht 850-keV Cockcroft-Walton generator and the 3-MV Van de Graaff generator. Targets were natural sulphur compound on tantalum backings.  $\gamma$  rays were detected using a cylindrical 10-cm-by-10-cm NaI crystal. Measured  $\sigma(E_p)$ ,  $E_\gamma$ . Deduced the resonance strength for  $E_p=1214$  keV.
- 1966Az01:** 1.214-MeV protons were produced from the Ontario Cancer Institute 3-MeV Van de Graaff accelerator. Targets were CdS with enriched  $^{34}\text{S}$  (46%).  $\gamma$  rays were detected using a Compton polarimeter. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ . Deduced levels, J,  $\pi$ , branchings.
- 1967Wa22:** 1.214- and 1.512-MeV protons were produced from the Aerospace Research Laboratories. Targets were enriched CdS (40%  $^{34}\text{S}$ ) and Ag<sub>2</sub>S (86%  $^{34}\text{S}$ ).  $\gamma$  rays were detected using two large volume NaI(Tl) detectors. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ ,  $\gamma$ -polarization. Deduced levels, J,  $\pi$ , branchings.
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**$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10 (continued)**

- 1967Ta08:** 1.020-, 1.213- and 1.513-MeV protons were produced from the 3-MV Van de Graaff accelerator of the Ontario Cancer Institute. Targets were CdS (46%  $^{34}\text{S}$ ) on thick metal foils.  $\gamma$  rays were detected using a 7.6-cm-long by 7.6-cm-diam NaI(Tl) detector. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ ,  $\gamma$ -polarization. Deduced levels,  $J$ ,  $\pi$ , branchings, mixing ratios.
- 1977Ko35:** 1.055, 1.183, 1.214, and 1.416 MeV protons were produced from the 4 MV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Target of  $^{34}\text{S}$ . A 40 cm<sup>3</sup> coaxial Ge(Li) detector. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $\gamma(\theta)$ . Deduced levels, branchings, mixing ratios.
- 1977Ko34:** 1.0-1.4 MeV protons. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ . Deduced levels,  $J^\pi$ , branchings, mixing ratios.
- 1970Ko33:** 2.317-, 2.223-, and 2.336-MeV protons were produced from the 4-MV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Targets were  $^{34}\text{S}$  on tantalum backings.  $\gamma$  rays were detected using a 12-cm<sup>3</sup> coaxial Ge(Li) detector. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ . Deduced 3 levels,  $J$ , and branching ratios.
- 1967Ko19,1967Ko20,1967Ko23:** 2.079-MeV protons were produced from the 4 MV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Targets were made by implanting  $^{34}\text{S}$  ions in a tantalum substrate.  $\gamma$  rays were detected using a 70-mm by 50-mm and a 70-mm by 60-mm NaI(Tl) detectors. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\text{p}\gamma\gamma(\theta)$ . Deduced levels,  $J$ ,  $\pi$ , branchings, mixing ratios.
- 1968Az02:** 1.214- and 1.513-MeV protons were produced from the Ontario Cancer Institute 3-MeV Van de Graaff accelerator. Targets were CdS (37%  $^{34}\text{S}$ ) evaporated onto gold backings.  $\gamma$  rays were detected using a 5-cm-diam by 5-cm-long NaI(Tl) crystal. Measured  $E_\gamma$ ,  $\gamma\gamma$ -delay. Deduced lifetimes of the level of 3163 keV using the delayed coincidence method and DSAM.
- 1969Mi23:** 2.079-MeV protons.  $^{34}\text{S}$  target. Ge(Li) detector. Measured  $E_\gamma$ ,  $I_\gamma$ . Deduced levels, branchings.
- 1971Ko32,1971Ko46:** 0.8-2.6-MeV protons. Targets were made by evaporating  $^{34}\text{S}$  ions onto tantalum backings.  $\gamma$  rays were detected using Ge(Li) and NaI(Tl) detectors. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ , and  $\text{p}\gamma\gamma(\theta)$ . Deduced levels,  $J$ ,  $\pi$ , level widths, and branchings.
- 1971Ba23:** 1.214-MeV protons were produced from the 3 MeV Dynamitron of Carleton and Ottawa Universities. Target of CdS (93%  $^{34}\text{S}$ ) on a gold backing. A 28 cm<sup>3</sup> Ge(Li) detector. Measured  $E_\gamma$ ,  $I_\gamma$ . Deduced lifetime of the 3163 keV level using the delayed coincidence method.
- 1974Al04:** 1.214-MeV protons. Targets were Ag<sub>2</sub>S, CdS, and ZnS of 300- $\mu\text{g}/\text{cm}^2$ . Ge(Li) detector. Measured  $\sigma(E_\gamma)$ . Deduced resonance strengths.
- 1975Ke11:** 1.214-MeV protons were produced from the Helsinki University 2.5-MV van de Graaff. Targets of CdS and ZnS on tantalum backings.  $\gamma$  rays were detected using a 120 cm<sup>3</sup> Ge(Li) detector with FWHM=2.9 keV at 2.6 MeV. Measured  $\sigma(E_\gamma)$ . Deduced resonance strengths.
- 1979Pa16:** 1.211-MeV protons were produced from the University of Melbourne 5U Pelletron accelerator. Natural sulphur target. NaI(Tl) and Ge(Li) detector. Measured  $\sigma(E_p)$ . Deduced resonance strength.
- 1988Va06:** 0.4-0.7 and 1.47 MeV protons were produced from the Utrecht 3 MV Van de Graaff accelerator. Targets of Ag<sub>2</sub>S (90%  $^{34}\text{S}$ ) of 10 and 34  $\mu\text{g}/\text{cm}^2$  on Ta backings. A 80-cm<sup>3</sup> coaxial Philips Ge(Li) or an  $\gamma$ -X intrinsic Ge detector, FWHM=1.85 and 1.75 keV at 1.33 MeV. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ . Deduced levels,  $J^\pi$ , branchings, mixing ratios.
- 1988Co03:** 1.211-MeV protons were produced from the Geel 7-MV CN-type Van de Graaff. Targets of  $^{34}\text{S}$ . Ge(Li) detector. Measured  $\gamma\gamma$ -coin. Deduced weighting function.
- 1996Vo20:** 1.891, 1.900 and 2.070-MeV protons were produced by an electrostatic accelerator ESU-4 at the Physical and Technical Institute. Target made by implanting  $^{34}\text{S}$  ions in a tantalum backing. A 63 cm<sup>3</sup> Ge(Li) detector. Measured  $E_\gamma$ ,  $I_\gamma$ . Deduced levels, gamma widths.
- 1996Ka21:** 1-3 MeV protons were produced from the electrostatic accelerator of the Institute of Physica and Technology (Kharkov). Targets were 20- $\mu\text{g}/\text{cm}^2$  Ag<sub>2</sub>S.  $\gamma$  rays were detected using a 60 cm<sup>3</sup> Ge(Li) detector and a 150×100-mm<sup>2</sup> NaI(Tl) as a monitor. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ . Deduced  $J^\pi$  for 3 resonances, resonance strengths.
- 2000Vo21:** 1.794 and 2.275 MeV protons. Measured  $E_\gamma$ ,  $I_\gamma$ . Deduced gamma widths, transition strengths.
- 2001Ka69:** 0.8-3 MeV protons produced from the ESU-4 accelerator (NSC KhPEI). Target of  $^{34}\text{S}$ . A 15-cm-diam by 19-cm-long NaI(Tl) detector. Measured  $E_\gamma$ ,  $I_\gamma$ . Deduced levels, transition strengths.
- Others: **1962Du05**, **1963Du08**, **1964Hy01**, **1966Ko22**, **1966Ko15**, **1967Ma48**, **1969Ko26**, **1969Ah02**, **1969KrZW**, **1972LeYW**, **1974Il01**, **1974Bi16**, **1981Va02**.
- Theoretical studies: **2020Ri06** (shell model).
- Standard resonance strength  $\omega\gamma=(2J_r+1)/[(2J_p+1)(2J_T+1)]\times\Gamma_\gamma\Gamma_p/\Gamma$ , where  $J_r$  is the spin of the resonance,  $J_p=1/2$  is the spin of the proton, and  $J_T=0$  is the spin of the target  $^{34}\text{S}$  ground state. The resonance strengths  $S$  from the original papers in this dataset are mainly obtained from resonance  $\gamma$ -ray yields and are defined without the  $(2J_p+1)$  factor. Therefore, to obtain the standard resonance strength, all the resonance strengths provided in the comments below should be divided by 2.
- $S=21$  eV 3 at  $E_p=1213$  measured by **1966En04** has been used as a normalization standard by **1971BrXT**, **1972Hu10**, **1973Fa07**,

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)**

1974Bi16, and 1976Me12. 1971BrXT measured  $^{34}\text{S}(\text{p},\text{p})$  elastic scattering and determined  $\Gamma_{\text{p}}=4.6$  eV 10 and  $\Gamma=5$  eV +4–2 from a  $Q^2$  minimization analysis of the excitation function. 1971BrXT added  $\Gamma_{\text{p}}\Gamma_{\gamma}/\Gamma=2.6$  eV 4 converted from the S measured by 1966En04 to constrain their  $Q^2$  analysis and reported a final  $\Gamma_{\text{p}}=5.8$  eV 7,  $\Gamma_{\gamma}=4.7$  eV 17, and  $\Gamma=10.5$  eV 15. 1974Bi16 measured  $^{34}\text{S}(\text{p},\text{p})$  and determined  $\Gamma_{\text{p}}=7.2$  eV 15 from a  $Q^2$  minimization analysis of the excitation function and quoted  $S=21$  eV 3 (1966En04) to deduce  $\Gamma_{\gamma}=3.8$  eV 15 and  $\Gamma=11.0$  eV 15. 1974AI04 determined an absolute  $S_{1213}=9.7$  eV 9. 1975Ke11 determined an absolute  $S_{1213}=9.8$  eV 10 and quoted 1974Bi16's relative ratio of  $\Gamma_{\text{p}}/\Gamma_{\gamma}=7.2/3.9=1.9$  to deduce  $\Gamma_{\text{p}}=3.8$  eV 13,  $\Gamma_{\gamma}=2.0$  eV 13, and  $\Gamma=5.8$  eV 13. 1976Sp09 measured the resonant absorption of the  $E_{\text{p}}=2791$  resonance, which indicates that normalization using  $S_{1213}=9.7$  eV gives a better agreement than that using  $S_{1213}=21$  eV. 1981Bi05 determined an absolute  $S=4.1$  eV 5 for the  $E_{\text{p}}=1892$  resonance and quoted the relative ratio of  $S_{1213}/S_{1892}$  from 1978En02 to deduce  $S=9.5$  eV 12 for  $E_{\text{p}}=1213$  resonance. The  $S_{1213}/S_{1892}=2.3$  quoted by 1981Bi05 was based on two S values: first, 1978En02 quoted  $S=4.2$  eV for a 1892+1893 resonance doublet, which was scaled down by a factor of 21/9.7 from 1976Me12's original  $S_{1892}=9.0$  eV; second, 1978En02 quoted  $S_{1213}=9.7$  eV 7 from 1976Sp08, 1974AI04, and 1975Ke11. 1979Pa16 determined  $S=9.0$  eV 10 using the  $^{27}\text{Al}(\text{p},\gamma)^{28}\text{Si}$  resonances as normalization standards. 1996Ka21 normalized all their S to  $S_{1213}=9.7$  eV 7 and cited 1976Sp08 but listed this value in the "this study" column. 2001Vo24 normalized all S to  $S_{1213}=9.7$  eV 7 and cited 1976Sp08 and 1975Ke11. However, 2001Vo24 also reported a different  $S_{1213}=9.9$  eV 7 without giving details. Although 1976Sp08 is frequently cited as the source of  $S_{1213}=9.7$  eV 7, they did not measure the absolute S value; instead, they calculated a weighted average of 1974AI04 and 1975Ke11.

In summary, the evaluators adopt  $S_{1213}=9.5$  eV 9: weighted average of all independently measured  $S_{1213}$ : 9.7 eV 9 (1974AI04), 9.8 eV 10 (1975Ke11), and 9.0 eV 10 (1979Pa16). All resonance strengths given in the comments below have been normalized to this value by the evaluators.  $\Gamma_{\text{p}}=4.6$  eV 10 and  $\Gamma=5$  eV +4–2 from (p,p) excitation function analysis in 1971BrXT are likely the only width data that are not influenced by the questionable  $S_{1213}=21$  eV 3.

 $^{35}\text{Cl}$  Levels

| E(level) <sup>†</sup> | $J^{\pi\ddagger}$ | $T_{1/2}$   | Comments                                                                                                                                                                                                                                                                                                                            |
|-----------------------|-------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                     | $3/2^+$           |             |                                                                                                                                                                                                                                                                                                                                     |
| 1219.39 7             | $1/2^+$           | 105 fs 14   | E(level): weighted average of 1219.1 5 (1972Hu10) and 1219.3 1 (1976Me12), and 1219.44 7 (1981Bi05).<br>$T_{1/2}$ : lifetime $\tau=152$ fs 20: weighted average of 80 fs 40 (1971Wi13, DSAM), 175 fs 20 (1972Hu11, DSAM), 150 fs 20 (1973Fa07, DSAM), and 145 fs 30 (1976Me12, DSAM).                                               |
| 1763.14 4             | $5/2^+$           | 0.347 ps 63 | E(level): weighted average of 1762.8 5 (1972Hu10), 1763.4 7 (1976Me12), and 1763.14 4 (1981Bi05).<br>$J^{\pi}$ : from $\gamma(\theta)$ in 1971Wi13.<br>$T_{1/2}$ : lifetime $\tau=0.50$ ps 9: weighted average of 0.50 ps 10 (1972Hu11, DSAM), 0.40 ps 14 (1973Fa07, DSAM), and 0.53 ps 9 (1976Me12, DSAM).                         |
| 2645.70 8             | $(5/2,7/2)$       | 0.146 ps 21 | E(level): weighted average of 2645.6 5 (1972Hu10), 2644.7 13 (1976Me12), and 2645.71 8 (1981Bi05).<br>$J^{\pi}$ : from $\gamma(\theta)$ in 1971Wi13. Other: $3/2$ from $\gamma(\theta)$ in 1967Ko20.<br>$T_{1/2}$ : lifetime $\tau=0.210$ ps 30: weighted average of 0.255 ps 65 (1972Hu11, DSAM) and 0.200 ps 30 (1976Me12, DSAM). |
| 2693.9 3              | $(3/2,5/2)$       | 13.9 fs 21  | E(level): weighted average of 2694.0 3 (1972Hu10), 2694.7 12 (1976Me12), and 2693.8 3 (1981Bi05).<br>$J^{\pi}$ : from $\gamma(\theta)$ in 1971Wi13.<br>$T_{1/2}$ : lifetime $\tau=20$ fs 3: weighted average of 21 fs 3 (1972Hu11, DSAM), 19 fs 4 (1973Fa07, DSAM), and 20 fs 4 (1976Me12, DSAM).                                   |
| 3002.75 8             | $5/2$             | 11.1 fs 21  | E(level): weighted average of 3002.7 7 (1972Hu10), 3003.7 8 (1976Me12), and 3002.74 8 (1981Bi05).<br>$J^{\pi}$ : $5/2$ from $\gamma(\theta)$ in 1967Ko20 and 1971Wi13.<br>$T_{1/2}$ : lifetime $\tau=16$ fs 3: weighted average of 22 fs 3 (1972Hu11, DSAM), 14 fs 2 (1973Fa07, DSAM), and 16 fs 4 (1976Me12, DSAM).                |
| 3162.99 7             | $7/2^-$           | 25.7 ps 28  | $T=1/2$<br>E(level): weighted average of 3162.5 5 (1972Hu10), 3163.9 7 (1976Me12), and 3162.99 6 (1981Bi05).<br>$J^{\pi}$ : spin from $\gamma(\theta)$ in 1966Wa09, 1967Wa22, 1967Ta08, and 1971Wi13; parity from $3163\gamma$ , M2+E3 to $3/2^+$ ground state in 1966Wa09 and 1967Wa22.                                            |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup> | T <sub>1/2</sub> | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------|-----------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                             |                  | T <sub>1/2</sub> : lifetime $\tau=37$ ps 4 (1971Ba23, delayed coincidence). Others: 140 ps 40 (1968Az02, delayed coincidence), >0.6 ps (1968Az02, DSAM), >0.2 ps (1971Wi13, DSAM), >2 ps (1972Hu11, DSAM), >0.9 ps (1973Fa07, DSAM), and >1 ps (1976Me12). 1971Ba23 states that the earlier 1968Az02 value was obtained using NaI(Tl) detectors with broad time resolution and the discrepancy between 1968Az02 and 1971Ba23 is not unreasonable. |
| 3918.48 10            | 3/2 <sup>+</sup>            | 4.9 fs 14        | E(level): weighted average of 3917.9 6 (1972Hu10), 3920.7 13 (1976Me12), and 3918.48 7 (1981Bi05).<br>J <sup>π</sup> : spin=3/2 determined by the feeding 5162 $\gamma(\theta)$ from 5/2 <sup>+</sup> , 9081 level in 1976Sp09; parity=+ determined by 3918 $\gamma$ , M1+E2 to 3/2 <sup>+</sup> ground state.                                                                                                                                    |
| 3942.9 2              |                             | 0.159 ps 42      | T <sub>1/2</sub> : lifetime $\tau=7$ fs 2 (1972Hu11, DSAM). Other: <6 fs (1976Me12, DSAM).<br>E(level): weighted average of 3942.6 9 (1972Hu10), 3944.1 11 (1976Me12), and 3942.9 2 (1981Bi05).                                                                                                                                                                                                                                                   |
| 3967.3 6              |                             | 11.8 fs 28       | T <sub>1/2</sub> : lifetime $\tau=0.23$ ps 6: weighted average of 0.39 ps 11 (1972Hu11, DSAM), 0.145 ps 50 (1973Fa07, DSAM), and 0.29 ps 5 (1976Me12, DSAM).<br>E(level): from 1972Hu10. Other: 3979.0 15 (1976Me12). 1976Me12 matched their 3979 level with the 3967 level observed in other work. No other work has observed a level at 3979 keV.                                                                                               |
| 4059.2 4              |                             | 13.9 fs 21       | T <sub>1/2</sub> : lifetime $\tau=17$ fs 4: weighted average of 13 fs 4 (1972Hu11, DSAM), 20 fs 5 (1973Fa07, DSAM), and 20 fs 4 (1976Me12, DSAM).<br>E(level): weighted average of 4058.4 8 (1972Hu10), 4059.4 4 (1976Me12), and 4059.2 3 (1981Bi05).                                                                                                                                                                                             |
| 4113.7 10             | 7/2                         |                  | T <sub>1/2</sub> : lifetime $\tau=20$ fs 3: weighted average of 22 fs 3 (1972Hu11, DSAM), 20 fs 4 (1973Fa07, DSAM), and 18 fs 3 (1976Me12, DSAM).<br>E(level): weighted average of 4111 3 (1972Hu10) and 4114.0 10 (1976Me12).<br>J <sup>π</sup> : spin=7/2 from feeding 4516 $\gamma(\theta)$ from 7/2, 8630 level in 1976Sp09.                                                                                                                  |
| 4173.45 19            |                             | 34.7 fs 63       | T=1/2<br>E(level): weighted average of 4173.0 6 (1972Hu10), 4174.7 10 (1976Me12), and 4173.45 19 (1981Bi05).                                                                                                                                                                                                                                                                                                                                      |
| 4177.9 2              | 3/2                         | 27.0 fs 28       | T <sub>1/2</sub> : lifetime $\tau=50$ fs 9: weighted average of 105 fs +35–30 (1971Wi13, DSAM), 55 fs 6 (1972Hu11, DSAM), and 34 fs 9 (1976Me12, DSAM).<br>E(level): weighted average of 4177.2 9 (1972Hu10), 4180.1 15 (1976Me12), and 4177.9 2 (1981Bi05).<br>J <sup>π</sup> : from $\gamma(\theta)$ in 1967Wa22.                                                                                                                               |
| 4347.76 6             | 5/2,9/2                     | 0.229 ps 42      | T <sub>1/2</sub> : lifetime $\tau=39$ fs 4: weighted average of 32 fs 5 (1972Hu11, DSAM), 42 fs 4 (1973Fa07, DSAM), and 47 fs 10 (1976Me12, DSAM).<br>E(level): weighted average of 4347.5 8 (1972Hu10), 4347.2 12 (1976Me12), and 4347.76 6 (1981Bi05).<br>J <sup>π</sup> : spin=5/2,9/2 based on feeding 4282 $\gamma(\theta)$ from 7/2, 8630 level in 1976Sp09.                                                                                |
| 4624.4 3              |                             |                  | T <sub>1/2</sub> : lifetime $\tau=0.33$ ps 6 (1976Me12, DSAM). Other: >0.6 ps (1972Hu11, DSAM).<br>E(level): weighted average of 4624.2 20 (1976Me12) and 4624.4 3 (1981Bi05). Other: 4625 5 (1972Hu10).<br>15% of I $\gamma$ from this level decays to unknown final levels (1976Me12).                                                                                                                                                          |
| 4769.86 23            |                             | 76 fs 20         | E(level): weighted average of 4768.9 12 (1972Hu10), 4766.9 15 (1976Me12), and 4769.90 15 (1981Bi05).<br>T <sub>1/2</sub> : lifetime $\tau=110$ fs 29: weighted average of 270 fs 95 (1972Hu11, DSAM) and 105 fs 17 (1976Me12, DSAM).                                                                                                                                                                                                              |
| 4839.10 11            |                             | 9.7 fs 35        | E(level): weighted average of 4838 3 (1972Hu10), 4841.7 19 (1976Me12), and 4839.09 11 (1981Bi05).<br>T <sub>1/2</sub> : lifetime $\tau=14$ fs 5 (1976Me12, DSAM).                                                                                                                                                                                                                                                                                 |
| 4854.4 19             |                             | 4.9 fs 14        | E(level): weighted average of 4853 2 (1972Hu10) and 4855.7 19 (1976Me12).<br>T <sub>1/2</sub> : lifetime $\tau=7$ fs 2 (1976Me12, DSAM). Other: <6 fs (1972Hu11, DSAM).                                                                                                                                                                                                                                                                           |
| 4880.96 15            |                             | 5.6 fs 14        | E(level): weighted average of 4880.9 1 (1972Hu10), 4885 2 (1976Me12), and 4881.14 20 (1981Bi05).<br>T <sub>1/2</sub> : lifetime $\tau=8$ fs 2: weighted average of 7 fs 2 (1972Hu11, DSAM), 7 fs 3 (1973Fa07, DSAM), and 10 fs 3 (1976Me12, DSAM).                                                                                                                                                                                                |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup> | T <sub>1/2</sub> | E <sub>p</sub> (lab) (keV) | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|-----------------------------|------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5010.1 18             |                             | 7.6 fs 21        |                            | E(level): weighted average of 5008 5 (1972Hu10) and 5010.4 18 (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                     |
| 5163.34 22            | 7/2                         |                  |                            | T <sub>1/2</sub> : lifetime $\tau=11$ fs 3 (1976Me12, DSAM).<br>E(level): weighted average of 5163 5 (1972Hu10), 5166.7 15 (1976Me12), and 5163.31 14 (1981Bi05).<br>J <sup>π</sup> : spin=3/2, 7/2 from feeding 3466 $\gamma$ ( $\theta$ ) from 7/2, 8630 level in 1976Sp09.<br>Spin=3/2 and $\delta(\text{O/Q})=0.12$ 1 would require a large E3 or M3 component, which is ruled out by RUL.<br>10% of I $\gamma$ transitions to unknown levels (1976Me12). |
| 5215.8 15             |                             | <4.85 fs         |                            | E(level): weighted average of 5214 3 (1972Hu10) and 5216.2 15 (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                     |
| 5403.5 10             |                             | 11.8 fs 28       |                            | T <sub>1/2</sub> : lifetime $\tau<7$ fs (1976Me12, DSAM).<br>E(level): weighted average of 5400 5 (1972Hu10) and 5403.6 10 (1976Me12).                                                                                                                                                                                                                                                                                                                        |
| 5520                  |                             |                  |                            | T <sub>1/2</sub> : lifetime $\tau=17$ fs 4 (1976Me12, DSAM).<br>20% of I $\gamma$ from this level decays (1976Me12). 50% of I $\gamma$ from this level decays to unknown levels (1972Hu10).                                                                                                                                                                                                                                                                   |
| 5586 2                |                             |                  |                            | E(level): from 1976Sp08.                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5599.7 15             |                             | 2.08 fs 69       |                            | E(level): weighted average of 5587 3 (1972Hu10) and 5586 2 (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                        |
| 5646 2                |                             | 2.77 fs 69       |                            | E(level): weighted average of 5598 3 (1972Hu10) and 5600.1 15 (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                     |
| 5655 2                |                             | 13.9 fs 35       |                            | T <sub>1/2</sub> : lifetime $\tau=3$ fs 1 (1976Me12, DSAM).<br>E(level): weighted average of 5645 3 (1972Hu10) and 5646 2 (1976Me12).                                                                                                                                                                                                                                                                                                                         |
|                       |                             |                  |                            | T <sub>1/2</sub> : lifetime $\tau=4$ fs 1 (1976Me12, DSAM).                                                                                                                                                                                                                                                                                                                                                                                                   |
|                       |                             |                  |                            | E(level): weighted average of 5651 3 (1972Hu10) and 5656 2 (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                        |
|                       |                             |                  |                            | J <sup>π</sup> : Mirror level of $^{35}\text{S}$ ground state and thus 3/2 <sup>+</sup> (2001Vo24). Other: spin=5/2 based on the feeding 2051g( $\theta$ ) from the 5/2, 7706 level in 1977Ko34.                                                                                                                                                                                                                                                              |
|                       |                             |                  |                            | T <sub>1/2</sub> : lifetime $\tau=20$ fs 5 (1976Me12, DSAM).                                                                                                                                                                                                                                                                                                                                                                                                  |
|                       |                             |                  |                            | $\Gamma_{\gamma}=0.033$ eV 10 (2001Vo24).                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5683 2                |                             |                  |                            | 14% of I $\gamma$ from this level decays to unknown final levels (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                  |
|                       |                             |                  |                            | E(level): weighted average of 5682 5 (1972Hu10) and 5683 2 (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                        |
| 5723.52 18            |                             |                  |                            | 40% of I $\gamma$ from this level decays to unknown final levels (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                  |
|                       |                             |                  |                            | E(level): from 1981Bi05. 38% of I $\gamma$ from this level decays to unknown final levels (1976Sp09).                                                                                                                                                                                                                                                                                                                                                         |
| 5758 3                |                             |                  |                            | E(level): weighted average of 5754 5 (1972Hu10) and 5759 3 (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                        |
|                       |                             |                  |                            | 25% of $\gamma$ from this level decays to unknown final levels and the other 75% is attributed to two $\gamma$ transitions (1976Me12). Other: 100% of $\gamma$ from this level decays to unknown final levels (1972Hu10).                                                                                                                                                                                                                                     |
| 5806 2                |                             | 3.47 fs 69       |                            | E(level): weighted average of 5802 5 (1972Hu10) and 5806 2 (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                        |
| 6106.2 21             |                             | 8.3 fs 21        |                            | T <sub>1/2</sub> : lifetime $\tau=5$ fs 1 (1976Me12, DSAM).                                                                                                                                                                                                                                                                                                                                                                                                   |
|                       |                             |                  |                            | E(level): weighted average of 6102 3 (1972Hu10) and 6107.2 15 (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                     |
| 6181 3                |                             |                  |                            | T <sub>1/2</sub> : lifetime $\tau=12$ fs 3 (1976Me12, DSAM).                                                                                                                                                                                                                                                                                                                                                                                                  |
|                       |                             |                  |                            | E(level): from 1976Me12.                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 6492 3                |                             |                  |                            | 55% of I $\gamma$ from this level decays to unknown final levels (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                  |
| 6866.7 6              |                             |                  | 510.6 6                    | E(level): weighted average of 6489 5 (1972Hu10) and 6493 3 (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                        |
| 7066.2 7              |                             |                  | 716.0 7                    | E <sub>p</sub> (lab) (keV): from 1988Va06.                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                       |                             |                  |                            | E <sub>p</sub> (lab) (keV): weighted average of 716.0 7 (1972Hu10) and 716.0 10 (1976Me12).                                                                                                                                                                                                                                                                                                                                                                   |
|                       |                             |                  |                            | $\omega\gamma=0.09$ eV 5 (1972Hu10).                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                       |                             |                  |                            | $\omega\gamma=0.14$ eV 4 (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 7103.4 7              | 3/2                         |                  | 754.3 7                    | J <sup>π</sup> : from $\gamma$ ( $\theta$ ) in 1976Me12.                                                                                                                                                                                                                                                                                                                                                                                                      |
|                       |                             |                  |                            | E <sub>p</sub> (lab) (keV): weighted average of 754.1 7 (1972Hu10) and 754.6 10 (1976Me12).                                                                                                                                                                                                                                                                                                                                                                   |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup> | E <sub>p</sub> (lab) (keV) | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------|-----------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7178.6 8              | 1/2                         | 831.8 8                    | $\omega\gamma=0.04$ eV (1963Ha32).<br>$\omega\gamma=0.23$ eV 11 (1972Hu10).<br>$\omega\gamma=0.23$ eV 7 (1976Me12).<br>J <sup>π</sup> : from $\gamma(\theta)$ in 1976Me12.<br>E <sub>p</sub> (lab) (keV): weighted average of 831.5 8 (1972Hu10), 832.2 15 (1976Me12), and 832 1 (2001Vo24).<br>$\omega\gamma=0.13$ eV 2 (2001Vo24).<br>$\Gamma_\gamma \geq 0.07$ eV (2001Vo24).<br>$\omega\gamma=0.18$ eV 9 (1972Hu10).<br>$\omega\gamma=0.14$ eV 4 (1976Me12). |
| 7185                  |                             | 838                        | E <sub>p</sub> (lab) (keV): from 1963Ha32.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7194.6 7              | 1/2,3/2                     | 848.2 7                    | $\omega\gamma=0.056$ eV (1963Ha32).<br>J <sup>π</sup> : from $\gamma(\theta)$ in 1960An06.<br>E <sub>p</sub> (lab) (keV): weighted average of 848.2 7 (1972Hu10) and 848.1 10 (1976Me12).<br>$\omega\gamma=0.17$ eV (1963Ha32).<br>$\omega\gamma=0.50$ eV 15 (1972Hu10).<br>$\omega\gamma=0.54$ eV 16 (1976Me12).                                                                                                                                                |
| 7225.5 8              | 5/2                         | 880.0 8                    | J <sup>π</sup> : from $\gamma(\theta)$ in 1976Me12.<br>E <sub>p</sub> (lab) (keV): weighted average of 880.5 8 (1972Hu10) and 879.3 10 (1976Me12).<br>$\omega\gamma=0.073$ eV (1963Ha32).<br>$\omega\gamma=0.18$ eV 9 (1972Hu10).<br>$\omega\gamma=0.14$ eV 4 (1976Me12).                                                                                                                                                                                        |
| 7234.0 8              | 5/2 <sup>+</sup>            | 888.8 8                    | J <sup>π</sup> : spin=5/2 from $\gamma(\theta)$ in 1960An06 and 6014 $\gamma$ to 1/2 <sup>+</sup> , 1219 level in 1972Hu10 and 1976Me12.<br>E <sub>p</sub> (lab) (keV): weighted average of 889.0 8 (1972Hu10) and 888.5 10 (1976Me12).<br>$\omega\gamma=0.23$ eV (1963Ha32).<br>$\omega\gamma=0.59$ eV 18 (1972Hu10).<br>$\omega\gamma=0.86$ eV 26 (1976Me12).                                                                                                  |
| 7269.2 1              |                             | 925.1 1                    | E <sub>p</sub> (lab) (keV): from 1976Sp08.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7272.7 9              | (1/2,3/2)                   | 928.6 9                    | J <sup>π</sup> : from $\gamma(\theta)$ in 1960An06.<br>E <sub>p</sub> (lab) (keV): weighted average of 928.2 9 (1972Hu10) and 929.0 10 (1976Me12).<br>$\omega\gamma=0.31$ eV (1963Ha32).<br>$\omega\gamma=0.63$ eV 19 (1972Hu10).<br>$\omega\gamma=1.00$ eV 30 (1976Me12).                                                                                                                                                                                       |
| 7362.0 8              | 3/2                         | 1020.6 8                   | J <sup>π</sup> : from $\gamma(\theta)$ in 1971Wi13 and 1977Ko34.<br>E <sub>p</sub> (lab) (keV): weighted average of 1020.4 8 (1972Hu10), 1020.5 11 (1973Fa07), and 1021.0 10 (1976Me12).<br>$\omega\gamma=0.48$ eV (1963Ha32).<br>$\omega\gamma=1.4$ eV 4 (1972Hu10).<br>$\omega\gamma=1.13$ eV (1973Fa07).<br>$\omega\gamma=1.4$ eV 4 (1976Me12).                                                                                                               |
| 7396.0 10             | 7/2                         | 1055.6 10                  | J <sup>π</sup> : from $\gamma(\theta)$ in 1977Ko35.<br>E <sub>p</sub> (lab) (keV): weighted average of 1054.9 10 (1972Hu10), 1055.8 12 (1973Fa07), and 1056.9 16 (1976Me12).<br>$\omega\gamma=0.67$ eV (1963Ha32).<br>$\omega\gamma=0.18$ eV 9 (1972Hu10).<br>$\omega\gamma=0.18$ eV (1973Fa07).<br>$\omega\gamma=0.23$ eV 7 (1976Me12).                                                                                                                         |
| 7451.1 6              | 3/2                         | 1112.3 6                   | J <sup>π</sup> : from $\gamma(\theta)$ in 1972LeYW.<br>E <sub>p</sub> (lab) (keV): weighted average of 1112.2 10 (1972Hu10), 1112.8 13 (1973Fa07), and 1112.2 6 (1976Me12).<br>$\omega\gamma=0.14$ eV 7 (1972Hu10).<br>$\omega\gamma=0.14$ eV (1973Fa07).<br>$\omega\gamma=0.18$ eV 5 (1976Me12).                                                                                                                                                                |
| 7496?                 |                             | 1158                       | E(level): from 1960An06.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 7501.1 8              |                             | 1163.8 8                   | E <sub>p</sub> (lab) (keV): from 1973Fa07. Other: 1165 (1972Hu10).                                                                                                                                                                                                                                                                                                                                                                                               |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$ | $T_{1/2}$ | $E_p(\text{lab})$ (keV) | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------|------------------|-----------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7502.9 7                  |                  |           | 1165.7 7                | $\omega\gamma=0.05$ eV (1973Fa07).<br>$\omega\gamma=0.27$ eV 14 for $E_p=1165+1166$ doublet (1972Hu10).<br>$E_p(\text{lab})$ (keV): from 1973Fa07. Other: 1165.4 7 (1976Me12), 1166 (1972Hu10).<br>$\omega\gamma=0.27$ eV 14 for $E_p=1165+1166$ doublet (1972Hu10).<br>$\omega\gamma=0.23$ eV (1973Fa07).<br>$\omega\gamma=0.36$ eV 11 (1976Me12), possible doublet.<br>$J^\pi$ : from $\gamma(\theta)$ in 1977Ko35.<br>$E_p(\text{lab})$ (keV): weighted average of 1183.2 11 (1972Hu10), 1182.2 12 (1973Fa07), and 1181.4 7 (1976Me12).<br>$\omega\gamma=0.09$ eV 5 (1972Hu10).<br>$\omega\gamma=0.14$ eV (1973Fa07).<br>$\omega\gamma=0.18$ eV 5 (1976Me12).<br>$T=3/2$<br>$J^\pi$ : spin from $\gamma(\theta)$ in 1966Wa09, 1967Wa22, and 1977Ko35; parity from 4386 $\gamma$ , M1+E2 to 7/2 $^-$ , 3163 level in 1966Wa09 and 1967Wa22.<br>$T_{1/2}$ : lifetime $\tau < 1$ fs (1973Fa07, DSAM). Others: $\tau < 13$ fs (1971Wi13, DSAM).<br>$E_p(\text{lab})$ (keV): weighted average of 1213.7 10 (1972Hu10), 1213.6 6 (1973Fa07), 1212.4 7 (1976Me12), and 1212 1 (2001Vo24).<br>The resonance strength of this level has been intensively studied. See the general comments at the beginning of this dataset for details.<br>1971Pr11 reports a 3166 $\gamma$ with $I_\gamma=0.37$ 13 to the 4382 final level.<br>$J^\pi$ : from $\gamma(\theta)$ in 1977Ko35.<br>$E_p(\text{lab})$ (keV): weighted average of 1225.6 10 (1972Hu10), 1226.3 6 (1973Fa07), and 1225.3 7 (1976Me12).<br>$\omega\gamma=0.68$ eV 20 (1972Hu10).<br>$\omega\gamma=0.50$ eV (1973Fa07).<br>$\omega\gamma=0.68$ eV 20 (1976Me12).<br>$\omega\gamma=0.7$ eV 2 (2001Vo24).<br>$A_2=2$ (2001Vo24).<br>$J^\pi$ : spin=5/2 from $\gamma(\theta)$ in 1971Wi13 and 1976Me12. Other: spin=3/2 from 1977Ko34.<br>$T_{1/2}$ : lifetime $\tau < 20$ fs (1971Wi13, DSAM).<br>$E_p(\text{lab})$ (keV): weighted average of 1266.5 10 (1972Hu10), 1267.9 11 (1973Fa07), and 1265.7 8 (1976Me12).<br>$\omega\gamma=3.4$ eV, $\Gamma_\gamma \approx 0.57$ eV (1971Wi13).<br>$\omega\gamma=1.4$ eV 4 (1972Hu10).<br>$\omega\gamma=1.31$ eV (1973Fa07).<br>$\omega\gamma=1.3$ eV 4 (1976Me12).<br>$J^\pi$ : from $\gamma(\theta)$ in 1977Ko34. Spin=3/2,5/2 from $\gamma(\theta)$ in 1976Me12.<br>$E_p(\text{lab})$ (keV): weighted average of 1285.5 11 (1972Hu10), 1286.1 11 (1973Fa07), and 1284.5 5 (1976Me12).<br>$\omega\gamma=0.54$ eV 16 (1972Hu10).<br>$\omega\gamma=0.50$ eV (1973Fa07).<br>$\omega\gamma=0.77$ eV 23 (1976Me12).<br>$E_p(\text{lab})$ (keV): weighted average of 1323.7 12 (1972Hu10), 1323.5 12 (1973Fa07), and 1324.2 8 (1976Me12).<br>$\omega\gamma=0.23$ eV 11 (1972Hu10).<br>$\omega\gamma=0.23$ eV (1973Fa07).<br>$\omega\gamma=0.36$ eV 11 (1976Me12).<br>$J^\pi$ : spin=5/2 from $\gamma(\theta)$ in 1977Ko34. Spin=5/2,7/2 from $\gamma(\theta)$ in 1976Me12.<br>$E_p(\text{lab})$ (keV): weighted average of 1339.8 11 (1972Hu10), 1339.4 11 (1973Fa07), and 1339.7 8 (1976Me12).<br>$\omega\gamma=0.68$ eV 20 (1972Hu10).<br>$\omega\gamma=0.63$ eV (1973Fa07). |
| 7518.8 7                  | 7/2              |           | 1182.0 7                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7548.9 6                  | 7/2 $^-$         | <0.69 fs  | 1213.0 6                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7561.3 6                  | 1/2,3/2          |           | 1225.8 6                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7600.8 8                  | 5/2              | <13.9 fs  | 1266.5 8                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7618.7 5                  | 5/2              |           | 1284.9 5                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7656.6 8                  |                  |           | 1323.9 8                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7671.9 8                  | 5/2              |           | 1339.7 8                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup> | T <sub>1/2</sub> | E <sub>p</sub> (lab) (keV) | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------|-----------------------------|------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7685.5 8              | 3/2 <sup>-</sup>            |                  | 1353.7 8                   | $\omega\gamma=0.59$ eV 18 (1976Me12).<br>J <sup>π</sup> : 3/2 <sup>-</sup> from $\sigma(E_p,\theta)$ in 1981Bi05. Spin=3/2 from $\gamma(\theta)$ in 1976Me12.<br>Spin=3/2,5/2 from $\gamma(\theta)$ in 1977Ko34.<br>E <sub>p</sub> (lab) (keV): weighted average of 1353.7 11 (1972Hu10), 1353.4 11 (1973Fa07), and 1353.8 8 (1976Me12).<br>$\Gamma_p=445$ eV 11 (1981Bi05).<br>$\omega\gamma=1.13$ eV 34 (1972Hu10).<br>$\omega\gamma=1.09$ eV (1973Fa07).<br>$\omega\gamma=1.3$ eV 4 (1976Me12).                                                           |
| 7693.9 8              |                             |                  | 1362.3 8                   | E <sub>p</sub> (lab) (keV): weighted average of 1362.4 12 (1972Hu10) and 1362.3 8 (1976Me12).<br>$\omega\gamma=0.18$ eV 9 (1972Hu10).<br>$\omega\gamma=0.27$ eV 8 (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                |
| 7706.4 8              | 5/2 <sup>+</sup>            |                  | 1375.2 8                   | J <sup>π</sup> : 5/2 <sup>+</sup> from $\sigma(E_p,\theta)$ in 1981Bi05. Spin=5/2 from $\gamma(\theta)$ in 1977Ko34.<br>Spin=3/2,5/2 from $\gamma(\theta)$ in 1976Me12 and 5/2 is favored.<br>E <sub>p</sub> (lab) (keV): weighted average of 1375.7 11 (1972Hu10), 1375.0 11 (1973Fa07), and 1375.1 8 (1976Me12).<br>$\omega\gamma=1.7$ eV 5 (1972Hu10).<br>$\omega\gamma=1.40$ eV (1973Fa07).<br>$\omega\gamma=2.0$ eV 6 (1976Me12).<br>$\Gamma_p=4$ eV 1 (1981Bi05).                                                                                      |
| 7744.8 9              | 7/2                         |                  | 1414.8 9                   | J <sup>π</sup> : from $\gamma(\theta)$ for J <sub>f</sub> (2645.7)=7/2, J <sub>f</sub> (3163)=7/2, J <sub>f</sub> (3942.9)=9/2 (1977Ko35).<br>E <sub>p</sub> (lab) (keV): weighted average of 1415.6 12 (1972Hu10), 1414.2 12 (1973Fa07), and 1414.7 9 (1976Me12).<br>$\omega\gamma=0.59$ eV 18 (1972Hu10).<br>$\omega\gamma=0.54$ eV (1973Fa07).<br>$\omega\gamma=0.77$ eV 23 (1976Me12).                                                                                                                                                                   |
| 7748                  |                             |                  | 1418                       | E(level): from 1960An06.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 7777.1 9              |                             |                  | 1448.0 9                   | E <sub>p</sub> (lab) (keV): weighted average of 1449.0 11 (1972Hu10), 1447.5 12 (1973Fa07), and 1447.5 9 (1976Me12).<br>$\omega\gamma=0.68$ eV 20 (1972Hu10).<br>$\omega\gamma=(0.63)$ eV (1973Fa07).<br>$\omega\gamma=0.68$ eV 20 (1976Me12).                                                                                                                                                                                                                                                                                                               |
| 7781.7 11             | 5/2 <sup>-</sup>            |                  | 1452.8 11                  | J <sup>π</sup> : proposed by 1976Me12 based on strengths of primary transitions.<br>E <sub>p</sub> (lab) (keV): weighted average of 1453.6 11 (1972Hu10), 1451.8 15 (1973Fa07), and 1452.3 13 (1976Me12).<br>$\omega\gamma=0.36$ eV 18 (1972Hu10).<br>$\omega\gamma=(0.41)$ eV (1973Fa07).<br>$\omega\gamma=0.41$ eV 12 (1976Me12).                                                                                                                                                                                                                          |
| 7797.1 9              | 1/2                         |                  | 1468.6 9                   | J <sup>π</sup> : from $\gamma(\theta)$ in 1988Va06, which rejects spin=3/2 from $\gamma(\theta)$ in 1976Me12.<br>E <sub>p</sub> (lab) (keV): weighted average of 1469.6 12 (1972Hu10), 1468.2 12 (1973Fa07), and 1468.2 9 (1976Me12).<br>$\omega\gamma=0.68$ eV 20 (1972Hu10).<br>$\omega\gamma=0.72$ eV (1973Fa07).<br>$\omega\gamma=0.63$ eV 19 (1976Me12).                                                                                                                                                                                                |
| 7837.2 10             | 3/2                         | <3.47 fs         | 1509.9 10                  | T=3/2<br>J <sup>π</sup> : spin from $\gamma(\theta)$ in 1967Wa22.<br>T <sub>1/2</sub> : lifetime $\tau<5$ fs (1971Wi13, DSAM).<br>E <sub>p</sub> (lab) (keV): weighted average of 1509.1 15 (1973Fa07), 1510.2 10 (1976Me12), and 1510 1 (2001Vo24).<br>$\omega\gamma=3.4$ eV 6 (1967Wa22).<br>$\omega\gamma\approx 7.5$ eV, $\Gamma_\gamma=1.9$ eV (1971Wi13).<br>$\omega\gamma=5.0$ eV 15 for E <sub>p</sub> =1510 doublet (1972Hu10).<br>$\omega\gamma=0.59$ eV (1973Fa07).<br>$\omega\gamma=2.2$ eV 7 (1976Me12).<br>$\omega\gamma=2.1$ eV 3 (2001Vo24). |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup> | E <sub>p</sub> (lab) (keV) | Comments                                                                                                                                                                                                                                                                                               |
|-----------------------|-----------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7839.0 9              |                             | 1511.8 9                   | $\Gamma_{\gamma}=0.55$ eV 6 (2001Vo24).<br>E <sub>p</sub> (lab) (keV): weighted average of 1512.3 15 (1973Fa07) and 1511.6 9 (1976Me12).<br>$\omega\gamma=5.0$ eV 15 for E <sub>p</sub> =1510 doublet (1972Hu10).<br>$\omega\gamma=3.94$ eV (1973Fa07).<br>$\omega\gamma=2.8$ eV 8 (1976Me12).         |
| 7868.7 10             |                             | 1542.3 10                  | E <sub>p</sub> (lab) (keV): weighted average of 1542.3 13 (1972Hu10), 1542.3 12 (1973Fa07), and 1542.2 10 (1976Me12).<br>$\omega\gamma=0.23$ eV 11 (1972Hu10).<br>$\omega\gamma=(0.23)$ eV (1973Fa07).<br>$\omega\gamma=0.45$ eV 14 (1976Me12).                                                        |
| 7873?                 |                             | 1547                       | E(level): from 1960An06.                                                                                                                                                                                                                                                                               |
| 7880.8 9              | 3/2 <sup>-</sup> ,5/2       | 1554.8 9                   | J <sup>π</sup> : from $\gamma(\theta)$ in 1976Me12.<br>E <sub>p</sub> (lab) (keV): weighted average of 1554.9 12 (1972Hu10), 1554.8 11 (1973Fa07), and 1554.7 9 (1976Me12).<br>$\omega\gamma=0.72$ eV 22 (1972Hu10).<br>$\omega\gamma=0.59$ eV (1973Fa07).<br>$\omega\gamma=1.00$ eV 30 (1976Me12).    |
| 7885?                 |                             | 1559                       | E(level): from 1960An06.                                                                                                                                                                                                                                                                               |
| 7889.1 15             |                             | 1563.3 15                  | E <sub>p</sub> (lab) (keV): from 1973Fa07.<br>$\omega\gamma=(0.05)$ eV (1973Fa07).                                                                                                                                                                                                                     |
| 7899.2 7              | (3/2 <sup>-</sup> ,5/2)     | 1573.7 7                   | E <sub>p</sub> (lab) (keV): weighted average of 1574.4 12 (1972Hu10), 1573.2 11 (1973Fa07), and 1573.7 7 (1976Me12).<br>$\omega\gamma=0.63$ eV 19 (1972Hu10).<br>$\omega\gamma=0.45$ eV (1973Fa07).<br>$\omega\gamma=1.00$ eV 30 (1976Me12).                                                           |
| 7903?                 |                             | 1578                       | E(level): from 1960An06.                                                                                                                                                                                                                                                                               |
| 7923.3 9              |                             | 1598.6 9                   | E <sub>p</sub> (lab) (keV): weighted average of 1598.9 13 (1972Hu10), 1598.7 12 (1973Fa07), and 1598.4 9 (1976Me12).<br>$\omega\gamma=0.36$ eV 18 (1972Hu10).<br>$\omega\gamma=(0.32)$ eV (1973Fa07).<br>$\omega\gamma=0.45$ eV 14 (1976Me12).                                                         |
| 7930?                 |                             | 1605                       | E(level): from 1960An06.                                                                                                                                                                                                                                                                               |
| 7949?                 |                             | 1625                       | E(level): from 1960An06.                                                                                                                                                                                                                                                                               |
| 7970.3 10             |                             | 1646.9 10                  | E <sub>p</sub> (lab) (keV): weighted average of 1647.3 12 (1972Hu10), 1645.9 13 (1973Fa07), and 1647.1 10 (1976Me12).<br>$\omega\gamma=0.54$ eV 16 (1972Hu10).<br>$\omega\gamma=(0.45)$ eV (1973Fa07).<br>$\omega\gamma=0.86$ eV 26 (1976Me12).                                                        |
| 7987.8 10             | 3/2                         | 1665.0 10                  | J <sup>π</sup> : from $\gamma(\theta)$ in 1976Me12.<br>E <sub>p</sub> (lab) (keV): weighted average of 1665.2 12 (1972Hu10), 1663.3 12 (1973Fa07), and 1666.1 10 (1976Me12).<br>$\omega\gamma=0.68$ eV 20 (1972Hu10).<br>$\omega\gamma=(0.59)$ eV (1973Fa07).<br>$\omega\gamma=0.54$ eV 16 (1976Me12). |
| 7995.6 10             | 5/2                         | 1673.0 10                  | E <sub>p</sub> (lab) (keV): weighted average of 1673.7 11 (1972Hu10), 1672.4 13 (1973Fa07), and 1672.7 10 (1976Me12).<br>$\omega\gamma=1.2$ eV 4 (1972Hu10).<br>$\omega\gamma=(1.04)$ eV (1973Fa07).<br>$\omega\gamma=0.95$ eV 28 (1976Me12).                                                          |
| 8001.0 9              | (7/2,9/2 <sup>+</sup> )     | 1678.6 9                   | E <sub>p</sub> (lab) (keV): weighted average of 1679.7 11 (1972Hu10), 1678.2 13 (1973Fa07), and 1678.1 9 (1976Me12).<br>$\omega\gamma=1.9$ eV 6 (1972Hu10).<br>$\omega\gamma=(1.18)$ eV (1973Fa07).<br>$\omega\gamma=1.3$ eV 4 (1976Me12).                                                             |
| 8005.0 10             | 5/2 <sup>+</sup>            | 1682.7 10                  | J <sup>π</sup> : from $\sigma(\text{E}_p,\theta)$ in 1981Bi05. Spin=5/2 from $\gamma(\theta)$ in 1981Bi05. Spin=3/2,5/2 from                                                                                                                                                                           |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup>                 | E <sub>p</sub> (lab) (keV) | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------|---------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                                             |                            | $\gamma(\theta)$ in <a href="#">1976Me12</a> .<br>E <sub>p</sub> (lab) (keV): weighted average of 1684.3 <i>I3</i> ( <a href="#">1972Hu10</a> ), 1681.9 <i>I1</i> ( <a href="#">1973Fa07</a> ), and 1682.5 <i>I0</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=(1.54)$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=2.2$ eV <i>7</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=2.0$ eV <i>6</i> ( <a href="#">1972Hu10</a> ).<br>$\Gamma_p=21$ eV <i>3</i> ( <a href="#">1981Bi05</a> ).<br>E <sub>p</sub> (lab) (keV): weighted average of 1714.0 <i>I1</i> ( <a href="#">1972Hu10</a> ), 1713.5 <i>I4</i> ( <a href="#">1973Fa07</a> ), and 1714.9 <i>I0</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=0.41$ eV <i>20</i> ( <a href="#">1972Hu10</a> ).<br>$\omega\gamma=(0.41)$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=0.36$ eV <i>11</i> ( <a href="#">1976Me12</a> ). |
| 8035.7 <i>I0</i>      |                                             | 1714.3 <i>I0</i>           | J <sup>π</sup> : from $\gamma(\theta)$ in <a href="#">1976Me12</a> .<br>E <sub>p</sub> (lab) (keV): weighted average of 1717.9 <i>I1</i> ( <a href="#">1972Hu10</a> ), 1716.3 <i>I1</i> ( <a href="#">1973Fa07</a> ), and 1717.3 <i>I1</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=1.6$ eV <i>5</i> ( <a href="#">1972Hu10</a> ).<br>$\omega\gamma=1.36$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=1.3$ eV <i>4</i> ( <a href="#">1976Me12</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 8038.5 <i>I1</i>      | 3/2                                         | 1717.2 <i>I1</i>           | E <sub>p</sub> (lab) (keV): weighted average of 1756.0 <i>I0</i> ( <a href="#">1972Hu10</a> ), 1755.4 <i>I1</i> ( <a href="#">1973Fa07</a> ), and 1755.7 <i>I1</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=0.72$ eV <i>22</i> ( <a href="#">1972Hu10</a> ).<br>$\omega\gamma=(0.59)$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=0.72$ eV <i>22</i> ( <a href="#">1976Me12</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 8075.9 <i>I0</i>      |                                             | 1755.7 <i>I0</i>           | E <sub>p</sub> (lab) (keV): from <a href="#">1960An06</a> .<br>E <sub>p</sub> (lab) (keV): weighted average of 1776.7 <i>I1</i> ( <a href="#">1972Hu10</a> ), 1776.9 <i>I1</i> ( <a href="#">1973Fa07</a> ), and 1776.0 <i>I1</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=1.13$ eV <i>34</i> ( <a href="#">1972Hu10</a> ).<br>$\omega\gamma=1.13$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=1.1$ eV <i>3</i> ( <a href="#">1976Me12</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 8080?                 |                                             | 1760                       | J <sup>π</sup> : from $\gamma(\theta)$ in <a href="#">1976Me12</a> .<br>E <sub>p</sub> (lab) (keV): weighted average of 1787.7 <i>I1</i> ( <a href="#">1972Hu10</a> ), 1786.6 <i>I1</i> ( <a href="#">1973Fa07</a> ), and 1786.9 <i>I1</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=1.9$ eV <i>6</i> ( <a href="#">1972Hu10</a> ).<br>$\omega\gamma=1.63$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=1.7$ eV <i>5</i> ( <a href="#">1976Me12</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 8096.1 <i>I1</i>      | (5/2, 7/2 <sup>+</sup> )                    | 1776.5 <i>I1</i>           | E <sub>p</sub> (lab) (keV): weighted average of 1794.8 <i>I3</i> ( <a href="#">1972Hu10</a> ), 1793.3 <i>I4</i> ( <a href="#">1973Fa07</a> ), and 1794.2 <i>I1</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=0.54$ eV <i>16</i> ( <a href="#">1972Hu10</a> ).<br>$\omega\gamma=(0.50)$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=0.72$ eV <i>22</i> ( <a href="#">1976Me12</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 8106.4 <i>I1</i>      | 3/2                                         | 1787.1 <i>I1</i>           | E <sub>p</sub> (lab) (keV): weighted average of 1829.1 <i>I3</i> ( <a href="#">1972Hu10</a> ), 1828.6 <i>I2</i> ( <a href="#">1973Fa07</a> ), and 1829.5 <i>I1</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=1.09$ eV <i>33</i> ( <a href="#">1972Hu10</a> ).<br>$\omega\gamma=0.68$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=0.77$ eV <i>23</i> ( <a href="#">1976Me12</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 8113.3 <i>I1</i>      | (1/2 <sup>+</sup> , 3/2, 5/2 <sup>+</sup> ) | 1794.2 <i>I1</i>           | E <sub>p</sub> (lab) (keV): weighted average of 1839.0 <i>I2</i> ( <a href="#">1972Hu10</a> ), 1838.8 <i>I2</i> ( <a href="#">1973Fa07</a> ), and 1839.1 <i>I1</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=0.95$ eV <i>28</i> ( <a href="#">1972Hu10</a> ).<br>$\omega\gamma=0.95$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=0.86$ eV <i>26</i> ( <a href="#">1976Me12</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 8147.2 <i>I1</i>      | 3/2 <sup>-</sup>                            | 1829.1 <i>I1</i>           | E <sub>p</sub> (lab) (keV): weighted average of 1862.7 <i>I3</i> ( <a href="#">1972Hu10</a> ), 1862.1 <i>I3</i> ( <a href="#">1973Fa07</a> ), and 1862.1 <i>I2</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=0.36$ eV <i>18</i> ( <a href="#">1972Hu10</a> ).<br>$\omega\gamma=(0.32)$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=0.41$ eV <i>12</i> ( <a href="#">1976Me12</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 8156.8 <i>I1</i>      | (5/2 <sup>+</sup> , 7/2 <sup>-</sup> )      | 1839.0 <i>I1</i>           | J <sup>π</sup> : from $\sigma(E_p, \theta)$ in <a href="#">1981Bi05</a> . Spin=5/2 from $\gamma(\theta)$ in <a href="#">1996Ka21</a> and <a href="#">2001Vo24</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 8179.4 <i>I2</i>      |                                             | 1862.3 <i>I2</i>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8208.2 <i>I0</i>      | 5/2 <sup>+</sup>                            | 1891.9 <i>I0</i>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sub>z</sub> | E <sub>p</sub> (lab) (keV) | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------|-----------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                             |                            | E <sub>p</sub> (lab) (keV): weighted average of 1893.2 11 (1972Hu10), 1891.3 12 (1976Me12), 1892.0 11 (1973Fa07), and 1891 1 (2001Vo24).<br>$\omega\gamma=3.6$ eV 11 (1972Hu10).<br>$\omega\gamma=2.71$ eV (1973Fa07).<br>$\omega\gamma=4.1$ eV 12 (1976Me12).<br>$\omega\gamma=4.1$ eV 5, $\Gamma_p=39$ eV 5 (1981Bi05).<br>$\omega\gamma=2.5$ eV 7 (1996Ka21).<br>$\omega\gamma=4.0$ eV 8 (2001Vo24), sum of $\omega\gamma$ for the resonances at E <sub>p</sub> =1891 and 1893, $\Gamma_\gamma=0.70$ eV 14 (2001Vo24). |
| 8209                  | 1/2 <sup>+</sup>            | 1893                       | E(level): from 1974Bi16; a weak resonance in the tail of the 1892 resonance.<br>J <sup>π</sup> : from $\sigma(E_p, \theta)$ in 1981Bi05.<br>$\Gamma_p=120$ eV 11 (1981Bi05).                                                                                                                                                                                                                                                                                                                                              |
| 8216.3 10             | 5/2                         | 1900.3 10                  | J <sup>π</sup> : spin=5/2 from $\gamma(\theta)$ in 2001Vo24 and 1996Ka21.<br>E <sub>p</sub> (lab) (keV): weighted average of 1901.8 11 (1972Hu10), 1899.7 12 (1973Fa07), 1899.6 11 (1976Me12), and 1900 1 (2001Vo24).<br>$\omega\gamma=2.8$ eV 8 (1972Hu10).<br>$\omega\gamma=2.13$ eV (1973Fa07).<br>$\omega\gamma=3.2$ eV 10 (1976Me12).<br>$\omega\gamma=1.7$ eV 5 (1996Ka21).<br>$\omega\gamma=3.2$ eV 5 (2001Vo24).<br>$\Gamma_\gamma=0.55$ eV 8 (2001Vo24).                                                         |
| 8242.4 12             |                             | 1927.1 12                  | E <sub>p</sub> (lab) (keV): weighted average of 1928.2 13 (1972Hu10), 1926.8 13 (1973Fa07), and 1926.5 12 (1976Me12).<br>$\omega\gamma=0.86$ eV 26 (1972Hu10).<br>$\omega\gamma=0.86$ eV (1973Fa07).<br>$\omega\gamma=1.00$ eV 30 (1976Me12).<br>$\omega\gamma=1.0$ eV 3 (2001Vo24).<br>$\Gamma_\gamma=0.25$ eV 8 (2001Vo24).<br>20% of I $\gamma$ transitions to unknown levels (1973Fa07).                                                                                                                              |
| 8251? 5               |                             | 1936 5                     | E <sub>p</sub> (lab) (keV): from 1967Da12.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 8268.9 5              |                             | 1954.4 5                   | E <sub>p</sub> (lab) (keV): weighted average of 1955.9 13 (1972Hu10), 1952 2 (1973Fa07), 1954.3 12 (1976Me12), and 1954.4 5 (1976Sp08).<br>$\omega\gamma=0.81$ eV 24 (1972Hu10).<br>$\omega\gamma=(0.86)$ eV (1973Fa07).<br>$\omega\gamma=0.90$ eV 27 (1976Me12).<br>$\omega\gamma=0.6$ eV 3 (1976Sp08).                                                                                                                                                                                                                  |
| 8277.2 5              |                             | 1963.0 5                   | E <sub>p</sub> (lab) (keV): weighted average of 1964.1 14 (1972Hu10), 1962 2 (1973Fa07), 1962.8 12 (1976Me12), and 1962.9 5 (1976Sp08).<br>$\omega\gamma=0.72$ eV 22 (1972Hu10).<br>$\omega\gamma=(0.77)$ eV (1973Fa07).<br>$\omega\gamma=0.68$ eV 20 (1976Me12).<br>$\omega\gamma=0.7$ eV 4 (1976Sp08).                                                                                                                                                                                                                  |
| 8282.4 5              | (3/2 <sup>-</sup> , 5/2)    | 1968.3 5                   | E <sub>p</sub> (lab) (keV): weighted average of 1970.0 13 (1972Hu10), 1971 2 (1973Fa07), 1967.8 12 (1976Me12), and 1967.9 5 (1976Sp08).<br>$\omega\gamma=1.09$ eV 33 (1972Hu10).<br>$\omega\gamma=(1.09)$ eV (1973Fa07).<br>$\omega\gamma=0.59$ eV 18 (1976Me12).<br>$\omega\gamma=0.4$ eV 2 (1976Sp08).                                                                                                                                                                                                                  |
| 8287.82 30            | 1/2 <sup>-</sup>            | 1973.89 30                 | J <sup>π</sup> : from $\sigma(E_p, \theta)$ in 1981Bi05, which also rejects spin=3/2 from $\gamma(\theta)$ in 1976Me12.<br>E <sub>p</sub> (lab) (keV): weighted average of 1975.6 13 (1972Hu10), 1973.8 12 (1976Me12), and 1973.80 30 (1976Sp08).<br>$\omega\gamma=1.3$ eV 4 (1972Hu10).<br>$\omega\gamma=0.72$ eV 22 (1976Me12).<br>$\omega\gamma=0.9$ eV 5 (1976Sp08).<br>$\Gamma_p=48$ eV 6 (1981Bi05).                                                                                                                |
| 8297.2 9              |                             | 1983.5 9                   | E <sub>p</sub> (lab) (keV): weighted average of 1983.7 5 (1976Sp08) and 1980 2 (1973Fa07).                                                                                                                                                                                                                                                                                                                                                                                                                                |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup> | E <sub>p</sub> (lab) (keV) | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|-----------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8298.2 5              | 3/2                         | 1984.6 5                   | $\omega\gamma=(1.36)$ eV (1973Fa07).<br>$\omega\gamma=0.6$ eV 3 (1976Sp08).<br>J <sup>π</sup> : from $\gamma(\theta)$ in 1976Me12.<br>E <sub>p</sub> (lab) (keV): weighted average of 1986.0 13 (1972Hu10), 1983.0 20 (1973Fa07), 1984.5 12 (1976Me12), and 1984.5 5 (1976Sp08).<br>$\omega\gamma=1.8$ eV 5 (1972Hu10).<br>$\omega\gamma=(0.77)$ eV (1973Fa07).<br>$\omega\gamma=1.6$ eV 5 (1976Me12).<br>$\omega\gamma=0.9$ eV 5 (1976Sp08). |
| 8318.1 3              |                             | 2005.1 3                   | E <sub>p</sub> (lab) (keV): weighted average of 2006.7 12 (1972Hu10), 2004.6 13 (1973Fa07), 2006.0 13 (1976Me12), and 2005.0 3 (1976Sp08).<br>$\omega\gamma=1.9$ eV 6 (1972Hu10).<br>$\omega\gamma=2.08$ eV (1973Fa07).<br>$\omega\gamma=1.9$ eV 6 (1976Me12).<br>$\omega\gamma=2.6$ eV 5 (1976Sp08).                                                                                                                                         |
| 8323.1 13             |                             | 2010.2 13                  | E <sub>p</sub> (lab) (keV): weighted average of 2010.5 13 (1972Hu10) and 2009.9 13 (1976Me12).<br>$\omega\gamma=0.45$ eV 14 (1972Hu10).<br>$\omega\gamma=0.72$ eV 22 (1976Me12).                                                                                                                                                                                                                                                              |
| 8346? 5               |                             | 2034 5                     | E <sub>p</sub> (lab) (keV): from 1967Da12.                                                                                                                                                                                                                                                                                                                                                                                                    |
| 8381.8 8              | 5/2 <sup>+</sup>            | 2070.7 8                   | J <sup>π</sup> : spin=5/2 from $\gamma(\theta)$ in 2001Vo24 and 1967Ko23.<br>E <sub>p</sub> (lab) (keV): weighted average of 2073.5 12 (1972Hu10), 2070.4 5 (1976Sp08), and 2070 1 (2001Vo24).<br>$\omega\gamma\approx 4$ eV (1967Ko23).<br>$\omega\gamma=6.4$ eV 19 (1972Hu10).<br>$\omega\gamma=6.0$ eV 12 (1976Sp08).<br>$\omega\gamma=5.9$ eV 6 (2001Vo24).<br>$\Gamma_\gamma=1.02$ eV 10 (2001Vo24).                                     |
| 8389.4 16             |                             | 2078.5 16                  | E <sub>p</sub> (lab) (keV): unweighted average of 2080.0 14 (1972Hu10) and 2076.9 5 (1976Sp08).<br>$\omega\gamma=0.27$ eV 14 (1972Hu10).<br>$\omega\gamma=0.4$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                               |
| 8402.9 5              |                             | 2092.4 5                   | E <sub>p</sub> (lab) (keV): from 1976Sp08.<br>$\omega\gamma=0.7$ eV 4 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                             |
| 8405.7 14             |                             | 2095.3 14                  | E <sub>p</sub> (lab) (keV): unweighted average of 2096.6 12 (1972Hu10) and 2093.9 5 (1976Sp08).<br>$\omega\gamma=2.9$ eV 9 (1972Hu10).<br>$\omega\gamma=2.3$ eV 5 (1976Sp08).                                                                                                                                                                                                                                                                 |
| 8409.6 18             |                             | 2099.3 18                  | E <sub>p</sub> (lab) (keV): unweighted average of 2101.0 14 (1972Hu10) and 2097.5 5 (1976Sp08).<br>$\omega\gamma=1.6$ eV 5 (1972Hu10).<br>$\omega\gamma=1.4$ eV 3 (1976Sp08).                                                                                                                                                                                                                                                                 |
| 8416.7 5              |                             | 2106.6 5                   | $\omega\gamma=0.5$ eV 3 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8430.3 5              |                             | 2120.6 5                   | $\omega\gamma=1.0$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8434.8 5              |                             | 2125.2 5                   | $\omega\gamma=0.5$ eV 3 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8464.3 5              |                             | 2155.6 5                   | $\omega\gamma=0.9$ eV 5 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8484.4 5              | 3/2 <sup>+</sup>            | 2176.3 5                   | J <sup>π</sup> : from 2001Vo24 based on proposed isobaric analog to 3/2 <sup>+</sup> in $^{35}\text{S}$ .<br>E <sub>p</sub> (lab) (keV): weighted average of 2176.4 5 (1976Sp08) and 2176 1 (2001Vo24).<br>$\omega\gamma=2.1$ eV 4 (1976Sp08).<br>$\omega\gamma=2.1$ eV 4 (2001Vo24).<br>$\Gamma_\gamma=0.36$ eV 6 (2001Vo24).                                                                                                                |
| 8486.1 5              |                             | 2178.0 5                   | $\omega\gamma=1.9$ eV 4 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8506.5 5              |                             | 2199.1 5                   | $\omega\gamma=0.3$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8514.3 5              |                             | 2207.1 5                   | $\omega\gamma=0.5$ eV 3 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8533.9 5              |                             | 2227.3 5                   | $\omega\gamma=0.10$ eV 5 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                          |
| 8571.9 5              |                             | 2266.4 5                   | $\omega\gamma=2.8$ eV 6 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8580.5 5              |                             | 2275.3 5                   | $\omega\gamma=1.0$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8585.8 5              |                             | 2280.7 5                   | $\omega\gamma=0.10$ eV 5 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                          |
| 8590.0 5              |                             | 2285.0 5                   | $\omega\gamma=0.7$ eV 4 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8612.0 5              |                             | 2307.7 5                   | $\omega\gamma=1.0$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup>            | E <sub>p</sub> (lab) (keV) | Comments                                                                                                                                                                                                           |
|-----------------------|----------------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8613.7 5              |                                        | 2309.4 5                   | $\omega\gamma=2.4$ eV 5 (1976Sp08).                                                                                                                                                                                |
| 8618.1 5              | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) | 2314.0 5                   | J <sup>π</sup> : most probable spin=5/2 from $\gamma(\theta)$ in 1970Ko33.                                                                                                                                         |
|                       |                                        |                            | $\omega\gamma=3.7$ eV 8 (1976Sp08).                                                                                                                                                                                |
| 8630.1 3              | 7/2                                    | 2326.3 3                   | J <sup>π</sup> : spin=3/2, 7/2 from $\gamma(\theta)$ in 1976Sp09; 1976Sp09 ruled out 3/2 based on 4687 $\gamma$ to 9/2 <sup>+</sup> , 3943 level. Other: most probable spin=3/2 from $\gamma(\theta)$ in 1970Ko33. |
|                       |                                        |                            | $\omega\gamma=2.6$ eV 5 (1976Sp08).                                                                                                                                                                                |
| 8640                  | 5/2                                    | 2336                       | J <sup>π</sup> : probable spin=5/2 from $\gamma(\theta)$ in 1970Ko33.                                                                                                                                              |
|                       |                                        |                            | E <sub>p</sub> (lab) (keV): from 1970Ko33.                                                                                                                                                                         |
| 8641.4 5              | 3/2, 5/2                               | 2338.0 5                   | J <sup>π</sup> : spin=3/2 from $\gamma(\theta)$ in 1976Sp09; spin=5/2 is also possible.                                                                                                                            |
|                       |                                        |                            | $\omega\gamma=0.9$ eV 5 (1976Sp08).                                                                                                                                                                                |
| 8686.2 5              |                                        | 2384.1 5                   | $\omega\gamma=2.2$ eV 4 (1976Sp08).                                                                                                                                                                                |
| 8688.3 5              |                                        | 2386.3 5                   | $\omega\gamma=0.7$ eV 4 (1976Sp08).                                                                                                                                                                                |
| 8696.9 5              |                                        | 2395.1 5                   | $\omega\gamma=0.6$ eV 3 (1976Sp08).                                                                                                                                                                                |
| 8706.3 5              |                                        | 2404.8 5                   | $\omega\gamma=0.2$ eV 1 (1976Sp08).                                                                                                                                                                                |
| 8717.7 5              |                                        | 2416.5 5                   | $\omega\gamma=0.5$ eV 3 (1976Sp08).                                                                                                                                                                                |
| 8750.9 5              |                                        | 2450.7 5                   | $\omega\gamma=1.7$ eV 3 (1976Sp08).                                                                                                                                                                                |
| 8767.0 5              |                                        | 2467.3 5                   | $\omega\gamma=0.3$ eV 2 (1976Sp08).                                                                                                                                                                                |
| 8772.9 5              |                                        | 2473.4 5                   | $\omega\gamma=0.6$ eV 3 (1976Sp08).                                                                                                                                                                                |
| 8779.8 5              |                                        | 2480.5 5                   | $\omega\gamma=1.7$ eV 3 (1976Sp08).                                                                                                                                                                                |
| 8787.2 5              |                                        | 2488.1 5                   | $\omega\gamma=1.2$ eV 2 (1976Sp08).                                                                                                                                                                                |
| 8798.4 5              |                                        | 2499.6 5                   | $\omega\gamma=1.7$ eV 3 (1976Sp08).                                                                                                                                                                                |
| 8820.9 5              |                                        | 2522.8 5                   | $\omega\gamma=0.7$ eV 4 (1976Sp08).                                                                                                                                                                                |
| 8824.2 5              |                                        | 2526.2 5                   | $\omega\gamma=3.6$ eV 7 (1976Sp08).                                                                                                                                                                                |
| 8829.3 5              |                                        | 2531.4 5                   | $\omega\gamma=2.1$ eV 4 (1976Sp08).                                                                                                                                                                                |
| 8833.1 5              |                                        | 2535.3 5                   | $\omega\gamma=0.7$ eV 4 (1976Sp08).                                                                                                                                                                                |
| 8838.1 5              | 7/2 <sup>-</sup>                       | 2540.5 5                   | J <sup>π</sup> : 5/2 <sup>-</sup> , 7/2 <sup>-</sup> from $\sigma(\text{E}_p, \theta)$ in 1981Bi05. Spin=7/2 from $\gamma(\theta)$ in 1981Bi05.                                                                    |
|                       |                                        |                            | $\omega\gamma=1.5$ eV 3 (1976Sp08).                                                                                                                                                                                |
|                       |                                        |                            | $\Gamma_p=4$ eV 1 (1981Bi05).                                                                                                                                                                                      |
| 8856.1 5              |                                        | 2559.0 5                   | $\omega\gamma=4.5$ eV 9 (1976Sp08).                                                                                                                                                                                |
| 8868.6 5              |                                        | 2571.9 5                   | $\omega\gamma=1.0$ eV 2 (1976Sp08).                                                                                                                                                                                |
| 8884.1 5              |                                        | 2587.9 5                   | $\omega\gamma=0.4$ eV 2 (1976Sp08).                                                                                                                                                                                |
| 8886.1 5              |                                        | 2589.9 5                   | $\omega\gamma=1.4$ eV 3 (1976Sp08).                                                                                                                                                                                |
| 8893.3 5              | (5/2, 7/2) <sup>+</sup>                | 2597.3 5                   | J <sup>π</sup> : from 2001Vo24 with no supporting arguments.                                                                                                                                                       |
|                       |                                        |                            | E <sub>p</sub> (lab) (keV): weighted average of 2597.4 5 (1976Sp08) and 2597 1 (2001Vo24).                                                                                                                         |
|                       |                                        |                            | $\omega\gamma=1.2$ eV 2 (1976Sp08).                                                                                                                                                                                |
|                       |                                        |                            | $\omega\gamma=1.2$ eV 4 (2001Vo24).                                                                                                                                                                                |
|                       |                                        |                            | $\Gamma_\gamma=0.20$ eV 6 (2001Vo24).                                                                                                                                                                              |
| 8904.8 5              |                                        | 2609.2 5                   | $\omega\gamma=1.0$ eV 2 (1976Sp08).                                                                                                                                                                                |
| 8906.8 5              | 5/2 <sup>+</sup>                       | 2611.2 5                   | J <sup>π</sup> : from 2001Vo24 with no supporting arguments.                                                                                                                                                       |
|                       |                                        |                            | E <sub>p</sub> (lab) (keV): weighted average of 2611.3 5 (1976Sp08) and 2611 1 (2001Vo24).                                                                                                                         |
|                       |                                        |                            | $\omega\gamma=1.5$ eV 3 (1976Sp08).                                                                                                                                                                                |
|                       |                                        |                            | $\omega\gamma=1.5$ eV 3 (2001Vo24).                                                                                                                                                                                |
|                       |                                        |                            | $\Gamma_\gamma=0.25$ eV 5 (2001Vo24).                                                                                                                                                                              |
| 8919.8 5              |                                        | 2624.6 5                   | $\omega\gamma=1.1$ eV 2 (1976Sp08).                                                                                                                                                                                |
| 8933.3 5              |                                        | 2638.5 5                   | $\omega\gamma=1.0$ eV 2 (1976Sp08).                                                                                                                                                                                |
| 8953.0 5              | 3/2                                    | 2658.8 5                   | J <sup>π</sup> : from $\gamma(\theta)$ in 1976Sp09.                                                                                                                                                                |
|                       |                                        |                            | $\omega\gamma=2.4$ eV 5 (1976Sp08).                                                                                                                                                                                |
| 8957.9 5              |                                        | 2663.8 5                   | $\omega\gamma=1.2$ eV 2 (1976Sp08).                                                                                                                                                                                |
| 8981.9 5              |                                        | 2688.6 5                   | $\omega\gamma=2.2$ eV 4 (1976Sp08).                                                                                                                                                                                |
| 8984.2 5              |                                        | 2690.9 5                   | $\omega\gamma=0.6$ eV 3 (1976Sp08).                                                                                                                                                                                |
| 8988.4 5              |                                        | 2695.3 5                   | $\omega\gamma=0.7$ eV 4 (1976Sp08).                                                                                                                                                                                |
| 8992.4 5              |                                        | 2699.4 5                   | $\omega\gamma=2.3$ eV 5 (1976Sp08).                                                                                                                                                                                |
| 8996.7 5              |                                        | 2703.8 5                   | $\omega\gamma=1.7$ eV 3 (1976Sp08).                                                                                                                                                                                |
| 9001.1 5              |                                        | 2708.3 5                   | $\omega\gamma=0.6$ eV 3 (1976Sp08).                                                                                                                                                                                |

Continued on next page (footnotes at end of table)



$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$ | $T_{1/2}$ | $E_p(\text{lab})$ (keV) | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|------------------|-----------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9019.3 5                  |                  | 3.1 keV 3 | 2727.1 5                | $T_{1/2}$ : width measured from the slope of the low-energy side of the resonance peak (1976Sp08).<br>$\omega\gamma=0.8$ eV 4 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                |
| 9024.5 5                  |                  |           | 2732.4 5                | $\omega\gamma=1.1$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9029.9 5                  |                  |           | 2738.0 5                | $\omega\gamma=2.5$ eV 5 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9033.1 5                  |                  |           | 2741.3 5                | $\omega\gamma=0.4$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9038.3 5                  |                  |           | 2746.6 5                | $\omega\gamma=0.8$ eV 4 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9048.4 5                  |                  |           | 2757.0 5                | $\omega\gamma=3.1$ eV 6 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9081.4 4                  | 5/2 <sup>+</sup> | 65 eV 20  | 2791.0 4                | $J^\pi$ : from $\sigma(E_p, \theta)$ in 1981Bi05. Spin=5/2 from $\gamma(\theta)$ in 1976Sp09.<br>$T_{1/2}$ : width $\Gamma_p=62$ eV 20, $\Gamma_\gamma=2.3$ eV 4 (1976Sp09, resonant absorption).<br>$E_p(\text{lab})$ (keV): weighted average of 2791.0 4 (1976Sp08) and 2791 1 (2001Vo24).<br>$\omega\gamma=15.7$ eV 32 (1976Sp08).<br>$\omega\gamma=16.0$ eV 4 (1996Ka21).<br>$\omega\gamma=16.0$ eV 3 (2001Vo24).<br>$\omega\gamma=13.4$ eV 22 (1976Sp09).<br>$\Gamma_p=59$ eV 7 (1981Bi05).<br>$\Gamma_\gamma=2.7$ eV 5 (2001Vo24). |
| 9088.5 5                  |                  |           | 2798.3 5                | $\omega\gamma=0.7$ eV 4 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9099.3 5                  |                  |           | 2809.4 5                | $\omega\gamma=1.2$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9100.4 5                  |                  |           | 2810.6 5                | $\omega\gamma=2.1$ eV 4 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9107.4 5                  |                  |           | 2817.8 5                | $\omega\gamma=0.5$ eV 3 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9110.0 5                  |                  |           | 2820.5 5                | $\omega\gamma=0.5$ eV 3 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9124.0 5                  |                  |           | 2834.9 5                | $\omega\gamma=1.2$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9135.1 5                  |                  | 2.0 keV 4 | 2846.3 5                | $T_{1/2}$ : width measured from the slope of the low-energy side of the resonance peak (1976Sp08).<br>$\omega\gamma=0.2$ eV 1 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                |
| 9138.3 5                  |                  |           | 2849.6 5                | $\omega\gamma=1.9$ eV 4 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9155.7 5                  |                  |           | 2867.5 5                | $\omega\gamma=2.0$ eV 4 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9157.1 5                  | 5/2              |           | 2869.0 5                | $J^\pi$ : spin=5/2 from $\gamma(\theta)$ in 2001Vo24 and 1996Ka21.<br>$\omega\gamma=6.6$ eV 13 (1976Sp08).<br>$\omega\gamma=6.3$ eV 1 (1996Ka21).                                                                                                                                                                                                                                                                                                                                                                                        |
| 9163.3 5                  |                  |           | 2875.3 5                | $\omega\gamma=0.4$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9184.2 5                  |                  |           | 2896.9 5                | $\omega\gamma=0.9$ eV 5 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9188.8 5                  |                  |           | 2901.6 5                | $\omega\gamma=0.9$ eV 5 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9194.1 5                  |                  |           | 2907.1 5                | $\omega\gamma=3.9$ eV 8 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

<sup>†</sup> For resonances above  $S(\text{p})(^{35}\text{Cl})=6370.81$  4 (2021Wa16),  $E(\text{level})=E_{\text{c.m.}}+S(\text{p})(^{35}\text{Cl})$ , where  $E_{\text{c.m.}}=E_p(\text{lab})\times m(^{34}\text{S})/[m_p+m(^{34}\text{S})]$ . Weighted average is taken when multiple  $E_p(\text{lab})$  measurements are available. For levels below 8.27 MeV ( $E_p<1.95$  MeV),  $E_p(\text{lab})$  is mainly from 1972Hu10, 1973Fa07, and 1976Me12; for levels above 8.27 MeV ( $E_p>1.95$  MeV),  $E_p(\text{lab})$  is mainly from 1976Sp08, unless otherwise noted.

<sup>‡</sup> From the comparison of experimental angular distributions of  $\gamma$ -rays with theoretical predictions or deduced from  $\gamma$ -feedings, unless otherwise noted.

 $\gamma(^{35}\text{Cl})$ 

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$        | $J_f^\pi$                            | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                                                        |
|---------------------|------------------|--------------------|---------------|--------------|--------------------------------------|--------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1219.39             | 1/2 <sup>+</sup> | 1219.37            | 100           | 0            | 3/2 <sup>+</sup>                     |                    |            | $E_\gamma$ : other: 1220 40 (1963Ha32).<br>$A_2=-0.03$ 3, $A_4=-0.002$ 50 (1971Wi13).<br>$A_2=-0.01$ 2, $A_4=+0.005$ 24 (1967Ta08).                             |
| 1763.14             | 5/2 <sup>+</sup> | 543.75<br>1763.09  | <0.2<br>100   | 1219.39<br>0 | 1/2 <sup>+</sup><br>3/2 <sup>+</sup> | D+Q                | -0.42 17   | $I_\gamma$ : from 1972Hu10 and 1976Me12.<br>$I_\gamma$ : from 1972Hu10 and 1976Me12.<br>Mult., $\delta$ : from $\gamma\gamma(\theta)$ in 1967Ko20 and 1967Ko23. |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------|------------------|--------------------|---------------|---------|------------------|--------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |                  |                    |               |         |                  |                    |            | Other: $+2.8 < \delta < +4.0$ (1971Wi13).<br>$A_2 = -0.24$ 6, $A_4 = -0.08$ 7 (1971Wi13).<br>$A_2 = +0.25$ 3, $A_4 = -0.49$ 8<br>(1967Ko20,1967Ko23,1970Ko33).<br>$I_\gamma$ : weighted average of 8 1 (1972Hu10)<br>and 9.4 10 (1976Me12).                                                                                                                                                                                                                                                                                                      |
| 2645.70             | (5/2,7/2)        | 882.55             | 8.7 10        | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : weighted average of 92 1 (1972Hu10)<br>and 90.6 10 (1976Me12).<br>Mult., $\delta$ : D+Q with $\delta = -0.62 + 38-36$ from<br>$\gamma(\theta)$ for $J_1=3/2$ (1967Ko20).<br>$A_2 = +0.124$ 17, $A_4 = -0.230$ 24<br>(1967Ko20,1967Ko23).                                                                                                                                                                                                                                                                                            |
|                     |                  | 1426.28            | <2            | 1219.39 | 1/2 <sup>+</sup> |                    |            | $E_\gamma$ : other: 933 7 (1970Ko33).<br>$I_\gamma$ : weighted average of 13 2 (1972Hu10)<br>and 13.0 10 (1976Me12).<br>$A_2 = -0.05$ 38 (1971Wi13).<br>$E_\gamma$ : other: 1475 7 (1970Ko33).<br>$I_\gamma$ : weighted average of 6 2 (1972Hu10)<br>and 8.0 10 (1976Me12). Other: 66 14<br>from 1963Ha32.                                                                                                                                                                                                                                       |
|                     |                  | 2645.59            | 91.3 10       | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : weighted average of 81 3 (1972Hu10)<br>and 79 2 (1976Me12). Other: 34 9<br>from 1963Ha32.                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2693.9              | (3/2,5/2)        | 930.8              | 13.0 10       | 1763.14 | 5/2 <sup>+</sup> |                    |            | $\delta$ : $+0.1 < \delta < +0.26$ or $-35 < \delta < -7$ for $J=3/2$ ,<br>$\delta < -5$ for $J=5/2$ (1971Wi13).<br>$A_2 = +0.20$ 10, $A_4 = -0.16$ 11 (1971Wi13).<br>$I_\gamma$ : from 1972Hu10. Other: $I_\gamma < 2$<br>(1976Me12).<br>$I_\gamma$ : other: $I_\gamma \leq 5$ (1972Hu10).<br>$E_\gamma$ : other: 3006 7 (1970Ko33).<br>$I_\gamma$ : from 1972Hu10 and 1976Me12.                                                                                                                                                                |
|                     |                  | 1474.5             | 7.6 10        | 1219.39 | 1/2 <sup>+</sup> |                    |            | Mult., $\delta$ : from $\gamma\gamma(\theta)$ in 1967Ko20.<br>$A_2 = +0.45$ 18, $A_4 = +0.05$ 19 (1971Wi13).<br>$A_2 = -0.025$ 11, $A_4 = -0.084$ 16<br>(1967Ko20).                                                                                                                                                                                                                                                                                                                                                                              |
|                     |                  | 2693.8             | 80 2          | 0       | 3/2 <sup>+</sup> | D+Q                |            | $I_\gamma$ : from 1971Pr11. Other: <3<br>(1976Me12).<br>$I_\gamma$ : from 1971Pr11. Others: 11 2<br>(1972Hu10), 8.9 20 (1976Me12).<br>$A_2 = +0.31$ 8, $A_4 = -0.06$ 8 (1971Wi13).<br>$I_\gamma$ : from 1971Pr11. Others: $\leq 0.2$<br>(1972Hu10), <0.2 (1976Me12).<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1971Pr11, which<br>states that the anomalous $\gamma(\theta)$ can be<br>understood if it occurs by a mixture of<br>M2 and E1 radiation. In view of the<br>weakness of this transition, such a<br>mixture is not unreasonable. |
| 3002.75             | 5/2              | 1239.59            | $\leq 1$      | 1763.14 | 5/2 <sup>+</sup> |                    |            | $E_\gamma$ : other: 3163 7 (1970Ko33).<br>$I_\gamma$ : from 1971Pr11. Others: 89 2<br>(1972Hu10), 91 2 (1976Me12).<br>Mult.: Q+O from $\gamma(\theta)$ in 1966Az01,<br>1966Wa09, 1967Wa22, and 1967Ta08;                                                                                                                                                                                                                                                                                                                                         |
|                     |                  | 1783.31            | <3            | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                     |                  | 3002.61            | 100           | 0       | 3/2 <sup>+</sup> | D+Q                | -0.22 1    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 3162.99             | 7/2 <sup>-</sup> | 160.24             | 1.7 2         | 3002.75 | 5/2              |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                     |                  | 517.29             | 8 1           | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                     |                  | 1399.82            | 0.30 4        | 1763.14 | 5/2 <sup>+</sup> | E1+M2              | +0.44 12   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                     |                  | 1943.54            | <0.2          | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                     |                  | 3162.84            | 90 1          | 0       | 3/2 <sup>+</sup> | M2+E3              | +0.25 4    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$  | $I_\gamma^\#$ | $E_f$                                         | $J_f^\pi$ | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                               |
|---------------------|------------------|---------------------|---------------|-----------------------------------------------|-----------|--------------------|------------|------------------------------------------------------------------------------------------------------------------------|
|                     |                  |                     |               |                                               |           |                    |            | magnetic character from $\gamma(\text{pol})$ in <b>1966Wa09</b> and <b>1967Wa22</b> .                                  |
|                     |                  |                     |               |                                               |           |                    |            | $\delta$ : from <b>1966Az01</b> . Others: +0.25 5 ( <b>1967Ta08</b> ); +0.16 1 ( <b>1966Wa09,1967Wa22</b> ).           |
|                     |                  |                     |               |                                               |           |                    |            | Asymmetry ratio $R=1.426$ 7, $N_{90}/N_0=1.25$ 11 ( <b>1971Wi13</b> ).                                                 |
|                     |                  |                     |               |                                               |           |                    |            | $\text{POL}=-0.75$ 20, $\Delta\pi=\text{yes}$ ( <b>1967Wa22</b> ).                                                     |
|                     |                  |                     |               |                                               |           |                    |            | $\text{POL}=-1.27$ 10, $\Delta\pi=\text{yes}$ ( <b>1966Wa09</b> ).                                                     |
|                     |                  |                     |               |                                               |           |                    |            | $0.00<\text{POL}<0.384$ ( <b>1966Az01,1967Ta08</b> ).                                                                  |
|                     |                  |                     |               |                                               |           |                    |            | $A_2=+0.40$ 8, $A_4=-0.23$ 8 ( <b>1971Wi13</b> ).                                                                      |
|                     |                  |                     |               |                                               |           |                    |            | $A_2=+0.25$ 3, $A_4=+0.02$ 5 ( <b>1967Ta08</b> ).                                                                      |
|                     |                  |                     |               |                                               |           |                    |            | $A_2=+0.55$ 3, $A_4=-0.008$ 38 ( <b>1966Az01</b> ).                                                                    |
| 3918.48             | 3/2 <sup>+</sup> | 1272.76<br>2155.27  | <3<br>19.5 23 | 2645.70 (5/2,7/2)<br>1763.14 5/2 <sup>+</sup> |           |                    |            | $I_\gamma$ : weighted average of 17 3 ( <b>1972Hu10</b> ), 25 3 ( <b>1976Me12</b> ), and 18.0 21 ( <b>1976Sp09</b> ).  |
|                     |                  | 3918.25             | 81.5 15       | 0 3/2 <sup>+</sup>                            |           | M1+E2              |            | $I_\gamma$ : weighted average of 83 3 ( <b>1972Hu10</b> ), 75 3 ( <b>1976Me12</b> ), and 82.0 10 ( <b>1976Sp09</b> ).  |
|                     |                  |                     |               |                                               |           |                    |            | Mult.: D+Q from $\gamma(\theta)$ in <b>1976Sp09</b> ;                                                                  |
|                     |                  |                     |               |                                               |           |                    |            | $\delta=-0.21$ 2 or +20 +10-5 ( <b>1976Sp09</b> ).                                                                     |
|                     |                  |                     |               |                                               |           |                    |            | E1+M2 ruled by RUL.                                                                                                    |
|                     |                  |                     |               |                                               |           |                    |            | $A_2=+0.07$ 2, $A_4=-0.01$ 3 ( <b>1976Sp09</b> ).                                                                      |
| 3942.9              |                  | 1297.2              | 8 2           | 2645.70 (5/2,7/2)                             |           |                    |            | $I_\gamma$ : weighted average of 7 3 ( <b>1972Hu10</b> ) and 8 2 ( <b>1976Me12</b> ).                                  |
|                     |                  | 2179.7              | 92 2          | 1763.14 5/2 <sup>+</sup>                      |           |                    |            | $I_\gamma$ : weighted average of 93 3 ( <b>1972Hu10</b> ) and 92 2 ( <b>1976Me12</b> ).                                |
|                     |                  | 2723.4 <sup>c</sup> | <6            | 1219.39 1/2 <sup>+</sup>                      |           |                    |            |                                                                                                                        |
|                     |                  | 3942.7 <sup>c</sup> | <10           | 0 3/2 <sup>+</sup>                            |           |                    |            |                                                                                                                        |
| 3967.3              |                  | 1273.4              | 2.1 5         | 2693.9 (3/2,5/2)                              |           |                    |            | $I_\gamma$ : weighted average of 5 3 ( <b>1972Hu10</b> ) and 2.0 5 ( <b>1976Me12</b> ).                                |
|                     |                  | 1321.6 <sup>c</sup> | <2            | 2645.70 (5/2,7/2)                             |           |                    |            |                                                                                                                        |
|                     |                  | 2204.1              | $\leq 1$      | 1763.14 5/2 <sup>+</sup>                      |           |                    |            | $I_\gamma$ : from <b>1972Hu10</b> . Other: <2 ( <b>1976Me12</b> ).                                                     |
|                     |                  | 2747.8              | 79 5          | 1219.39 1/2 <sup>+</sup>                      |           |                    |            | $I_\gamma$ : weighted average of 74 5 ( <b>1972Hu10</b> ) and 83 5 ( <b>1976Me12</b> ).                                |
|                     |                  | 3967.1              | 19 3          | 0 3/2 <sup>+</sup>                            |           |                    |            | $I_\gamma$ : weighted average of 21 3 ( <b>1972Hu10</b> ) and 15 5 ( <b>1976Me12</b> ).                                |
| 4059.2              |                  | 1365.3              | 1.2 4         | 2693.9 (3/2,5/2)                              |           |                    |            |                                                                                                                        |
|                     |                  | 1413.5 <sup>c</sup> | <1            | 2645.70 (5/2,7/2)                             |           |                    |            |                                                                                                                        |
|                     |                  | 2296.0              | 4.8 15        | 1763.14 5/2 <sup>+</sup>                      |           |                    |            | $E_\gamma$ : other: 2296 7 ( <b>1970Ko33</b> ).                                                                        |
|                     |                  |                     |               |                                               |           |                    |            | $I_\gamma$ : other: 3 2 ( <b>1972Hu10</b> ).                                                                           |
|                     |                  | 2839.7              | 94 2          | 1219.39 1/2 <sup>+</sup>                      |           |                    |            | $E_\gamma$ : other: 2838 7 ( <b>1970Ko33</b> ).                                                                        |
|                     |                  |                     |               |                                               |           |                    |            | $I_\gamma$ : other: 97 2 ( <b>1972Hu10</b> ). 45 5 ( <b>1971Wi13</b> ).                                                |
|                     |                  | 4059.0              | <2            | 0 3/2 <sup>+</sup>                            |           |                    |            | $E_\gamma$ : other: 4058 7 ( <b>1970Ko33</b> ).                                                                        |
|                     |                  |                     |               |                                               |           |                    |            | $I_\gamma$ : other: 20 2 ( <b>1971Wi13</b> ).                                                                          |
| 4113.7              | 7/2              | 2350.5              | 47 3          | 1763.14 5/2 <sup>+</sup>                      |           | D+Q                | -0.16 2    | $I_\gamma$ : weighted average of 50 20 ( <b>1972Hu10</b> ), 44 7 ( <b>1976Me12</b> ), and 48 3 ( <b>1976Sp09</b> ).    |
|                     |                  |                     |               |                                               |           |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in <b>1976Sp09</b> .                                                           |
|                     |                  |                     |               |                                               |           |                    |            | $A_2=+0.56$ 3, $A_4=+0.05$ 4 ( <b>1976Sp09</b> ).                                                                      |
|                     |                  | 2894.2 <sup>c</sup> | <10           | 1219.39 1/2 <sup>+</sup>                      |           |                    |            |                                                                                                                        |
|                     |                  | 4113.4              | 53 3          | 0 3/2 <sup>+</sup>                            |           | Q(+O)              | 0.00 10    | $I_\gamma$ : weighted average of 50 20 ( <b>1972Hu10</b> ) and 56 7 ( <b>1976Me12</b> ), and 52 3 ( <b>1976Sp09</b> ). |
|                     |                  |                     |               |                                               |           |                    |            | Mult., $\delta$ : Q(+O) from $\gamma(\theta)$ in <b>1976Sp09</b> .                                                     |
|                     |                  |                     |               |                                               |           |                    |            | $A_2=+0.46$ 25, $A_4=-0.23$ 25 ( <b>1976Sp09</b> ).                                                                    |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$ | $E_\gamma^\dagger$   | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                 |
|---------------------|-----------|----------------------|---------------|---------|------------------|--------------------|------------|------------------------------------------------------------------------------------------|
| 4173.45             |           | 1479.5               | 26 8          | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : weighted average of 27 8 (1972Hu10) and 25 8 (1976Me12).                    |
|                     |           | 1527.71              | <10           | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : weighted average of 18 6 (1972Hu10) and 15 5 (1976Me12).                    |
|                     |           | 2410.22              | 16 5          | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10. Other: <4 (1976Me12); 32 3 (1971Wi13).                       |
|                     |           | 2953.93              | $\leq 3$      | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : weighted average of 55 10 (1972Hu10) and 60 10 (1976Me12).                  |
|                     |           | 4173.18              | 58 10         | 0       | 3/2 <sup>+</sup> |                    |            | $A_2=-0.01$ 3, $A_4=-0.006$ 40 (1967Ta08).                                               |
| 4177.9              | 3/2       | 1175.1               | <0.5          | 3002.75 | 5/2              |                    |            | $I_\gamma$ : weighted average of 7 2 (1972Hu10) and 10 3 (1976Me12).                     |
|                     |           | 1484.0               | 8 2           | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : other $\leq 1$ (1972Hu10).                                                  |
|                     |           | 1532.2               | <0.5          | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : weighted average of 31 5 (1972Hu10) and 30 5 (1976Me12).                    |
|                     |           | 2414.7               | <1            | 1763.14 | 5/2 <sup>+</sup> | D+Q                | +0.11 4    | Mult., $\delta$ : from $\gamma(\theta)$ in 1967Wa22.                                     |
|                     |           | 2958.4               | 31 5          | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : weighted average of 62 6 (1972Hu10) and 60 5 (1976Me12).                    |
|                     |           | 4177.6               | 61 5          | 0       | 3/2 <sup>+</sup> | D+Q                | +0.06 3    | Mult., $\delta$ : from $\gamma(\theta)$ in 1967Wa22.                                     |
| 4347.76             | 5/2,9/2   | 1184.75              | 70 2          | 3162.99 | 7/2 <sup>-</sup> |                    |            | POL=+0.12 31 (1967Wa22).                                                                 |
|                     |           | 1344.98 <sup>c</sup> | <4            | 3002.75 | 5/2              |                    |            | $I_\gamma$ : weighted average of 70 5 (1972Hu10), 72 5 (1976Me12), and 69 2 (1976Sp09).  |
|                     |           | 1653.8 <sup>c</sup>  | <4            | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : weighted average of 30 5 (1972Hu10), 28 5 (1976Me12), and 31 2 (1976Sp09).  |
|                     |           | 1702.02              | 31 2          | 2645.70 | (5/2,7/2)        | D+Q                | -0.018 12  | Mult., $\delta$ : D+Q from $\gamma(\theta)$ in 1976Sp09.                                 |
|                     |           | 2584.52              | $\leq 3$      | 1763.14 | 5/2 <sup>+</sup> |                    |            | $A_2=-0.34$ 3, $A_4=-0.00$ 4 (1976Sp09).                                                 |
|                     |           | 3128.22 <sup>c</sup> | <4            | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10. Other: <5 (1976Me12).                                        |
|                     |           | 4347.47              | $\leq 5$      | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10. Other: <10 (1976Me12).                                       |
| 4624.4              |           | 4624.1               | 85 10         | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 100 (1972Hu10).                                                      |
| 4769.86             |           | 1606.83              | 65 10         | 3162.99 | 7/2 <sup>-</sup> |                    |            | $I_\gamma$ : other: 53 10 (1972Hu10) and 47% of $I_\gamma$ to unknown levels (1972Hu10). |
|                     |           | 1767.06              | 35 10         | 3002.75 | 5/2              |                    |            |                                                                                          |
|                     |           | 2075.9               | <10           | 2693.9  | (3/2,5/2)        |                    |            |                                                                                          |
|                     |           | 3550.28              | <10           | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                          |
|                     |           | 4769.51              | <10           | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                          |
| 4839.10             |           | 1836.30              | <7            | 3002.75 | 5/2              |                    |            |                                                                                          |
|                     |           | 2145.1               | <4            | 2693.9  | (3/2,5/2)        |                    |            |                                                                                          |
|                     |           | 2193.33              | <3            | 2645.70 | (5/2,7/2)        |                    |            |                                                                                          |
|                     |           | 3075.82              | 59 5          | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : weighted average of 50 20 (1972Hu10) and 60 5 (1976Me12).                   |
|                     |           | 3619.51              | <7            | 1219.39 | 1/2 <sup>+</sup> |                    |            | Other: 48 4 (1976Sp09).                                                                  |
|                     |           | 4838.74              | 41 5          | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : weighted average of 50 20 (1972Hu10) and 40 5 (1976Me12).                   |
| 4854.4              |           | 3634.8               | 75 5          | 1219.39 | 1/2 <sup>+</sup> |                    |            | Other: 18 3 (1976Sp09).                                                                  |
|                     |           | 4854.0               | 25 5          | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                          |
| 4880.96             |           | 1878.16              | 9 3           | 3002.75 | 5/2              |                    |            |                                                                                          |
|                     |           | 2187.0               | <10           | 2693.9  | (3/2,5/2)        |                    |            |                                                                                          |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$ | $E_\gamma^\dagger$   | $I_\gamma^\#$    | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                  |
|---------------------|-----------|----------------------|------------------|---------|------------------|--------------------|------------|-------------------------------------------------------------------------------------------|
| 4880.96             |           | 2235.18              | 29 5             | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : other: 30 10 (1972Hu10).                                                     |
|                     |           | 3117.67              | 62 5             | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 70 10 (1972Hu10).                                                     |
|                     |           | 3661.36 <sup>c</sup> | <4               | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                           |
|                     |           | 4880.60              | <5               | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                           |
| 5010.1              |           | 836.6                | <25 <sup>a</sup> | 4173.45 |                  |                    |            |                                                                                           |
|                     |           | 950.9                | <10 <sup>a</sup> | 4059.2  |                  |                    |            |                                                                                           |
|                     |           | 1847.1               | <50 <sup>a</sup> | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                           |
|                     |           | 2007.3               | <20 <sup>a</sup> | 3002.75 | 5/2              |                    |            |                                                                                           |
|                     |           | 2316.1               | <25 <sup>a</sup> | 2693.9  | (3/2,5/2)        |                    |            |                                                                                           |
|                     |           | 2364.3               | <5               | 2645.70 | (5/2,7/2)        |                    |            |                                                                                           |
|                     |           | 3246.8               | <10              | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                           |
|                     |           | 3790.5               | <7               | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                           |
|                     |           | 5009.7               | 100              | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1976Me12 and 1972Hu10.                                                  |
| 5163.34             | 7/2       | 2000.29              | 40 5             | 3162.99 | 7/2 <sup>-</sup> | D+Q                | +0.44 20   | $I_\gamma$ : weighted average of 40 20 (1972Hu10), 36 5 (1976Me12), and 54 10 (1976Sp09). |
|                     |           |                      |                  |         |                  |                    |            | Mult., $\delta$ : D+Q from $\gamma(\theta)$ in 1976Sp09.                                  |
|                     |           |                      |                  |         |                  |                    |            | $A_2=+0.52$ 5, $A_4=-0.02$ 6 (1976Sp09).                                                  |
|                     |           | 2160.52              | 10 4             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : weighted average of 10 5 (1972Hu10), 10 4 (1976Me12), and 10 5 (1976Sp09).   |
|                     |           | 3400.02              | 43 5             | 1763.14 | 5/2 <sup>+</sup> | D(+Q)              | 0.00 3     | $I_\gamma$ : weighted average of 50 20 (1972Hu10), 44 5 (1976Me12), and 36 10 (1976Sp09). |
|                     |           |                      |                  |         |                  |                    |            | Mult., $\delta$ : D(+Q) from $\gamma(\theta)$ in 1976Sp09.                                |
|                     |           |                      |                  |         |                  |                    |            | $A_2=-0.23$ 6, $A_4=-0.08$ 7 (1976Sp09).                                                  |
|                     |           | 3943.71              | <10              | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                           |
|                     |           | 5162.93              | <10              | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                           |
| 5215.8              |           | 205.7                | <23 <sup>a</sup> | 5010.1  |                  |                    |            |                                                                                           |
|                     |           | 1042.3               | <60 <sup>a</sup> | 4173.45 |                  |                    |            |                                                                                           |
|                     |           | 1156.6               | <10 <sup>a</sup> | 4059.2  |                  |                    |            |                                                                                           |
|                     |           | 2052.8               | <15 <sup>a</sup> | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                           |
|                     |           | 2213.0               | <25 <sup>a</sup> | 3002.75 | 5/2              |                    |            |                                                                                           |
|                     |           | 2521.8               | <15 <sup>a</sup> | 2693.9  | (3/2,5/2)        |                    |            |                                                                                           |
|                     |           | 2570.0               | <15 <sup>a</sup> | 2645.70 | (5/2,7/2)        |                    |            |                                                                                           |
|                     |           | 3452.5               | <6               | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                           |
|                     |           | 3996.2               | <4               | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                           |
|                     |           | 5215.4               | 100              | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                           |
| 5403.5              |           | 2400.7               | 25 5             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 25 10 (1972Hu10).                                                     |
|                     |           | 3640.2               | <10              | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 25 10 from 1972Hu10.                                                  |
|                     |           | 4183.8               | 55 10            | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                           |
|                     |           | 5403.1               | <7               | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                           |
| 5586                |           | 2940                 | 40 10            | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : weighted average of 40 20 (1972Hu10) and 40 10 (1976Me12).                   |
|                     |           | 3823                 | 60 10            | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : weighted average of 60 20 (1972Hu10) and 60 10 (1976Me12).                   |
|                     |           | 4366                 | <10              | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                           |
|                     |           | 5586                 | <10              | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                           |
| 5599.7              |           | 3836.3               | 27 10            | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : weighted average of 20 10 (1972Hu10) and 33 10 (1976Me12).                   |
|                     |           | 5599.2               | 74 10            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : weighted average of 80 10 (1972Hu10) and 67 10 (1976Me12).                   |
| 5646                |           | 2483                 | 100              | 3162.99 | 7/2 <sup>-</sup> |                    |            | $I_\gamma$ : from 1972Hu10, 1976Me12, 1971Pr11.                                           |
|                     |           | 2643                 |                  | 3002.75 | 5/2              |                    |            | $I_\gamma$ : 80 from 2001Vo24 only; not observed by others.                               |
|                     |           | 3883                 |                  | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : 6 from 2001Vo24 only; not observed by others.                                |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$ | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                 |
|---------------------|-----------|--------------------|---------------|---------|------------------|--------------------|------------|--------------------------------------------------------------------------|
| 5646                |           | 5646               | <8            | 0       | 3/2 <sup>+</sup> |                    |            |                                                                          |
| 5655                |           | 2652               | 80 5          | 3002.75 | 5/2              |                    |            |                                                                          |
|                     |           | 3892               | 6 2           | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                          |
|                     |           | 4435               | <6            | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                          |
| 5683                |           | 1505               | 60 10         | 4177.9  | 3/2              |                    |            |                                                                          |
| 5723.52             |           | 3029.5             | 24 3          | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : from 1976Sp09.                                              |
|                     |           | 3960.14            | 38 4          | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1976Sp09.                                              |
| 5758                |           | 1580               | 30 10         | 4177.9  | 3/2              |                    |            |                                                                          |
|                     |           | 4538               | <6            | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                          |
|                     |           | 5758               | 45 10         | 0       | 3/2 <sup>+</sup> |                    |            |                                                                          |
| 5806                |           | 5806               | 100           | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10 and 1976Me12.                                 |
| 6106.2              |           | 4342.8             | 6 2           | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                          |
|                     |           | 4886.4             | 36 10         | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 40 10 (1972Hu10).                                    |
|                     |           | 6105.6             | 58 10         | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 60 10 (1972Hu10).                                    |
| 6181                |           | 6180               | 45 10         | 0       | 3/2 <sup>+</sup> |                    |            |                                                                          |
| 6492                |           | 6491               | 100           | 0       | 3/2 <sup>+</sup> |                    |            |                                                                          |
| 7066.2              |           | 1902.8             | 0.9           | 5163.34 | 7/2              |                    |            |                                                                          |
|                     |           | 2892.6             | 0.5           | 4173.45 |                  |                    |            |                                                                          |
|                     |           | 2952.4             | 0.6           | 4113.7  | 7/2              |                    |            |                                                                          |
|                     |           | 3006.9             | 1.4           | 4059.2  |                  |                    |            | $I_\gamma$ : other: 1.0 5 from 1972Hu10.                                 |
|                     |           | 3903.0             | 6             | 3162.99 | 7/2 <sup>-</sup> |                    |            | $I_\gamma$ : other: 6.0 6 from 1972Hu10.                                 |
|                     |           | 4063.2             | 3             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 2 1 from 1972Hu10.                                   |
|                     |           | 4372.0             | 5             | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : other: 6.0 6 from 1972Hu10.                                 |
|                     |           | 4420.2             | 1.6           | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : other: 3.0 15 from 1972Hu10.                                |
|                     |           | 5302.6             | 14            | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 16.0 16 from 1972Hu10.                               |
|                     |           | 5846.3             | 18            | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 18.0 18 from 1972Hu10.                               |
|                     |           | 7065.4             | 49            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 48.0 48 from 1972Hu10.                               |
| 7103.4              | 3/2       | 2925.4             | 1.5           | 4177.9  | 3/2              |                    |            |                                                                          |
|                     |           | 3184.8             | 1.5           | 3918.48 | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                      |
|                     |           | 4100.4             | 3             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 3.0 15 from 1972Hu10.                                |
|                     |           | 4409.2             | 11            | 2693.9  | (3/2,5/2)        | D(+Q)              | 0.00 3     | $I_\gamma$ , Mult., $\delta$ : from 1976Me12. Other: 13.0 13 (1972Hu10). |
|                     |           |                    |               |         |                  |                    |            | $A_2=+0.34$ 7, $A_4=+0.03$ 11 (1976Me12).                                |
|                     |           | 5339.8             | 4             | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 4 2 from 1972Hu10.                                   |
|                     |           | 5883.5             | 64            | 1219.39 | 1/2 <sup>+</sup> | D(+Q)              | 0.00 3     | $I_\gamma$ , Mult., $\delta$ : from 1976Me12. Other: 67.0 67 (1972Hu10). |
|                     |           |                    |               |         |                  |                    |            | $A_2=-0.53$ 8, $A_4=+0.01$ 11 (1976Me12).                                |
|                     |           | 7102.6             | 15            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 11.0 11 (1972Hu10).                                  |
|                     |           |                    |               |         |                  |                    |            | $A_2=+0.54$ 5, $A_4=-0.01$ 8 (1976Me12).                                 |
| 7178.6              | 1/2       | 1072.4             | 3             | 6106.2  |                  |                    |            | $I_\gamma$ : other: 4 2 (1972Hu10).                                      |
|                     |           | 1775.1             | 1.7           | 5403.5  |                  |                    |            |                                                                          |
|                     |           | 2339.4             | 1.5           | 4839.10 |                  |                    |            | $I_\gamma$ : other: 2 (2001Vo24).                                        |
|                     |           | 3005.0             | 0.8           | 4173.45 |                  |                    |            | $I_\gamma$ : other: 1 (2001Vo24).                                        |
|                     |           | 3119.3             | 22            | 4059.2  |                  |                    |            | $I_\gamma$ : others: 22 (2001Vo24), 24.0 24 (1972Hu10).                  |
|                     |           |                    |               |         |                  |                    |            | $A_2=-0.02$ 4, $A_4=+0.10$ 66 (1976Me12).                                |
|                     |           |                    |               |         |                  |                    |            | $A_2=-0.01$ 4, $A_4=+0.09$ 24 (2001Vo24).                                |
|                     |           | 3211.1             | 9             | 3967.3  |                  |                    |            | $I_\gamma$ : others: 9 (2001Vo24), 10.0 10 (1972Hu10).                   |
|                     |           | 3260.0             | 3             | 3918.48 | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : others: 3 (2001Vo24), 2 1 (1972Hu10).                       |
|                     |           | 4484.4             | 4             | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : others: 4 (2001Vo24), 4 2 (1972Hu10).                       |
|                     |           | 5958.7             | 17            | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : others: 17 (2001Vo24), 16.0 16 (1972Hu10).                  |
|                     |           |                    |               |         |                  |                    |            | $A_2=-0.01$ 3, $A_4=+0.06$ 7 (2001Vo24).                                 |
|                     |           |                    |               |         |                  |                    |            | $A_2=-0.01$ 5, $A_4=+0.07$ 8 (1976Me12).                                 |
|                     |           | 7177.8             | 38            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : others: 38 (2001Vo24), 40.0 40 (1972Hu10).                  |

Continued on next page (footnotes at end of table)



$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$    | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$    | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                             |
|---------------------|--------------|--------------------|---------------|---------|--------------|--------------------|------------|--------------------------------------------------------------------------------------|
|                     |              |                    |               |         |              |                    |            | $A_2=-0.01$ 6, $A_4=-0.02$ 7 (1976Me12).<br>$A_2=-0.02$ 4, $A_4=-0.01$ 5 (2001Vo24). |
| 7185                |              | 1969               | $<0.5^a$      | 5215.8  |              |                    |            |                                                                                      |
|                     |              | 2175               | $<0.5^a$      | 5010.1  |              |                    |            |                                                                                      |
|                     |              | 3011               | $<5^a$        | 4173.45 |              |                    |            |                                                                                      |
|                     |              | 3126               | $26^a$ 6      | 4059.2  |              |                    |            |                                                                                      |
|                     |              | 4022               | $<3^a$        | 3162.99 | $7/2^-$      |                    |            |                                                                                      |
|                     |              | 4182               | $<3^a$        | 3002.75 | $5/2$        |                    |            |                                                                                      |
|                     |              | 4491               | $2^a$ 1       | 2693.9  | $(3/2, 5/2)$ |                    |            |                                                                                      |
|                     |              | 4539               | $6^a$ 2       | 2645.70 | $(5/2, 7/2)$ |                    |            |                                                                                      |
|                     |              | 5421               | $4^a$ 3       | 1763.14 | $5/2^+$      |                    |            |                                                                                      |
|                     |              | 5965               | $16^a$ 7      | 1219.39 | $1/2^+$      |                    |            |                                                                                      |
| 7194.6              | $1/2, 3/2$   | 7184               | $46^a$ 10     | 0       | $3/2^+$      |                    |            |                                                                                      |
|                     |              | 2184.4             | 0.8           | 5010.1  |              |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                |
|                     |              | 2340.1             | 2             | 4854.4  |              |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                  |
|                     |              | 2355.4             | 1.8           | 4839.10 |              |                    |            |                                                                                      |
|                     |              | 3016.6             | 16            | 4177.9  | $3/2$        |                    |            | $I_\gamma$ : other: 15.0 15 (1972Hu10).                                              |
|                     |              | 3135.3             | 3             | 4059.2  |              |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                               |
|                     |              | 3227.1             | 1.8           | 3967.3  |              |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                  |
|                     |              | 3276.0             | 4             | 3918.48 | $3/2^+$      |                    |            | $I_\gamma$ : other: 5.0 25 (1972Hu10).                                               |
|                     |              | 4500.4             | 1.1           | 2693.9  | $(3/2, 5/2)$ |                    |            |                                                                                      |
|                     |              | 5431.0             | 0.5           | 1763.14 | $5/2^+$      |                    |            |                                                                                      |
| 7225.5              | $5/2$        | 5974.7             | 67            | 1219.39 | $1/2^+$      |                    |            | $I_\gamma$ : other: 72.0 72 (1972Hu10).                                              |
|                     |              | 7193.8             | 2             | 0       | $3/2^+$      |                    |            |                                                                                      |
|                     |              | 1119.3             | 1.8           | 6106.2  |              |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                               |
|                     |              | 1420               |               | 5806    |              |                    |            | $I_\gamma$ : 1.0 5 (1972Hu10).                                                       |
|                     |              | 1543               | 2.0           | 5683    |              |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                  |
|                     |              | 1571               | 0.8           | 5655    |              |                    |            |                                                                                      |
|                     |              | 1822.0             | 0.9           | 5403.5  |              |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                |
|                     |              | 2009.6             | 7             | 5215.8  |              |                    |            | $I_\gamma$ : other: 6.0 30 (1972Hu10).                                               |
|                     |              | 3047.5             | 8             | 4177.9  | $3/2$        |                    |            | $I_\gamma$ : other: 7.0 35 (1972Hu10).                                               |
|                     |              | 3166.2             | 2             | 4059.2  |              |                    |            |                                                                                      |
| 7234.0              | $5/2^+$      | 3306.9             | 1.5           | 3918.48 | $3/2^+$      |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                |
|                     |              | 4222.5             |               | 3002.75 | $5/2$        |                    |            | $I_\gamma$ : 3.0 15 (1972Hu10).                                                      |
|                     |              | 4531.3             | 10            | 2693.9  | $(3/2, 5/2)$ | D+Q                | +0.36 4    | $I_\gamma$ , Mult., $\delta$ : from 1976Me12. Other: 10.0 10 (1972Hu10).             |
|                     |              |                    |               |         |              |                    |            | $A_2=+0.43$ 8, $A_4=-0.21$ 9 (1976Me12).                                             |
|                     |              | 5461.9             | 7             | 1763.14 | $5/2^+$      | D+Q                | -0.36 7    | $I_\gamma$ , Mult., $\delta$ : from 1976Me12. Other: 8.0 8 (1972Hu10).               |
|                     |              |                    |               |         |              |                    |            | $A_2=+0.02$ 10, $A_4=-0.09$ 17 (1976Me12).                                           |
|                     |              | 7224.7             | 59            | 0       | $3/2^+$      | D+Q                | +0.36 4    | $I_\gamma$ , Mult., $\delta$ : from 1976Me12. Other: 58.0 58 (1972Hu10).             |
|                     |              |                    |               |         |              |                    |            | $A_2=+0.40$ 7, $A_4=+0.04$ 12 (1976Me12).                                            |
|                     |              | 1579               | 0.2           | 5655    |              |                    |            |                                                                                      |
|                     |              | 3056.0             |               | 4177.9  | $3/2$        |                    |            | $I_\gamma$ : 1.0 5 (1972Hu10).                                                       |
| 7272.7              | $(1/2, 3/2)$ | 4539.8             | 1.8           | 2693.9  | $(3/2, 5/2)$ |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                  |
|                     |              | 4588.0             | 3             | 2645.70 | $(5/2, 7/2)$ |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                  |
|                     |              | 5470.4             | 0.2           | 1763.14 | $5/2^+$      |                    |            |                                                                                      |
|                     |              | 6014.1             | 1.8           | 1219.39 | $1/2^+$      |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                  |
|                     |              | 7233.2             | 93            | 0       | $3/2^+$      | D+Q                | -0.10      | $I_\gamma$ : other: 93.0 93 (1972Hu10).                                              |
|                     |              |                    |               |         |              |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 1960An06.                                 |
| 7272.7              | $(1/2, 3/2)$ | 1515               | 1.2           | 5758    |              |                    |            | $I_\gamma$ : other: 1.5 8 (1972Hu10).                                                |
|                     |              | 1869.2             | 0.2           | 5403.5  |              |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                |
|                     |              | 2262.5             | 1.4           | 5010.1  |              |                    |            | $I_\gamma$ : other: 1.5 8 (1972Hu10).                                                |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$ | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                    |
|---------------------|-----------|--------------------|---------------|---------|------------------|--------------------|------------|-------------------------------------------------------------------------------------------------------------|
| 7272.7              | (1/2,3/2) | 2433.5             | 1.0           | 4839.10 |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                       |
|                     |           | 3099.1             | 0.4           | 4173.45 |                  |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                       |
|                     |           | 3213.3             | 0.9           | 4059.2  |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                       |
|                     |           | 3305.2             | 0.8           | 3967.3  |                  |                    |            | $I_\gamma$ : other: 1.0 5 from 1972Hu10.                                                                    |
|                     |           | 4578.5             | 1.1           | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                       |
|                     |           | 6052.8             | 24            | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 23.0 23 (1972Hu10).                                                                     |
|                     |           | 7271.9             | 69            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 69.0 69 (1972Hu10).                                                                     |
| 7362.0              | 3/2       | 1707               | 0.3           | 5655    |                  |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                       |
|                     |           | 2351.8             | 0.6           | 5010.1  |                  |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                       |
|                     |           | 2522.8             | 1.0           | 4839.10 |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                       |
|                     |           | 3183.9             | 0.5           | 4177.9  | 3/2              |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                       |
|                     |           | 3302.6             | 2             | 4059.2  |                  |                    |            | $I_\gamma$ : others: 3.0 15 (1972Hu10); 4 (1971Wi13).                                                       |
|                     |           | 3394.5             | 3             | 3967.3  |                  |                    |            | $I_\gamma$ : others: 3.0 15 (1972Hu10); 4 (1971Wi13).                                                       |
|                     |           | 3443.3             | 0.8           | 3918.48 | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                       |
|                     |           | 4359.0             | 0.3           | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                       |
|                     |           | 4667.8             | 0.5           | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                             |
|                     |           | 5598.4             | 8             | 1763.14 | 5/2 <sup>+</sup> | D(+Q)              |            | $I_\gamma$ : others: 10.0 10 (1972Hu10); 9 (1971Wi13).                                                      |
|                     |           |                    |               |         |                  |                    |            | Mult., $\delta$ : $\delta=-0.01$ 10 or $\delta>+2.5$ (1977Ko34).                                            |
|                     |           | 6142.0             | 73            | 1219.39 | 1/2 <sup>+</sup> | D+Q                |            | $A_2=-0.20$ 24, $A_4=+0.92$ 25 (1977Ko34).                                                                  |
|                     |           |                    |               |         |                  |                    |            | $I_\gamma$ : others: 70.0 70 (1972Hu10); 72 (1971Wi13).                                                     |
|                     |           |                    |               |         |                  |                    |            | Mult., $\delta$ : $\delta=-0.05$ 3 or $+2.5$ 3 (1977Ko34); $\delta=-20$ or $+0.22<\delta<+0.26$ (1971Wi13). |
|                     |           |                    |               |         |                  |                    |            | $A_2=-0.05$ 3, $A_4=-0.04$ 5 (1971Wi13).                                                                    |
|                     |           |                    |               |         |                  |                    |            | $A_2=-0.25$ 11, $A_4=+0.31$ 11 (1977Ko34).                                                                  |
|                     |           | 7361.2             | 10            | 0       | 3/2 <sup>+</sup> | D+Q                |            | $I_\gamma$ : others: 10.0 10 (1972Hu10); 11 (1971Wi13).                                                     |
|                     |           |                    |               |         |                  |                    |            | Mult., $\delta$ : $-0.16<\delta<-0.12$ or $+7.5<\delta<+10$ (1971Wi13).                                     |
|                     |           |                    |               |         |                  |                    |            | $A_2=+0.19$ 3, $A_4=-0.08$ 5 (1971Wi13).                                                                    |
| 7396.0              | 7/2       | 3048.1             | 11            | 4347.76 | 5/2,9/2          |                    |            | $I_\gamma$ : other: 10.0 10 (1972Hu10).                                                                     |
|                     |           | 3282.1             | 8             | 4113.7  | 7/2              |                    |            | $I_\gamma$ : other: 9.0 9 (1972Hu10).                                                                       |
|                     |           | 3452.9             | 5             | 3942.9  |                  |                    |            | $I_\gamma$ : other: 8.0 8 (1972Hu10).                                                                       |
|                     |           | 4232.7             | 47            | 3162.99 | 7/2 <sup>-</sup> | D+Q                | +0.28 25   | $I_\gamma$ : other: 49.0 49 (1972Hu10).                                                                     |
|                     |           |                    |               |         |                  |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko35.                                                        |
|                     |           | 4393.0             | 14            | 3002.75 | 5/2              | D(+Q)              | +0.011 18  | $I_\gamma$ : other: 14.0 14 (1972Hu10).                                                                     |
|                     |           |                    |               |         |                  |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko35.                                                        |
|                     |           |                    |               |         |                  |                    |            | $I_\gamma$ : other: 10.0 10 (1972Hu10).                                                                     |
|                     |           | 4750.0             | 13            | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                             |
|                     |           | 5632.4             | 2             | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                                             |
| 7451.1              | 3/2       | 3532.4             | 5.0 25        | 3918.48 | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                 |
|                     |           | 4448.1             | 10 1          | 3002.75 | 5/2              |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                 |
|                     |           | 4756.9             | 8.0 8         | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                 |
|                     |           | 5687.5             | 4 2           | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                 |
|                     |           | 6231.1             | 73 7          | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                 |
| 7501.1              |           | 3558.0             | 25 3          | 3942.9  |                  |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                 |
|                     |           | 4498.0             | 13.0 13       | 3002.75 | 5/2              |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                 |
|                     |           | 4855.0             | 55 6          | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                 |
|                     |           | 5737.5             | 7.0 7         | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                 |
| 7502.9              |           | 1396.7             | 7             | 6106.2  |                  |                    |            | $I_\gamma$ : from 1976Me12 containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                |
|                     |           |                    |               |         |                  |                    |            | Other: 10.0 10 for 7503 level only (1972Hu10).                                                              |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------|------------------|--------------------|---------------|---------|------------------|--------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7502.9              |                  | 1903.1             | 2             | 5599.7  |                  |                    |            | $I_\gamma$ : from <b>1976Me12</b> containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     |                  | 2492.7             | 2             | 5010.1  |                  |                    |            | $I_\gamma$ : from <b>1976Me12</b> containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     |                  | 3324.8             | 3             | 4177.9  | 3/2              |                    |            | $I_\gamma$ : from <b>1976Me12</b> containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     |                  | 3443.5             | 50            | 4059.2  |                  |                    |            | $I_\gamma$ : from <b>1976Me12</b> containing $\gamma$ rays of 7501 and 7503 doublet initial levels. Other: 70.0 70 for 7503 level only ( <b>1972Hu10</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                     |                  | 3559.8             | 2             | 3942.9  |                  |                    |            | $I_\gamma$ : from <b>1976Me12</b> containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     |                  | 4339.6             | 4             | 3162.99 | 7/2 <sup>-</sup> |                    |            | $I_\gamma$ : from <b>1976Me12</b> containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     |                  | 4499.8             | 4             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : from <b>1976Me12</b> containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     |                  | 4856.8             | 7             | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : from <b>1976Me12</b> containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     |                  | 5739.3             | 0.7           | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : from <b>1976Me12</b> containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     |                  | 6282.9             | 18            | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : from <b>1976Me12</b> containing $\gamma$ rays of 7501 and 7503 doublet initial levels. Other: 20.0 20 for 7503 level only ( <b>1972Hu10</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                     |                  | 7502.0             | 0.3           | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : from <b>1976Me12</b> containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 7518.8              | 7/2              | 1919.0             | 2 1           | 5599.7  |                  |                    |            | $I_\gamma$ : from <b>1972Hu10</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     |                  | 1933               | 3.0 15        | 5586    |                  |                    |            | $I_\gamma$ : from <b>1972Hu10</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     |                  | 3170.9             | 2 1           | 4347.76 | 5/2,9/2          |                    |            | $I_\gamma$ : from <b>1972Hu10</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     |                  | 3575.7             | 2 1           | 3942.9  |                  |                    |            | $I_\gamma$ : from <b>1972Hu10</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     |                  | 4355.5             | 68 7          | 3162.99 | 7/2 <sup>-</sup> | D(+Q)              | +0.1 2     | $I_\gamma$ : from <b>1972Hu10</b> .<br>Mult., $\delta$ : from <b>1977Ko35</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                     |                  | 4872.7             | 3.0 15        | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : from <b>1972Hu10</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     |                  | 5755.2             | 20 2          | 1763.14 | 5/2 <sup>+</sup> | D+Q                | +0.098 22  | $I_\gamma$ : from <b>1972Hu10</b> .<br>Mult., $\delta$ : from $\gamma(\theta)$ in <b>1977Ko35</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 7548.9              | 7/2 <sup>-</sup> | 1903               | 0.55 6        | 5646    |                  |                    |            | $I_\gamma$ : from <b>1971Pr11</b> . Others: 0.6 ( <b>1976Me12</b> ); 0.5 3 ( <b>1972Hu10</b> ); 1 ( <b>2001Vo24</b> , <b>2002Vo17</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                  | 1963               | 0.1           | 5586    |                  |                    |            | $I_\gamma$ : other: 0.2 1 ( <b>1972Hu10</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                     |                  | 2778.9             | 1.4 1         | 4769.86 |                  |                    |            | $I_\gamma$ : from <b>1971Pr11</b> . Others: 1.3 ( <b>1976Me12</b> ); 1.5 8 ( <b>1972Hu10</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                     |                  | 4385.6             | 94 1          | 3162.99 | 7/2 <sup>-</sup> | M1+E2              | -0.07 2    | $I_\gamma$ : from <b>1971Pr11</b> . Others: 95 ( <b>1976Me12</b> ); 95.0 95 ( <b>1972Hu10</b> ); $\geq 93$ ( <b>1971Wi13</b> ).<br>Mult.: D+Q from $\gamma(\theta)$ in <b>1967Wa22</b> and <b>1966Wa09</b> ; magnetic character from $\gamma(\text{pol})$ in <b>1967Wa22</b> .<br>$\delta$ : from <b>1966Wa09</b> and <b>1967Wa22</b> . Others: $\delta=+0.08$ 10 ( <b>1977Ko35</b> ), $\delta=-0.03$ 3 from <b>2001Vo24</b> and <b>2002Vo17</b> .<br>$N_{90}/N_0=0.72$ 6 ( <b>1971Wi13</b> ).<br>POL=+0.76 6, $\Delta\pi=\text{no}$ ( <b>1966Wa09</b> ).<br>POL=+0.93 10, $\Delta\pi=\text{no}$ ( <b>1967Wa22</b> ).<br>0.021<POL<0.813 ( <b>1966Az01</b> ).<br>$A_2=+0.42$ 3, $A_4=0.00$ 3 ( <b>2001Vo24</b> , <b>2002Vo17</b> ).<br>$A_2=+0.42$ 3, $A_4=-0.001$ 27 ( <b>1966Az01</b> ).<br>$A_2=+0.42$ 3, $A_4=+0.00$ 3 ( <b>1967Ta08</b> ). |
|                     |                  | 4545.8             | 2.0 1         | 3002.75 | 5/2              | D(+Q)              | +0.6 4     | $I_\gamma$ : from <b>1971Pr11</b> . Others: 2 ( <b>1976Me12</b> ); 1.8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$ | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------|------------------|--------------------|---------------|---------|-----------|--------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7548.9              | 7/2 <sup>-</sup> | 4902.8             | 0.68 7        | 2645.70 | (5/2,7/2) |                    |            | 9 (1972Hu10); 1.3 (1971Wi13).<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko35.<br>I $_\gamma$ : from 1971Pr11. Others: 0.5 (1976Me12); 0.5 3 (1972Hu10); $\leq 5$ (1971Wi13).<br>5785.3 0.33 4 1763.14 5/2 <sup>+</sup><br>I $_\gamma$ : from 1971Pr11. Others: 0.3 (1976Me12); 0.3 2 (1972Hu10); $\approx 0.4$ (1971Wi13).<br>7548.0 0.27 3 0 3/2 <sup>+</sup> Q(+O) +0.1 2<br>I $_\gamma$ : from 1971Pr11. Others: 0.2 (1976Me12); 0.2 1 (1972Hu10); $\approx 0.3$ (1971Wi13).<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko35.<br>7561.3 1/2,3/2 3383.2 0.9 4177.9 3/2<br>I $_\gamma$ : other: 1.0 5 (1972Hu10).<br>3501.9 6 4059.2<br>I $_\gamma$ : other: 5.0 25 (1972Hu10).<br>3593.8 0.5 3967.3<br>I $_\gamma$ : other: 1.0 5 (1972Hu10).<br>3642.6 1.6 3918.48 3/2 <sup>+</sup><br>I $_\gamma$ : other: 21.0 21 (1972Hu10).<br>4867.0 22 2693.9 (3/2,5/2)<br>A <sub>2</sub> =+0.04 3, A <sub>4</sub> =-0.05 3 (1976Me12).<br>I $_\gamma$ : other: 36.0 36 (1972Hu10).<br>6341.3 35 1219.39 1/2 <sup>+</sup><br>A <sub>2</sub> =-0.02 2, A <sub>4</sub> =-0.05 3 (1976Me12).<br>I $_\gamma$ : other: 36.0 36 (1972Hu10).<br>7560.4 34 0 3/2 <sup>+</sup><br>A <sub>2</sub> =+0.01 2, A <sub>4</sub> =-0.03 3 (1976Me12).<br>7600.8 5/2 2437.4 1.9 5163.34 7/2<br>2719.7 2 4880.96<br>2761.6 2 4839.10<br>2830.8 1.3 4769.86<br>2976.3 1.8 4624.4<br>3422.7 3 4177.9 3/2<br>I $_\gamma$ : other: 2 1 (1972Hu10).<br>3427.2 4173.45<br>I $_\gamma$ : 2 1 from 1972Hu10.<br>3541.4 2 4059.2<br>I $_\gamma$ : other: 1.5 8 (1972Hu10).<br>4437.5 5 3162.99 7/2 <sup>-</sup><br>I $_\gamma$ : others: 4 2 (1972Hu10); 8 (1971Wi13).<br>Mult., $\delta$ : Q+O with $\delta=-0.25$ 15 for J <sub>i</sub> =3/2 (1977Ko34).<br>A <sub>2</sub> =+0.19 2, A <sub>4</sub> =+0.08 2 (1977Ko34).<br>A <sub>2</sub> =+0.35 32, A <sub>4</sub> =-0.44 32 (1971Wi13).<br>4597.7 8 3002.75 5/2 D+Q +0.50 22<br>I $_\gamma$ : others: 9.0 9 (1972Hu10); 12 (1971Wi13).<br>Mult., $\delta$ : from 1971Wi13; $\delta=+0.85$ 42 for J <sub>i</sub> =3/2 (1977Ko34).<br>A <sub>2</sub> =+0.75 7, A <sub>4</sub> =-0.06 7 (1971Wi13).<br>A <sub>2</sub> =+0.53 21, A <sub>4</sub> =-0.04 25 (1976Me12).<br>A <sub>2</sub> =+0.42 3, A <sub>4</sub> =-0.10 3 (1977Ko34).<br>4906.5 18 2693.9 (3/2,5/2) D+Q +0.32 7<br>I $_\gamma$ : others: 19.0 19 (1972Hu10); 24 (1971Wi13).<br>$\delta$ : from 1971Wi13; other: $\delta=-5.3$ 9 or +0.01 5 for J <sub>i</sub> =3/2 (1977Ko34).<br>A <sub>2</sub> =+0.33 1, A <sub>4</sub> =-0.05 1 (1977Ko34).<br>A <sub>2</sub> =+0.18 4, A <sub>4</sub> =-0.02 5 (1976Me12).<br>A <sub>2</sub> =+0.26 10, A <sub>4</sub> =-0.05 10 (1971Wi13).<br>4954.7 1.9 2645.70 (5/2,7/2)<br>I $_\gamma$ : other: 1.5 8 (1972Hu10).<br>5837.1 19 1763.14 5/2 <sup>+</sup> D+Q +0.31 7<br>I $_\gamma$ : others: 20 2 (1971Wi13), 2 1 (1972Hu10); 25 (1971Wi13).<br>Mult., $\delta$ : from 1971Wi13. $\delta=+0.2$ 22 for J <sub>i</sub> =3/2 (1977Ko34).<br>A <sub>2</sub> =+0.10 7, A <sub>4</sub> =-0.10 8 (1971Wi13).<br>A <sub>2</sub> =+0.12 2, A <sub>4</sub> =-0.10 3 (1976Me12).<br>A <sub>2</sub> =+0.30 9, A <sub>4</sub> =-0.17 9 (1977Ko34).<br>I $_\gamma$ : others: 1.0 5 (1972Hu10); $\leq 1.5$ (1971Wi13).<br>6380.8 1.1 1219.39 1/2 <sup>+</sup> |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)**

| $\gamma(^{35}\text{Cl})$ (continued) |                  |                    |               |         |                  |                    |            | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------|------------------|--------------------|---------------|---------|------------------|--------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 7600.8                               | 5/2              | 7599.9             | 33            | 0       | 3/2 <sup>+</sup> | D+Q                | -0.19 4    | $I_\gamma$ : others: 31.0 3I (1972Hu10); 31 (1971Wi13).<br>Mult., $\delta$ : from 1976Me12. Others: $\delta=+0.17$ 3 (1971Wi13).<br>$A_2=-0.73$ 7, $A_4=+0.04$ 6 (1971Wi13).<br>$A_2=-0.62$ 5, $A_4=-0.06$ 5 (1976Me12).<br>$A_2=-0.56$ 3, $A_4=-0.05$ 2 (1977Ko34).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 7618.7                               | 5/2              | 1438               | 0.6           | 6181    |                  |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                      |                  | 2402.8             | 2             | 5215.8  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 2455.3             | 0.7           | 5163.34 | 7/2              |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 2779.5             | 0.7           | 4839.10 |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 3440.6             | 3             | 4177.9  | 3/2              |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 3559.3             | 2             | 4059.2  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 4455.4             | 3             | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 5855.0             | 10            | 1763.14 | 5/2 <sup>+</sup> | D+Q                | +0.48 11   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 7656.6                               |                  | 1550.4             | 3.0 15        | 6106.2  |                  |                    |            | $I_\gamma$ : other: 10.0 10 (1972Hu10).<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko34.<br>$A_2=+0.38$ 5, $A_4=-0.01$ 6 (1976Me12).<br>$A_2=-0.11$ 3, $A_4=-0.06$ 3 (1977Ko34).<br>$I_\gamma$ : other: 80 8 (1972Hu10).<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko34.<br>$A_2=-0.46$ 4, $A_4=+0.03$ 4 (1976Me12).<br>$A_2=-0.57$ 3, $A_4=+0.27$ 3 (1977Ko34).<br>$I_\gamma$ : from 1972Hu10.<br>$I_\gamma$ : from 1972Hu10.<br>$I_\gamma$ : from 1972Hu10.<br>$I_\gamma$ : from 1972Hu10.<br>$I_\gamma$ : from 1972Hu10.<br>$I_\gamma$ : from 1972Hu10.<br>$I_\gamma$ : from 1972Hu10.<br>$I_\gamma$ : from 1972Hu10.                                                                                                                                                                                                                                                                                                                                            |
|                                      |                  | 1974               | 1.0 5         | 5683    |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 2002               | 1.0 5         | 5655    |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 2817.4             | 3.0 15        | 4839.10 |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 3478.5             | 1.0 5         | 4177.9  | 3/2              |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 3597.2             | 32.0 32       | 4059.2  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 3689.1             | 2.0 10        | 3967.3  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 7655.7             | 57 6          | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 7671.9                               | 5/2              | 2072.1             | 1.2           | 5599.7  |                  |                    |            | $I_\gamma$ : other: 4 2 (1972Hu10).<br>$I_\gamma$ : other: 1.0 5 (1972Hu10).<br>$I_\gamma$ : other: 6.0 6 (1972Hu10).<br>$I_\gamma$ : other: 4 2 (1972Hu10).<br>$I_\gamma$ : other: 4 2 (1972Hu10).<br>$I_\gamma$ : other: 8.0 8 (1972Hu10).<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko34.<br>$A_2=-0.35$ 4, $A_4=+0.01$ 5 (1976Me12).<br>$A_2=+0.52$ 1, $A_4=-0.18$ 1 (1977Ko34).<br>$I_\gamma$ : other: 14.0 14 (1972Hu10).<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko34.<br>$A_2=+0.41$ 3, $A_4=-0.03$ 4 (1976Me12).<br>$A_2=+0.33$ 8, $A_4=-0.12$ 8 (1977Ko34).<br>$I_\gamma$ : other: 57.0 57 (1972Hu10).<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko34.<br>$A_2=-0.38$ 4, $A_4=-0.06$ 5 (1976Me12).<br>$A_2=-0.40$ 9, $A_4=+0.07$ 8 (1977Ko34).<br>$I_\gamma$ : other: 2 1 (1972Hu10).<br>$I_\gamma$ : 0.5 3 (1972Hu10).<br>$I_\gamma$ : other: 2 1 (1972Hu10).<br>$I_\gamma$ : other: 2 1 (1972Hu10).<br>$I_\gamma$ : other: 0.5 3 (1972Hu10). |
|                                      |                  | 2508.5             | 4             | 5163.34 | 7/2              |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 2790.8             | 1.2           | 4880.96 |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 2901.9             | 1.0           | 4769.86 |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 3324.0             | 6             | 4347.76 | 5/2,9/2          |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 3498.3             | 5             | 4173.45 |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 4508.6             | 1.8           | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 4668.8             | 9             | 3002.75 | 5/2              | D+Q                | -1.4 5     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 7685.5                               | 3/2 <sup>-</sup> | 5025.8             | 13            | 2645.70 | (5/2,7/2)        | D(+Q)              | +0.3 22    | $I_\gamma$ : other: 2 1 (1972Hu10).<br>$I_\gamma$ : 0.5 3 (1972Hu10).<br>$I_\gamma$ : other: 2 1 (1972Hu10).<br>$I_\gamma$ : other: 2 1 (1972Hu10).<br>$I_\gamma$ : other: 0.5 3 (1972Hu10).<br>$I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                      |                  | 5908.2             | 57            | 1763.14 | 5/2 <sup>+</sup> | D+Q                | +0.86 14   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 7685.5                               | 3/2 <sup>-</sup> | 7671.0             | 0.8           | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                      |                  | 1927               |               | 5758    |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 2085.7             | 2             | 5599.7  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 2281.9             | 2             | 5403.5  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 2469.6             | 0.7           | 5215.8  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 2675.3             | 0.6           | 5010.1  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                  | 3507.4             | 3             | 4177.9  | 3/2              |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)**

| $\gamma(^{35}\text{Cl})$ (continued) |           |                    |                  |         |                  |                    |            |                                                                         |
|--------------------------------------|-----------|--------------------|------------------|---------|------------------|--------------------|------------|-------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$ | $E_\gamma^\dagger$ | $I_\gamma^\#$    | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                |
| 7685.5                               | $3/2^-$   | 3511.9             | 4                | 4173.45 |                  |                    |            | $I_\gamma$ : other: 5.0 25 (1972Hu10).                                  |
|                                      |           |                    |                  |         |                  |                    |            | $A_2=+0.01$ 14, $A_4=+0.23$ 16 (1977Ko34).                              |
|                                      |           | 3626.1             | 6                | 4059.2  |                  |                    |            | $I_\gamma$ : other: 6.0 6 (1972Hu10).                                   |
|                                      |           | 4682.4             | 10               | 3002.75 | 5/2              | D(+Q)              | -0.01 5    | $I_\gamma$ : other: 9.0 9 (1972Hu10).                                   |
|                                      |           |                    |                  |         |                  |                    |            | Mult., $\delta$ : other: $\delta=-2.4$ +20-4 (1976Me12);                |
|                                      |           |                    |                  |         |                  |                    |            | $\delta=+0.01$ 7 (1977Ko34).                                            |
|                                      |           |                    |                  |         |                  |                    |            | $A_2=-0.04$ 6, $A_4=-0.09$ 7 (1976Me12).                                |
|                                      |           |                    |                  |         |                  |                    |            | $A_2=-0.06$ 12, $A_4=+0.08$ 13 (1977Ko34).                              |
|                                      |           |                    |                  |         |                  |                    |            |                                                                         |
|                                      |           |                    |                  |         |                  |                    |            |                                                                         |
| 7693.9                               | $5/2^+$   | 5921.8             | 0.7              | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                   |
|                                      |           | 6465.5             | 1                | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ , Mult., $\delta$ : from 1976Me12. Other: $I_\gamma=72.0$ 72 |
|                                      |           | 7684.6             | 70               | 0       | 3/2 <sup>+</sup> | D+Q                | +6.0 7     | (1972Hu10); $\delta<-3.5$ for $J_i=3/2$ (1977Ko34).                     |
|                                      |           |                    |                  |         |                  |                    |            | $A_2=+0.25$ 1, $A_4=-0.03$ 1 (1976Me12).                                |
|                                      |           |                    |                  |         |                  |                    |            | $A_2=+0.18$ 11, $A_4=-0.04$ 11 (1977Ko34).                              |
|                                      |           | 2094.1             | 1.0 5            | 5599.7  |                  |                    |            | $I_\gamma$ : from 1972Hu10.                                             |
|                                      |           | 3634.5             | 1.0 5            | 4059.2  |                  |                    |            | $I_\gamma$ : from 1972Hu10.                                             |
|                                      |           | 4690.8             | 2.0 10           | 3002.75 | 5/2              |                    |            | $I_\gamma$ : from 1972Hu10.                                             |
|                                      |           | 7693.0             | 96 10            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10.                                             |
|                                      |           | 7706.4             | 5/2 <sup>+</sup> | 2051    | 4                | 5655               |            |                                                                         |
|                                      | $5/2^+$   |                    |                  |         |                  |                    |            | Mult., $\delta$ : D+Q with $\delta\leq-2$ or $\geq 13$ for              |
|                                      |           |                    |                  |         |                  |                    |            | $J_f(5655)=5/2$ (1977Ko34), but $J_f(5655)=(3/2)^+$                     |
|                                      |           |                    |                  |         |                  |                    |            | from the Adopted Levels.                                                |
|                                      |           |                    |                  |         |                  |                    |            | $A_2=+0.14$ 15, $A_4=-0.39$ 17 (1977Ko34).                              |
|                                      |           |                    |                  |         |                  |                    |            |                                                                         |
|                                      |           | 2825.3             | 0.4              | 4880.96 |                  |                    |            |                                                                         |
|                                      |           | 2867.2             | 1.3              | 4839.10 |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                   |
|                                      |           | 3081.9             | 0.8              | 4624.4  |                  |                    |            |                                                                         |
|                                      |           | 3528.3             | 2                | 4177.9  | 3/2              |                    |            |                                                                         |
|                                      |           | 3592.5             | 2                | 4113.7  | 7/2              |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                   |
|                                      | $7/2$     | 3787.7             | 0.6              | 3918.48 | 3/2 <sup>+</sup> |                    |            |                                                                         |
|                                      |           | 4703.3             | 1                | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                   |
|                                      |           | 5060.3             | 4                | 2645.70 | (5/2,7/2)        | D+Q                | -1.01 10   | $I_\gamma$ : other: 4 2 (1972Hu10).                                     |
|                                      |           |                    |                  |         |                  |                    |            | Mult., $\delta$ : or $\geq 13$ (1977Ko34).                              |
|                                      |           |                    |                  |         |                  |                    |            | $A_2=-0.03$ 6, $A_4=+0.08$ 8 (1976Me12).                                |
|                                      |           |                    |                  |         |                  |                    |            | $A_2=-0.18$ 17, $A_4=+0.41$ 18 (1977Ko34).                              |
|                                      |           |                    |                  |         |                  |                    |            |                                                                         |
|                                      |           | 5942.7             | 0.4              | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                   |
|                                      |           | 6486.4             | 0.5              | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 88.0 88 (1972Hu10).                                 |
|                                      |           | 7705.5             | 83               | 0       | 3/2 <sup>+</sup> | D(+Q)              | +0.1 1     | Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko34.                    |
|                                      | $7/2$     |                    |                  |         |                  |                    |            | $A_2=-0.54$ 3, $A_4=-0.07$ 3 (1976Me12).                                |
|                                      |           |                    |                  |         |                  |                    |            | $A_2=-0.63$ 4, $A_4=+0.15$ 3 (1977Ko34).                                |
|                                      |           | 2863.7             | 20               | 4880.96 |                  |                    |            | $I_\gamma$ : other: 19.0 19 (1972Hu10).                                 |
|                                      |           | 2974.8             | 2                | 4769.86 |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                   |
|                                      |           | 3630.9             | 6                | 4113.7  | 7/2              |                    |            | $I_\gamma$ : other: 5.0 25 (1972Hu10).                                  |
|                                      |           | 3801.7             | 10               | 3942.9  |                  | D(+Q)              | -0.16 17   | $I_\gamma$ : other: 12.0 12 (1972Hu10).                                 |
|                                      |           |                    |                  |         |                  |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko35.                    |
|                                      |           | 4581.5             | 16               | 3162.99 | 7/2 <sup>-</sup> | D+Q                | +0.18 8    | Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko35.                    |
|                                      |           | 5098.7             | 43               | 2645.70 | (5/2,7/2)        | D(+Q)              | +0.1 30    | Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko35.                    |
|                                      |           | 5981.1             | 2.4              | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                         |
| 7777.1                               | $7/2$     | 6524.8             | 0.6              | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                         |
|                                      |           | 2122               | 0.5              | 5655    |                  |                    |            |                                                                         |
|                                      |           | 2613.7             | 1.1              | 5163.34 | 7/2              |                    |            |                                                                         |
|                                      |           | 2937.9             | 2                | 4839.10 |                  |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                     |
|                                      |           | 3007.1             | 3                | 4769.86 |                  |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                     |
|                                      |           | 3599.0             | 7                | 4177.9  | 3/2              |                    |            | $I_\gamma$ : other: 7.0 7 (1972Hu10).                                   |
|                                      |           | 3717.7             | 1.1              | 4059.2  |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                   |
|                                      |           |                    |                  |         |                  |                    |            |                                                                         |
|                                      |           |                    |                  |         |                  |                    |            |                                                                         |
|                                      |           |                    |                  |         |                  |                    |            |                                                                         |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                                                     |
|---------------------|------------------|--------------------|---------------|---------|------------------|--------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7777.1              |                  | 4613.8             | 2             | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                                              |
|                     |                  | 4774.0             | 4             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 4 2 (1972Hu10).                                                                                                                          |
|                     |                  | 5082.8             | 9             | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : other: 7.0 7 (1972Hu10).                                                                                                                        |
|                     |                  | 5131.0             | 1.3           | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                          |
|                     |                  | 6013.4             | 40            | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 46.0 46 (1972Hu10).                                                                                                                      |
|                     |                  | 6557.1             | 11            | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 13.0 13 (1972Hu10).                                                                                                                      |
|                     |                  | 7776.2             | 18            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 16.0 16 (1972Hu10).                                                                                                                      |
| 7781.7              | 5/2 <sup>-</sup> | 1675.5             | 1.4           | 6106.2  |                  |                    |            |                                                                                                                                                              |
|                     |                  | 2181.9             |               | 5599.7  |                  |                    |            | $I_\gamma$ : 1.0 5 from 1972Hu10.                                                                                                                            |
|                     |                  | 2378.1             |               | 5403.5  |                  |                    |            | $I_\gamma$ : 1.0 5 from 1972Hu10.                                                                                                                            |
|                     |                  | 2565.8             | 1.5           | 5215.8  |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                        |
|                     |                  | 3011.7             | 7             | 4769.86 |                  |                    |            | $I_\gamma$ : other: 7.0 7 (1972Hu10).                                                                                                                        |
|                     |                  | 3433.8             | 2             | 4347.76 | 5/2,9/2          |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                          |
|                     |                  | 3608.1             | 36            | 4173.45 |                  |                    |            | $I_\gamma$ : other: 40.0 40 (1972Hu10).                                                                                                                      |
|                     |                  | 3667.8             | 6             | 4113.7  | 7/2              |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                                                       |
|                     |                  | 3722.3             | 1.2           | 4059.2  |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                        |
|                     |                  | 3863.0             | 11            | 3918.48 | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 10.0 10 (1972Hu10).                                                                                                                      |
|                     |                  | 4618.4             | 5             | 3162.99 | 7/2 <sup>-</sup> |                    |            | $I_\gamma$ : other: 5.0 25 (1972Hu10).                                                                                                                       |
|                     |                  | 5135.6             | 8             | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : other: 9.0 9 (1972Hu10).                                                                                                                        |
|                     |                  | 6018.0             | 18            | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 20.0 20 (1972Hu10).                                                                                                                      |
|                     |                  | 6561.7             | 1.2           | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                              |
|                     |                  | 7780.8             | 1.7           | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                              |
| 7797.1              | 1/2              | 2039               |               | 5758    |                  |                    |            | $I_\gamma$ : 0.5 3 from 1972Hu10.                                                                                                                            |
|                     |                  | 2142               | 1.3           | 5655    |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                        |
|                     |                  |                    |               |         |                  |                    |            | $A_2=+0.03$ 8 (1988Va06).                                                                                                                                    |
|                     |                  | 2393.5             | 2             | 5403.5  |                  |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                          |
|                     |                  |                    |               |         |                  |                    |            | $A_2=+0.04$ 5 (1988Va06).                                                                                                                                    |
|                     |                  | 2786.9             | 2             | 5010.1  |                  |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                          |
|                     |                  |                    |               |         |                  |                    |            | $A_2=-0.07$ 5 (1988Va06).                                                                                                                                    |
|                     |                  | 2942.6             | 1.3           | 4854.4  |                  |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                                                                        |
|                     |                  | 2957.9             | 1.1           | 4839.10 |                  |                    |            |                                                                                                                                                              |
|                     |                  | 3737.7             | 0.3           | 4059.2  |                  |                    |            |                                                                                                                                                              |
|                     |                  | 6577.1             | 9             | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 9.0 9 (1972Hu10).                                                                                                                        |
|                     |                  |                    |               |         |                  |                    |            | $\delta$ : $\delta(Q/D)=+0.21$ 4 or $-1.9$ 2 for $J=3/2$ (1976Me12), which is invalidated by the $J^\pi=1/2^-$ assignment for the 7797 level in 1988Va06.    |
|                     |                  |                    |               |         |                  |                    |            | $A_2=-0.02$ 6, $A_4=-0.14$ 7 (1976Me12).                                                                                                                     |
|                     |                  |                    |               |         |                  |                    |            | $A_2=-0.03$ 3 (1988Va06).                                                                                                                                    |
|                     |                  | 7796.2             | 83            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 85.0 85 (1972Hu10).                                                                                                                      |
|                     |                  |                    |               |         |                  |                    |            | $\delta$ : $\delta(Q/D)=-0.33$ 2 or $-19$ +5-10 for $J=3/2$ (1976Me12), which is invalidated by the $J^\pi=1/2^-$ assignment for the 7797 level in 1988Va06. |
|                     |                  |                    |               |         |                  |                    |            | $A_2=-0.06$ 2, $A_4=-0.03$ 2 (1976Me12).                                                                                                                     |
|                     |                  |                    |               |         |                  |                    |            | $A_2=+0.006$ 14 (1988Va06).                                                                                                                                  |
| 7837.2              | 3/2              | 1656               | 0.5           | 6181    |                  |                    |            | $I_\gamma$ : for 7837 and 7839 doublet initial levels (1976Me12). Other: 1 from 2001Vo24.                                                                    |
|                     |                  | 2237.4             | 0.5           | 5599.7  |                  |                    |            | $I_\gamma$ : for 7837 and 7839 doublet initial levels (1976Me12). Other: 1 (2001Vo24).                                                                       |
|                     |                  | 2621.3             | 1.4           | 5215.8  |                  |                    |            | $I_\gamma$ : for 7837 and 7839 doublet initial levels (1976Me12). Others: 1 (2001Vo24); 1.0 5 (1972Hu10).                                                    |
|                     |                  | 3659.1             | 28            | 4177.9  | 3/2              | M1+E2              | -0.05 3    | $I_\gamma$ : for 7837 and 7839 doublet initial levels (1976Me12). Others: 28 (2001Vo24); 35.0 35 (1972Hu10).                                                 |
|                     |                  |                    |               |         |                  |                    |            | Mult., $\delta$ : D+Q from $\gamma(\theta)$ in 1967Wa22; magnetic                                                                                            |

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$\gamma(^{35}\text{Cl})$  (continued)Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$      | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$    | Comments                                 |
|---------------------|----------------|--------------------|---------------|---------|--------------|------------------------------------------|
| 7868.7              |                | 7867.8             | 61            | 0       | $3/2^+$      | $I_\gamma$ : other: 62.0 62 (1972Hu10).  |
| 7880.8              | $3/2, 5/2$     | 2226               | 12            | 5655    |              | $I_\gamma$ : other: 9.0 9 (1972Hu10).    |
|                     |                | 3041.6             | 1.7           | 4839.10 |              | $I_\gamma$ : other: 3.0 15 (1972Hu10).   |
|                     |                | 3256.2             | 2             | 4624.4  |              | $I_\gamma$ : other: 1.0 5 (1972Hu10).    |
|                     |                | 3702.7             | 3             | 4177.9  | $3/2$        |                                          |
|                     |                | 3766.9             | 1.3           | 4113.7  | $7/2$        | $I_\gamma$ : other: 4 2 (1972Hu10).      |
|                     |                | 3821.4             | 2             | 4059.2  |              | $I_\gamma$ : other: 1.0 5 (1972Hu10).    |
|                     |                | 3962.1             | 2             | 3918.48 | $3/2^+$      | $I_\gamma$ : other: 2 1 (1972Hu10).      |
|                     |                | 4877.7             | 11            | 3002.75 | $5/2$        | $I_\gamma$ : other: 10.0 10 (1972Hu10).  |
|                     |                | 5186.5             | 25            | 2693.9  | $(3/2, 5/2)$ | $I_\gamma$ : other: 30.0 30 (1972Hu10).  |
|                     |                |                    |               |         |              | $A_2=+0.50$ 3, $A_4=+0.09$ 4 (1976Me12). |
|                     |                | 6117.1             | 30            | 1763.14 | $5/2^+$      | $I_\gamma$ : other: 31.0 31 (1972Hu10).  |
|                     |                |                    |               |         |              | $A_2=+0.08$ 3, $A_4=-0.03$ 4 (1976Me12). |
|                     |                | 7879.9             | 10            | 0       | $3/2^+$      | $I_\gamma$ : other: 9.0 9 (1972Hu10).    |
|                     |                |                    |               |         |              | $A_2=-0.06$ 4, $A_4=+0.04$ 5 (1976Me12). |
| 7899.2              | $(3/2^-, 5/2)$ | 2253               | 1.5           | 5646    |              |                                          |
|                     |                | 2735.8             | 0.9           | 5163.34 | $7/2$        |                                          |
|                     |                | 3018.1             | 0.8           | 4880.96 |              |                                          |
|                     |                | 3274.6             | 0.5           | 4624.4  |              |                                          |
|                     |                | 3721.1             | 8             | 4177.9  | $3/2$        | $I_\gamma$ : other: 8.0 8 (1972Hu10).    |
|                     |                | 3725.5             | 8             | 4173.45 |              | $I_\gamma$ : other: 6.0 6 (1972Hu10).    |
|                     |                | 3839.8             | 3             | 4059.2  |              | $I_\gamma$ : other: 2 1 (1972Hu10).      |
|                     |                | 4735.9             | 4             | 3162.99 | $7/2^-$      | $I_\gamma$ : other: 4 2 (1972Hu10).      |
|                     |                | 5204.9             | 1.3           | 2693.9  | $(3/2, 5/2)$ |                                          |
|                     |                | 6135.5             | 68            | 1763.14 | $5/2^+$      | $I_\gamma$ : other: 77.0 77 (1972Hu10).  |
|                     |                | 7898.2             | 4             | 0       | $3/2^+$      | $I_\gamma$ : other: 3.0 15 (1972Hu10).   |
| 7923.3              |                | 3084.1             | 1.4           | 4839.10 |              |                                          |
|                     |                | 3153.3             | 2             | 4769.86 |              |                                          |
|                     |                | 3298.7             | 1.9           | 4624.4  |              |                                          |
|                     |                | 3745.2             | 3             | 4177.9  | $3/2$        | $I_\gamma$ : other: 2 1 (1972Hu10).      |
|                     |                | 3749.6             | 5             | 4173.45 |              | $I_\gamma$ : other: 6.0 6 (1972Hu10).    |
|                     |                | 3863.9             | 1.7           | 4059.2  |              | $I_\gamma$ : other: 1.0 5 (1972Hu10).    |
|                     |                | 3955.8             | 1.5           | 3967.3  |              | $I_\gamma$ : other: 1.0 5 (1972Hu10).    |
|                     |                | 4004.6             | 1.2           | 3918.48 | $3/2^+$      | $I_\gamma$ : other: 1.0 5 (1972Hu10).    |
|                     |                | 4920.2             | 0.9           | 3002.75 | $5/2$        |                                          |
|                     |                | 5229.0             | 11            | 2693.9  | $(3/2, 5/2)$ | $I_\gamma$ : other: 12.0 12 (1972Hu10).  |
|                     |                | 5277.2             | 0.9           | 2645.70 | $(5/2, 7/2)$ |                                          |
|                     |                | 6159.6             | 1.5           | 1763.14 | $5/2^+$      | $I_\gamma$ : other: 1.0 5 (1972Hu10).    |
|                     |                | 6703.2             | 27            | 1219.39 | $1/2^+$      | $I_\gamma$ : other: 28.0 28 (1972Hu10).  |
|                     |                | 7922.3             | 41            | 0       | $3/2^+$      | $I_\gamma$ : other: 48.0 48 (1972Hu10).  |
| 7970.3              |                | 2324               | 3             | 5646    |              | $I_\gamma$ : other: 3.0 15 (1972Hu10).   |
|                     |                | 2384               | 3             | 5586    |              |                                          |
|                     |                | 2754.4             | 3             | 5215.8  |              | $I_\gamma$ : other: 3.0 15 (1972Hu10).   |
|                     |                | 3089.2             | 10            | 4880.96 |              | $I_\gamma$ : other: 9.0 9 (1972Hu10).    |
|                     |                | 3200.3             | 5             | 4769.86 |              | $I_\gamma$ : other: 6.0 6 (1972Hu10).    |
|                     |                | 3622.3             | 8             | 4347.76 | $5/2, 9/2$   | $I_\gamma$ : other: 8.0 8 (1972Hu10).    |
|                     |                | 3792.2             | 6             | 4177.9  | $3/2$        | $I_\gamma$ : other: 6.0 6 (1972Hu10).    |
|                     |                | 3856.4             | 6             | 4113.7  | $7/2$        | $I_\gamma$ : other: 2 1 (1972Hu10).      |
|                     |                | 4807.0             | 10            | 3162.99 | $7/2^-$      | $I_\gamma$ : other: 13.0 13 (1972Hu10).  |
|                     |                | 4967.2             | 5             | 3002.75 | $5/2$        | $I_\gamma$ : other: 5.0 25 (1972Hu10).   |
|                     |                | 5324.2             | 20            | 2645.70 | $(5/2, 7/2)$ | $I_\gamma$ : other: 22.0 22 (1972Hu10).  |
|                     |                | 6206.6             | 20            | 1763.14 | $5/2^+$      | $I_\gamma$ : other: 21.0 21 (1972Hu10).  |
|                     |                | 6750.2             | 1             | 1219.39 | $1/2^+$      |                                          |
| 7987.8              | $3/2$          | 1881.6             | 0.5           | 6106.2  |              | $I_\gamma$ : other: 1.0 5 (1972Hu10).    |
|                     |                | 2333               | 1.7           | 5655    |              | $I_\gamma$ : other: 1.0 5 (1972Hu10).    |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$               | $E_\gamma^\dagger$ | $I_\gamma^\#$           | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                       |
|---------------------|-------------------------|--------------------|-------------------------|---------|------------------|--------------------|------------|--------------------------------------------------------------------------------------------------------------------------------|
| 7987.8              | 3/2                     | 3809.7             | <0.8                    | 4177.9  | 3/2              |                    |            | $I_\gamma$ : total $I_\gamma=0.8$ for $\gamma$ decays to 4173 and 4178 doublet final levels (1976Me12).                        |
|                     |                         | 3814.1             | <0.8                    | 4173.45 |                  |                    |            | $I_\gamma$ : total $I_\gamma=0.8$ for $\gamma$ decays to 4173 and 4178 doublet final levels (1976Me12).                        |
|                     |                         | 3928.4             | 1                       | 4059.2  |                  |                    |            |                                                                                                                                |
|                     |                         | 6767.7             | 60                      | 1219.39 | 1/2 <sup>+</sup> | D+Q                | +0.21 4    | $I_\gamma$ : other: 62.0 62 (1972Hu10).<br>Mult., $\delta$ : or -3.1 4 (1976Me12).<br>$A_2=-0.07$ 2, $A_4=-0.05$ 3 (1976Me12). |
|                     |                         | 7986.8             | 36                      | 0       | 3/2 <sup>+</sup> | D+Q                | -0.36 2    | $I_\gamma$ : other: 36.0 36 (1972Hu10).<br>Mult., $\delta$ : or -11 3 (1976Me12).<br>$A_2=-0.10$ 2, $A_4=-0.06$ 2 (1976Me12).  |
| 7995.6              | 5/2                     | 2190               | 0.6                     | 5806    |                  |                    |            |                                                                                                                                |
|                     |                         | 2395.8             | 1.1                     | 5599.7  |                  |                    |            |                                                                                                                                |
|                     |                         | 2592.0             | 3                       | 5403.5  |                  |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                         |
|                     |                         | 2832.1             | 1.5                     | 5163.34 | 7/2              |                    |            |                                                                                                                                |
|                     |                         | 3817.5             | 4                       | 4177.9  | 3/2              |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                         |
|                     |                         | 3936.2             | 1.1                     | 4059.2  |                  |                    |            |                                                                                                                                |
|                     |                         | 4832.3             | 7                       | 3162.99 | 7/2 <sup>-</sup> |                    |            | $I_\gamma$ : other: 6.0 6 (1972Hu10).                                                                                          |
|                     |                         | 4992.5             | 2                       | 3002.75 | 5/2              |                    |            |                                                                                                                                |
|                     |                         | 5301.3             | 0.2                     | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                                |
|                     |                         | 5349.5             | 0.6                     | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                                |
|                     |                         | 6231.9             | 0.9                     | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                          |
|                     |                         | 7994.6             | 78                      | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 87.0 87 (1972Hu10).                                                                                        |
|                     |                         | 8001.0             | (7/2,9/2 <sup>+</sup> ) | 2355    | 1.8              | 5646               |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                         |
|                     |                         | 2785.1             | 0.8                     | 5215.8  |                  |                    |            |                                                                                                                                |
|                     |                         | 3119.9             | 0.7                     | 4880.96 |                  |                    |            |                                                                                                                                |
| 8001.0              | (7/2,9/2 <sup>+</sup> ) | 3231.0             | 8                       | 4769.86 |                  |                    |            | $I_\gamma$ : other: 8.0 8 (1972Hu10).                                                                                          |
|                     |                         | 3653.0             | 1.9                     | 4347.76 | 5/2,9/2          |                    |            |                                                                                                                                |
|                     |                         | 3827.3             | 10                      | 4173.45 |                  |                    |            | $I_\gamma$ : other: 8.0 8 (1972Hu10).                                                                                          |
|                     |                         | 4057.9             | 3                       | 3942.9  |                  |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                            |
|                     |                         | 4837.7             | 0.8                     | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                |
|                     |                         | 5354.9             | 1.6                     | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                            |
|                     |                         | 6237.3             | 70                      | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 76.0 76 (1972Hu10).                                                                                        |
|                     |                         | 8000.0             | 1.4                     | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                          |
|                     |                         | 2350               | 1.8                     | 5655    |                  |                    |            |                                                                                                                                |
|                     |                         | 3123.9             | 0.6                     | 4880.96 |                  |                    |            |                                                                                                                                |
|                     |                         | 3165.8             | 3                       | 4839.10 |                  |                    |            |                                                                                                                                |
|                     |                         | 3826.9             | <1.3                    | 4177.9  | 3/2              |                    |            | $I_\gamma$ : total $I_\gamma=1.3$ for $\gamma$ decays to 4173 and 4178 doublet final levels (1976Me12).                        |
|                     |                         | 3831.3             | <1.3                    | 4173.45 |                  |                    |            | $I_\gamma$ : total $I_\gamma=1.3$ for $\gamma$ decays to 4173 and 4178 doublet final levels (1976Me12).                        |
|                     |                         | 4841.7             | 15                      | 3162.99 | 7/2 <sup>-</sup> |                    |            | $I_\gamma$ : other: 17.0 17 (1972Hu10).<br>$A_2=-0.084$ 14, $A_4=-0.0041$ 15 (1981Bi05).                                       |
|                     |                         | 5001.9             | 2                       | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                         |
| 8005.0              | 5/2 <sup>+</sup>        | 5310.7             | 1.7                     | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                                |
|                     |                         | 5358.9             | 4                       | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                                |
|                     |                         | 6241.3             | 15                      | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 20.0 20 (1972Hu10).<br>$A_2=-0.11$ 2, $A_4=-0.09$ 3 (1976Me12).                                            |
|                     |                         | 6784.9             | 0.6                     | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                |
|                     |                         | 8004.0             | 55                      | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 60.0 60 (1972Hu10).<br>$A_2=-0.03$ 1, $A_4=-0.02$ 2 (1976Me12).                                            |
|                     |                         | 8035.7             |                         | 2230    | 6.0 6            | 5806               |            | $I_\gamma$ : from 1972Hu10.                                                                                                    |
|                     |                         | 3857.6             | 15.0 15                 | 4177.9  | 3/2              |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                                    |
|                     |                         | 3976.3             | 11.0 11                 | 4059.2  |                  |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                                    |
|                     |                         | 5341.4             | 5.0 25                  | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                                    |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$               | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                                                                                                                                                                                        |
|---------------------|-------------------------|--------------------|---------------|---------|------------------|--------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8035.7              |                         | 6272.0             | 5.0 25        | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                                                                                                                                                                                                     |
|                     |                         | 6815.6             | 56 6          | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                                                                                                                                                                                                     |
|                     |                         | 8034.7             | 2.0 10        | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                                                                                                                                                                                                     |
| 8038.5              | 3/2                     | 3183.9             | 0.6           | 4854.4  |                  |                    |            |                                                                                                                                                                                                                                                                                                 |
|                     |                         | 3199.2             | 1.6           | 4839.10 |                  |                    |            |                                                                                                                                                                                                                                                                                                 |
|                     |                         | 3413.9             | 3             | 4624.4  |                  |                    |            |                                                                                                                                                                                                                                                                                                 |
|                     |                         | 3860.4             | 1.6           | 4177.9  | 3/2              |                    |            |                                                                                                                                                                                                                                                                                                 |
|                     |                         | 3979.1             | 3             | 4059.2  |                  |                    |            | $I_\gamma$ : other: 4 2 (1972Hu10).                                                                                                                                                                                                                                                             |
|                     |                         | 4071.0             | 0.8           | 3967.3  |                  |                    |            |                                                                                                                                                                                                                                                                                                 |
|                     |                         | 4119.8             | 1.4           | 3918.48 | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                                                 |
|                     |                         | 5035.4             | 4             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                                                                                                                                                                                          |
|                     |                         | 5344.2             | 57            | 2693.9  | (3/2,5/2)        | D+Q                | -0.20 2    | $I_\gamma$ : other: 61.0 61 (1972Hu10).<br>Mult., $\delta$ : or +19 +10-5 (1976Me12).<br>$A_2=+0.08$ 2, $A_4=-0.03$ 2 (1976Me12).<br>$I_\gamma$ : other: 17.0 17 (1972Hu10).<br>$A_2=+0.24$ 2, $A_4=+0.01$ 3 (1976Me12).                                                                        |
|                     |                         | 6274.8             | 15            | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 6.0 6 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 6818.4             | 4             | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 9.0 9 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 8037.5             | 8             | 0       | 3/2 <sup>+</sup> | D+Q                | -0.13 4    | Mult., $\delta$ : or +8.1 +33-18 (1976Me12).<br>$A_2=+0.20$ 3, $A_4=-0.08$ 3 (1976Me12).<br>$I_\gamma$ : other: 10.0 10 (1972Hu10).<br>$I_\gamma$ : other: 11.0 11 (1972Hu10).<br>$I_\gamma$ : other: 7.0 7 (1972Hu10).<br>$I_\gamma$ : 1.0 5 from 1972Hu10.<br>$I_\gamma$ : 2 1 from 1972Hu10. |
| 8075.9              |                         | 2430               | 11            | 5646    |                  |                    |            | $I_\gamma$ : other: 4 2 (1972Hu10).                                                                                                                                                                                                                                                             |
|                     |                         | 2490               | 11            | 5586    |                  |                    |            | $I_\gamma$ : other: 7.0 7 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 2860.0             | 8             | 5215.8  |                  |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                                                                                                                                                             |
|                     |                         | 2912.4             |               | 5163.34 | 7/2              |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                                                                                                                                                             |
|                     |                         | 3194.8             |               | 4880.96 |                  |                    |            | $I_\gamma$ : other: 53.0 53 (1972Hu10).                                                                                                                                                                                                                                                         |
|                     |                         | 3305.9             | 4             | 4769.86 |                  |                    |            | $I_\gamma$ : 1.0 5 from 1972Hu10.                                                                                                                                                                                                                                                               |
|                     |                         | 3897.8             | 8             | 4177.9  | 3/2              |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 3962.0             | 2             | 4113.7  | 7/2              |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 5072.8             | 2             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                                                                                                                                                             |
|                     |                         | 6312.2             | 56            | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 53.0 53 (1972Hu10).                                                                                                                                                                                                                                                         |
|                     |                         | 8074.9             |               | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : 1.0 5 from 1972Hu10.                                                                                                                                                                                                                                                               |
| 8096.1              | (5/2,7/2 <sup>+</sup> ) | 2290               | 2             | 5806    |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 2450               | 5             | 5646    |                  |                    |            | $I_\gamma$ : other: 5.0 25 (1972Hu10).                                                                                                                                                                                                                                                          |
|                     |                         | 2496.3             | 5             | 5599.7  |                  |                    |            | $I_\gamma$ : other: 6.0 6 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 2880.2             | 1.3           | 5215.8  |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 2932.6             | 2             | 5163.34 | 7/2              |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                                                                                                                                                             |
|                     |                         | 3256.8             | 3             | 4839.10 |                  |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                                                                                                                                                                                          |
|                     |                         | 3326.1             | 2             | 4769.86 |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 3918.0             | 1.9           | 4177.9  | 3/2              |                    |            |                                                                                                                                                                                                                                                                                                 |
|                     |                         | 4036.7             | 2             | 4059.2  |                  |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                                                                                                                                                             |
|                     |                         | 4177.4             | 1.8           | 3918.48 | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                                                                                                                                                                                          |
|                     |                         | 4932.7             | 5             | 3162.99 | 7/2 <sup>-</sup> |                    |            | $I_\gamma$ : other: 5.0 25 (1972Hu10).                                                                                                                                                                                                                                                          |
|                     |                         | 5401.8             | 5             | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : other: 5.0 25 (1972Hu10).                                                                                                                                                                                                                                                          |
|                     |                         | 5449.9             | 17            | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : other: 17.0 17 (1972Hu10).                                                                                                                                                                                                                                                         |
|                     |                         | 6332.4             | 8             | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 9.0 9 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 8095.1             | 39            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 40.0 40 (1972Hu10).                                                                                                                                                                                                                                                         |
| 8106.4              | 3/2                     | 2506.6             | 0.5           | 5599.7  |                  |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 3251.8             | 0.6           | 4854.4  |                  |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 3267.1             | 0.5           | 4839.10 |                  |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 3928.3             | 1.6           | 4177.9  | 3/2              |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 4047.0             | 1.3           | 4059.2  |                  |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 4138.8             | 3             | 3967.3  |                  |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                                                                                                                                                                                          |
|                     |                         | 5103.3             | 9             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 8.0 8 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 5412.1             | 9             | 2693.9  | (3/2,5/2)        | D+Q                | +4.9 10    | $I_\gamma$ : other: 8.0 8 (1972Hu10).                                                                                                                                                                                                                                                           |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)**

| $\gamma(^{35}\text{Cl})$ (continued) |                                           |                    |                                       |         |                  |                           |            |                                                    |                                        |
|--------------------------------------|-------------------------------------------|--------------------|---------------------------------------|---------|------------------|---------------------------|------------|----------------------------------------------------|----------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$                                 | $E_\gamma^\dagger$ | $I_\gamma^\#$                         | $E_f$   | $J_f^\pi$        | Mult. <sup><i>b</i></sup> | $\delta^b$ | Comments                                           |                                        |
| 8106.4                               | 3/2                                       | 6342.6             | 3                                     | 1763.14 | 5/2 <sup>+</sup> |                           |            | Mult., $\delta$ : -0.05 4 (1976Me12).              |                                        |
|                                      |                                           | 6886.3             | 35                                    | 1219.39 | 1/2 <sup>+</sup> |                           |            | $A_2=+0.32$ 4, $A_4=-0.08$ 5 (1976Me12).           |                                        |
|                                      |                                           |                    |                                       |         |                  |                           |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).             |                                        |
|                                      |                                           |                    |                                       |         |                  |                           |            | $I_\gamma$ : other: 35.0 35 (1972Hu10).            |                                        |
| 8113.3                               | (1/2 <sup>+</sup> ,3/2,5/2 <sup>+</sup> ) | 8105.4             | 37                                    | 0       | 3/2 <sup>+</sup> | D+Q                       | +1.2 1     | $A_2=-0.46$ 2, $A_4=-0.05$ 3 (1976Me12).           |                                        |
|                                      |                                           |                    |                                       |         |                  |                           |            | $I_\gamma$ : other: 40.0 40 (1972Hu10).            |                                        |
|                                      |                                           |                    |                                       |         |                  |                           |            | Mult., $\delta$ : or -0.46 4 (1976Me12).           |                                        |
|                                      |                                           |                    |                                       |         |                  |                           |            | $A_2=+0.83$ 3, $A_4=+0.01$ 2 (1976Me12).           |                                        |
|                                      |                                           |                    |                                       |         |                  |                           |            | $I_\gamma$ : other: 6.0 6 (1972Hu10).              |                                        |
|                                      |                                           | 1621               | 6                                     | 6492    |                  |                           |            |                                                    |                                        |
|                                      |                                           | 2007.0             | 6                                     | 6106.2  |                  |                           |            |                                                    |                                        |
|                                      |                                           | 2307               | 1.1                                   | 5806    |                  |                           |            |                                                    |                                        |
|                                      |                                           | 2458               | 0.9                                   | 5655    |                  |                           |            |                                                    |                                        |
|                                      |                                           | 2513.5             |                                       | 5599.7  |                  |                           |            |                                                    | $I_\gamma$ : 3.0 15 (1972Hu10).        |
|                                      |                                           | 2897.4             | 1.8                                   | 5215.8  |                  |                           |            |                                                    |                                        |
|                                      |                                           | 3939.6             | 7                                     | 4173.45 |                  |                           |            |                                                    | $I_\gamma$ : other: 5.0 25 (1972Hu10). |
|                                      |                                           | 4053.9             | 1.3                                   | 4059.2  |                  |                           |            |                                                    |                                        |
|                                      |                                           | 4145.7             | 5                                     | 3967.3  |                  |                           |            |                                                    | $I_\gamma$ : other: 5.0 25 (1972Hu10). |
|                                      |                                           | 5110.2             | 1.9                                   | 3002.75 | 5/2              |                           |            |                                                    |                                        |
| 8147.2                               | 3/2 <sup>-</sup>                          | 5419.0             | 13                                    | 2693.9  | (3/2,5/2)        |                           |            | $I_\gamma$ : other: 17.0 17 (1972Hu10).            |                                        |
|                                      |                                           | 6349.5             | 5                                     | 1763.14 | 5/2 <sup>+</sup> |                           |            | $I_\gamma$ : other: 6.0 6 (1972Hu10).              |                                        |
|                                      |                                           | 6893.2             | 32                                    | 1219.39 | 1/2 <sup>+</sup> |                           |            | $I_\gamma$ : other: 32.0 32 (1972Hu10).            |                                        |
|                                      |                                           | 8112.3             | 25                                    | 0       | 3/2 <sup>+</sup> |                           |            | $I_\gamma$ : other: 26.0 26 (1972Hu10).            |                                        |
|                                      |                                           | 2464               | 1.0                                   | 5683    |                  |                           |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).              |                                        |
|                                      |                                           | 3137.0             | 3                                     | 5010.1  |                  |                           |            |                                                    |                                        |
|                                      |                                           | 3969.1             | 8                                     | 4177.9  | 3/2              |                           |            |                                                    |                                        |
|                                      |                                           | 4087.7             | 3                                     | 4059.2  |                  |                           |            |                                                    |                                        |
|                                      |                                           | 4179.6             | 2                                     | 3967.3  |                  |                           |            |                                                    |                                        |
|                                      |                                           | 4228.5             | 3                                     | 3918.48 | 3/2 <sup>+</sup> |                           |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).             |                                        |
|                                      |                                           | 5452.8             | 3                                     | 2693.9  | (3/2,5/2)        |                           |            | $I_\gamma$ : other: 4 2 (1972Hu10).                |                                        |
|                                      |                                           | 6927.1             | 15                                    | 1219.39 | 1/2 <sup>+</sup> |                           |            | $I_\gamma$ : other: 20.0 20 (1972Hu10).            |                                        |
|                                      |                                           | 8146.2             | 62                                    | 0       | 3/2 <sup>+</sup> |                           |            | $I_\gamma$ : other: 75.0 75 (1972Hu10).            |                                        |
|                                      |                                           | 8156.8             | (5/2 <sup>+</sup> ,7/2 <sup>-</sup> ) | 2511    | 4                | 5646                      |            |                                                    |                                        |
| 2571                                 | 8                                         |                    |                                       | 5586    |                  |                           |            | $I_\gamma$ : other: 7.0 7 (1972Hu10).              |                                        |
| 2940.9                               | 6                                         |                    |                                       | 5215.8  |                  |                           |            | $I_\gamma$ : other: 8.0 8 (1972Hu10).              |                                        |
| 3386.8                               | 1.0                                       |                    |                                       | 4769.86 |                  |                           |            |                                                    |                                        |
| 3978.7                               | 5                                         |                    |                                       | 4177.9  | 3/2              |                           |            | $I_\gamma$ : other: 8.0 8 (1972Hu10).              |                                        |
| 4042.9                               | 7                                         |                    |                                       | 4113.7  | 7/2              |                           |            | $I_\gamma$ : other: 7.0 7 (1972Hu10).              |                                        |
| 4097.3                               | 1                                         |                    |                                       | 4059.2  |                  |                           |            |                                                    |                                        |
| 4213.6                               | 2                                         |                    |                                       | 3942.9  |                  |                           |            |                                                    |                                        |
| 4993.4                               | 5                                         |                    |                                       | 3162.99 | 7/2 <sup>-</sup> |                           |            | $I_\gamma$ : other: 9.0 9 (1972Hu10).              |                                        |
| 5153.6                               | 6                                         |                    |                                       | 3002.75 | 5/2              |                           |            | $I_\gamma$ : other: 4 2 (1972Hu10).                |                                        |
| 5510.6                               | 51                                        |                    |                                       | 2645.70 | (5/2,7/2)        |                           |            | $I_\gamma$ : other: 50.0 50 (1972Hu10).            |                                        |
| 6393.0                               | 1                                         |                    |                                       | 1763.14 | 5/2 <sup>+</sup> |                           |            |                                                    |                                        |
| 8155.8                               | 3                                         |                    |                                       | 0       | 3/2 <sup>+</sup> |                           |            | $I_\gamma$ : other: 4 2 (1972Hu10).                |                                        |
| 3554.8                               | 9.0 9                                     |                    |                                       | 4624.4  |                  |                           |            | $I_\gamma$ : from 1972Hu10.                        |                                        |
| 8179.4                               |                                           | 5016.0             | 8.0 8                                 | 3162.99 | 7/2 <sup>-</sup> |                           |            | $I_\gamma$ : from 1972Hu10.                        |                                        |
|                                      |                                           | 5176.2             | 4.0 20                                | 3002.75 | 5/2              |                           |            | $I_\gamma$ : from 1972Hu10.                        |                                        |
|                                      |                                           | 5485.0             | 3.0 15                                | 2693.9  | (3/2,5/2)        |                           |            | $I_\gamma$ : from 1972Hu10.                        |                                        |
|                                      |                                           | 6415.6             | 68 7                                  | 1763.14 | 5/2 <sup>+</sup> |                           |            | $I_\gamma$ : from 1972Hu10.                        |                                        |
|                                      |                                           | 8178.4             | 8.0 8                                 | 0       | 3/2 <sup>+</sup> |                           |            | $I_\gamma$ : from 1972Hu10.                        |                                        |
|                                      |                                           |                    |                                       |         |                  |                           |            |                                                    |                                        |
| 8208.2                               | 5/2 <sup>+</sup>                          | 2553               | 0.7                                   | 5655    |                  |                           |            | $I_\gamma$ : other: 1 (2001Vo24).                  |                                        |
|                                      |                                           | 3327.1             | 0.8                                   | 4880.96 |                  |                           |            | $I_\gamma$ : other: 1 (2001Vo24).                  |                                        |
|                                      |                                           | 3368.9             | 0.4                                   | 4839.10 |                  |                           |            | $I_\gamma$ : other: 1 (2001Vo24).                  |                                        |
|                                      |                                           | 4148.7             | 1.5                                   | 4059.2  |                  |                           |            | $I_\gamma$ : others: 2 (2001Vo24); 2 1 (1972Hu10). |                                        |

Continued on next page (footnotes at end of table)



$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10** (continued) $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$ | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$    | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                |
|---------------------|-----------|--------------------|---------------|---------|--------------|--------------------|------------|---------------------------------------------------------------------------------------------------------|
| 8208.2              | $5/2^+$   | 4289.4             |               | 3918.48 | $3/2^+$      |                    |            | $I_\gamma$ : 1.0 5 (1972Hu10).                                                                          |
|                     |           | 5205.0             | 1.1           | 3002.75 | $5/2$        |                    |            | $I_\gamma$ : others: 1 (2001Vo24); 1.0 5 (1972Hu10).                                                    |
|                     |           | 5513.8             | 0.5           | 2693.9  | $(3/2, 5/2)$ |                    |            | $I_\gamma$ : other: 1 (2001Vo24); 1.0 5 (1972Hu10).                                                     |
|                     |           | 6444.4             | 14            | 1763.14 | $5/2^+$      | D+Q                | +0.33 4    | $I_\gamma$ : other: 14 (2001Vo24); 16.0 16 (1972Hu10).                                                  |
|                     |           |                    |               |         |              |                    |            | Mult., $\delta$ : from 2001Vo24.                                                                        |
|                     |           |                    |               |         |              |                    |            | $A_2=+0.51$ 13, $A_4=-0.21$ 11, $A_6=+0.0004$ (1996Ka21).                                               |
|                     |           |                    |               |         |              |                    |            | $A_2=+0.51$ 13, $A_4=-0.21$ 11 (2001Vo24).                                                              |
|                     |           | 6988.1             | 3             | 1219.39 | $1/2^+$      |                    |            | $I_\gamma$ : other: 3 (2001Vo24); 3.0 15 (1972Hu10).                                                    |
|                     |           | 8207.2             | 78            | 0       | $3/2^+$      | D+Q                | -0.36 4    | $I_\gamma$ : other: 78 (2001Vo24); 76.0 76 (1972Hu10).                                                  |
|                     |           |                    |               |         |              |                    |            | Mult., $\delta$ : from 2001Vo24. Other: $\delta(\text{Q/D})=0.52$ 42 from 1996Ka21.                     |
| 8216.3              | $5/2$     |                    |               |         |              |                    |            | $A_2=-0.55$ 8, $A_4=+0.004$ 78, $A_6=+0.006$ 72 (1996Ka21).                                             |
|                     |           |                    |               |         |              |                    |            | $A_2=-0.55$ 8, $A_4=0.00$ 8 (2001Vo24).                                                                 |
|                     |           | 2561               | 0.8           | 5655    |              |                    |            | $I_\gamma$ : other: 1 (2001Vo24).                                                                       |
|                     |           | 3446.3             | 0.8           | 4769.86 |              |                    |            | $I_\gamma$ : other: 1 (2001Vo24).                                                                       |
|                     |           | 4038.2             | <3            | 4177.9  | $3/2$        |                    |            | $I_\gamma$ : other: 3 (2001Vo24).                                                                       |
|                     |           | 4042.6             | <3            | 4173.45 |              |                    |            |                                                                                                         |
|                     |           | 5052.9             | 41            | 3162.99 | $7/2^-$      | D+Q                | +0.11 2    | $I_\gamma$ : other: 41 (2001Vo24); 44.0 44 (1972Hu10).                                                  |
|                     |           |                    |               |         |              |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 2001Vo24.                                                    |
|                     |           |                    |               |         |              |                    |            | $A_2=-0.32$ 7, $A_4=-0.11$ 7, $A_6=+0.05$ 6 (1996Ka21).                                                 |
|                     |           |                    |               |         |              |                    |            | $A_2=-0.32$ 7, $A_4=-0.11$ 7 (2001Vo24).                                                                |
|                     |           | 5213.1             | 5             | 3002.75 | $5/2$        |                    |            | $I_\gamma$ : other: 5 (2001Vo24); 5.0 25 (1972Hu10).                                                    |
|                     |           | 5521.9             | 3             | 2693.9  | $(3/2, 5/2)$ |                    |            | $I_\gamma$ : other: 3 (2001Vo24); 4 2 (1972Hu10).                                                       |
|                     |           | 6452.5             | 1.4           | 1763.14 | $5/2^+$      |                    |            | $I_\gamma$ : other: 1 (2001Vo24).                                                                       |
|                     |           | 8215.3             | 45            | 0       | $3/2^+$      | D+Q                | +0.06 3    | $I_\gamma$ : other: 45 (2001Vo24); 47.0 47 (1972Hu10).                                                  |
|                     |           |                    |               |         |              |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 2001Vo24. Other: $\delta(\text{Q/D})=0.59$ 56 from 1996Ka21. |
|                     |           |                    |               |         |              |                    |            | $A_2=-0.560$ 8, $A_4=+0.017$ 71, $A_6=+0.066$ 75 (1996Ka21).                                            |
|                     |           |                    |               |         |              |                    |            | $A_2=-0.56$ 1, $A_4=+0.02$ 7 (2001Vo24).                                                                |
| 8242.4              |           | 4068.7             | 7             | 4173.45 |              |                    |            |                                                                                                         |
|                     |           | 5239.2             | 2             | 3002.75 | $5/2$        |                    |            |                                                                                                         |
|                     |           | 7022.3             | 91            | 1219.39 | $1/2^+$      |                    |            |                                                                                                         |
|                     |           |                    |               |         |              |                    |            |                                                                                                         |
| 8251?               |           | 4137               | 4&            | 4113.7  | $7/2$        |                    |            |                                                                                                         |
|                     |           | 5248               | 6&            | 3002.75 | $5/2$        |                    |            |                                                                                                         |
|                     |           | 7031               | 82&           | 1219.39 | $1/2^+$      |                    |            |                                                                                                         |
|                     |           | 8250               | 8&            | 0       | $3/2^+$      |                    |            |                                                                                                         |
| 8268.9              |           | 3498.9             | 2             | 4769.86 |              |                    |            | $I_\gamma$ : from 1976Sp08.                                                                             |
|                     |           | 4090.7             | 12            | 4177.9  | $3/2$        |                    |            |                                                                                                         |
|                     |           | 4350.1             | 3             | 3918.48 | $3/2^+$      |                    |            |                                                                                                         |
|                     |           | 5105.5             | 7             | 3162.99 | $7/2^-$      |                    |            |                                                                                                         |
|                     |           | 5265.7             | 13            | 3002.75 | $5/2$        |                    |            |                                                                                                         |
|                     |           | 5574.5             | 6             | 2693.9  | $(3/2, 5/2)$ |                    |            |                                                                                                         |
|                     |           | 5622.7             | 4             | 2645.70 | $(5/2, 7/2)$ |                    |            |                                                                                                         |
|                     |           | 6505.1             | 8             | 1763.14 | $5/2^+$      |                    |            |                                                                                                         |
|                     |           | 8267.9             | 45            | 0       | $3/2^+$      |                    |            |                                                                                                         |
|                     |           |                    |               |         |              |                    |            |                                                                                                         |
| 8277.2              |           | 2622               | 1.5           | 5655    |              |                    |            |                                                                                                         |
|                     |           | 3396.1             | 5             | 4880.96 |              |                    |            |                                                                                                         |
|                     |           | 3437.9             | 11            | 4839.10 |              |                    |            |                                                                                                         |
|                     |           | 3507.2             | 1.2           | 4769.86 |              |                    |            |                                                                                                         |
|                     |           | 3652.6             | 3             | 4624.4  |              |                    |            |                                                                                                         |
|                     |           | 3929.2             | 1.8           | 4347.76 | $5/2, 9/2$   |                    |            |                                                                                                         |
|                     |           |                    |               |         |              |                    |            |                                                                                                         |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$               | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | Comments                                                                   |
|---------------------|-------------------------|--------------------|---------------|---------|------------------|--------------------|----------------------------------------------------------------------------|
| 8277.2              |                         | 4103.5             | 2             | 4173.45 |                  |                    |                                                                            |
|                     |                         | 4217.7             | 1.5           | 4059.2  |                  |                    |                                                                            |
|                     |                         | 4358.4             | 5             | 3918.48 | 3/2 <sup>+</sup> |                    |                                                                            |
|                     |                         | 5274.0             | 10            | 3002.75 | 5/2              |                    |                                                                            |
|                     |                         | 5582.8             | 3             | 2693.9  | (3/2,5/2)        |                    |                                                                            |
|                     |                         | 5631.0             | 3             | 2645.70 | (5/2,7/2)        |                    |                                                                            |
|                     |                         | 6513.4             | 23            | 1763.14 | 5/2 <sup>+</sup> |                    |                                                                            |
|                     |                         | 8276.2             | 29            | 0       | 3/2 <sup>+</sup> |                    |                                                                            |
| 8282.4              | (3/2 <sup>-</sup> ,5/2) | 2696               | 8             | 5586    |                  |                    |                                                                            |
|                     |                         | 3066.5             | 8             | 5215.8  |                  |                    |                                                                            |
|                     |                         | 3118.9             | 3             | 5163.34 | 7/2              |                    |                                                                            |
|                     |                         | 4104.2             | 14            | 4177.9  | 3/2              |                    |                                                                            |
|                     |                         | 4222.9             | 6             | 4059.2  |                  |                    |                                                                            |
|                     |                         | 5119.0             | 26            | 3162.99 | 7/2 <sup>-</sup> |                    |                                                                            |
|                     |                         | 6518.6             | 7             | 1763.14 | 5/2 <sup>+</sup> |                    |                                                                            |
|                     |                         | 8281.4             | 28            | 0       | 3/2 <sup>+</sup> |                    |                                                                            |
| 8287.82             | 1/2 <sup>-</sup>        | 2633               | 2             | 5655    |                  |                    |                                                                            |
|                     |                         | 3277.6             | 3             | 5010.1  |                  |                    |                                                                            |
|                     |                         | 4109.7             | 9             | 4177.9  | 3/2              |                    |                                                                            |
|                     |                         | 4228.4             | 2             | 4059.2  |                  |                    |                                                                            |
|                     |                         | 4320.2             | 3             | 3967.3  |                  |                    |                                                                            |
|                     |                         | 4369.05            | 1             | 3918.48 | 3/2 <sup>+</sup> |                    |                                                                            |
|                     |                         | 5593.4             | 23            | 2693.9  | (3/2,5/2)        |                    | $A_2=-0.01$ 4, $A_4=-0.02$ 4 (1976Me12).                                   |
|                     |                         | 7067.66            | 44            | 1219.39 | 1/2 <sup>+</sup> |                    | $\delta: \delta(\text{Q/D})=+0.21$ 2 or $-3.1$ 2 for $J_i=3/2$ (1976Me12). |
|                     |                         |                    |               |         |                  |                    | $A_2=-0.08$ 2, $A_4=-0.01$ 3 (1976Me12).                                   |
|                     |                         | 8286.77            | 13            | 0       | 3/2 <sup>+</sup> |                    | $\delta: \delta(\text{Q/D})=-0.35$ 4 or $-11$ 3 for $J_i=3/2$ (1976Me12).  |
|                     |                         |                    |               |         |                  |                    | $A_2=-0.11$ 3, $A_4=-0.01$ 4 (1976Me12).                                   |
| 8298.2              | 3/2                     | 3287.9             | 3             | 5010.1  |                  |                    |                                                                            |
|                     |                         | 3417.1             | 4             | 4880.96 |                  |                    |                                                                            |
|                     |                         | 3443.6             | 5             | 4854.4  |                  |                    |                                                                            |
|                     |                         | 4124.5             | 6             | 4173.45 |                  |                    |                                                                            |
|                     |                         | 4238.7             | 10            | 4059.2  |                  |                    |                                                                            |
|                     |                         | 5134.8             | 1             | 3162.99 | 7/2 <sup>-</sup> |                    |                                                                            |
|                     |                         | 5295.0             | 1             | 3002.75 | 5/2              |                    |                                                                            |
|                     |                         | 5603.8             | 10            | 2693.9  | (3/2,5/2)        |                    | $A_2=+0.09$ 6, $A_4=-0.07$ 7 (1976Me12).                                   |
|                     |                         | 5652.0             | 6             | 2645.70 | (5/2,7/2)        |                    |                                                                            |
|                     |                         | 6534.4             | 8             | 1763.14 | 5/2 <sup>+</sup> |                    | $A_2=+0.43$ 11, $A_4=-0.51$ 12 (1976Me12).                                 |
|                     |                         | 7078.0             | 24            | 1219.39 | 1/2 <sup>+</sup> | D+Q                | Mult., $\delta$ : $-1.7$ 1 or $-0.02$ 2 (1976Me12).                        |
|                     |                         |                    |               |         |                  |                    | $A_2=-0.53$ 3, $A_4=+0.11$ 3 (1976Me12).                                   |
|                     |                         | 8297.1             | 13            | 0       | 3/2 <sup>+</sup> | D+Q                | Mult., $\delta$ : $-0.15$ 4 or $+9.5$ +48-24 (1976Me12).                   |
|                     |                         |                    |               |         |                  |                    | $A_2=+0.18$ 3, $A_4=-0.10$ 4 (1976Me12).                                   |
| 8318.1              |                         | 2512               | 0.7           | 5806    |                  |                    |                                                                            |
|                     |                         | 2672               | 0.7           | 5646    |                  |                    |                                                                            |
|                     |                         | 2718.3             | 1.5           | 5599.7  |                  |                    |                                                                            |
|                     |                         | 3548.1             | 0.3           | 4769.86 |                  |                    |                                                                            |
|                     |                         | 3693.5             | 1.8           | 4624.4  |                  |                    |                                                                            |
|                     |                         | 4139.9             | <3            | 4177.9  | 3/2              |                    |                                                                            |
|                     |                         | 4144.4             | <3            | 4173.45 |                  |                    |                                                                            |
|                     |                         | 4258.6             | 1.2           | 4059.2  |                  |                    |                                                                            |
|                     |                         | 4350.5             | 3             | 3967.3  |                  |                    |                                                                            |
|                     |                         | 5154.7             | 0.5           | 3162.99 | 7/2 <sup>-</sup> |                    |                                                                            |
|                     |                         | 5314.9             | 3             | 3002.75 | 5/2              |                    |                                                                            |
|                     |                         | 5623.7             | 1.6           | 2693.9  | (3/2,5/2)        |                    |                                                                            |
|                     |                         | 5671.9             | 0.9           | 2645.70 | (5/2,7/2)        |                    |                                                                            |
|                     |                         | 6554.3             | 1.8           | 1763.14 | 5/2 <sup>+</sup> |                    |                                                                            |
|                     |                         | 7097.9             | 2             | 1219.39 | 1/2 <sup>+</sup> |                    |                                                                            |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                       |
|---------------------|------------------|--------------------|---------------|---------|------------------|--------------------|------------|----------------------------------------------------------------|
| 8318.1              |                  | 8317.0             | 78            | 0       | 3/2 <sup>+</sup> |                    |            |                                                                |
| 8323.1              |                  | 4209.1             | 1&            | 4113.7  | 7/2              |                    |            |                                                                |
|                     |                  | 5319.9             | 3&            | 3002.75 | 5/2              |                    |            |                                                                |
|                     |                  | 5676.9             | 4&            | 2645.70 | (5/2,7/2)        |                    |            |                                                                |
|                     |                  | 6559.3             | <2&           | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                |
|                     |                  | 7102.9             | 8&            | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                |
|                     |                  | 8322.0             | 82&           | 0       | 3/2 <sup>+</sup> |                    |            |                                                                |
| 8381.8              | 5/2 <sup>+</sup> | 3611.7             |               | 4769.86 |                  |                    |            | $I_\gamma$ : 1 (2001Vo24).                                     |
|                     |                  | 3757.2             |               | 4624.4  |                  |                    |            | $I_\gamma$ : 1 (2001Vo24).                                     |
|                     |                  | 4267.8             | 8.0 8         | 4113.7  | 7/2              |                    |            | $I_\gamma$ : from 1972Hu10. Other: 7 (2001Vo24).               |
|                     |                  | 4463.0             | 24.0 24       | 3918.48 | 3/2 <sup>+</sup> | D+Q                | +0.15 3    | $I_\gamma$ : from 1972Hu10. Other: 24 (2001Vo24).              |
|                     |                  |                    |               |         |                  |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 2001Vo24.           |
|                     |                  |                    |               |         |                  |                    |            | $A_2=+0.25$ 3, $A_4=-0.04$ 2 (2001Vo24).                       |
|                     |                  | 5378.6             | 25.0 25       | 3002.75 | 5/2              | D(+Q)              | +0.02 3    | $I_\gamma$ : from 1972Hu10. Other: 25 (2001Vo24).              |
|                     |                  |                    |               |         |                  |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 2001Vo24. Other:    |
|                     |                  |                    |               |         |                  |                    |            | $\delta=-0.11$ 1 from 1967Ko20, 1967Ko23.                      |
|                     |                  |                    |               |         |                  |                    |            | $A_2=+0.60$ 7, $A_4=-0.02$ 7 (1967Ko20,1967Ko23).              |
|                     |                  | 5687.4             | 1.0 5         | 2693.9  | (3/2,5/2)        |                    |            | $A_2=+0.51$ 23, $A_4=-0.05$ 24 (2001Vo24).                     |
|                     |                  |                    |               |         |                  |                    |            | $I_\gamma$ : from 1972Hu10. Other: 1 (2001Vo24).               |
|                     |                  | 5735.6             | 5.0 25        | 2645.70 | (5/2,7/2)        |                    |            | $A_2=+0.22$ 9, $A_4=-0.02$ 11 (1967Ko20,1967Ko23).             |
|                     |                  |                    |               |         |                  |                    |            | $I_\gamma$ : from 1972Hu10. Others: 5 (2001Vo24).              |
|                     |                  |                    |               |         |                  |                    |            | Mult., $\delta$ : D+Q with $\delta=+0.72$ +42-20 and $J_f=3/2$ |
|                     |                  |                    |               |         |                  |                    |            | for level 2645 (1967Ko20).                                     |
|                     |                  | 6618.0             | 34.0 34       | 1763.14 | 5/2 <sup>+</sup> | D+Q                | +0.06 3    | $I_\gamma$ : from 1972Hu10. Other: 34 (2001Vo24).              |
|                     |                  |                    |               |         |                  |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 2001Vo24. Other:    |
|                     |                  |                    |               |         |                  |                    |            | $\delta=-0.12$ 10 from $\gamma(\theta)$ in 1967Ko23.           |
|                     |                  |                    |               |         |                  |                    |            | $A_2=+0.79$ 15, $A_4=-0.08$ 15                                 |
|                     |                  |                    |               |         |                  |                    |            | (1967Ko20,1967Ko23).                                           |
|                     |                  |                    |               |         |                  |                    |            | $A_2=+0.02$ 13, $A_4=-0.08$ 13 (2001Vo24).                     |
|                     |                  | 8380.7             | 3.0 15        | 0       | 3/2 <sup>+</sup> | D+Q                | 1.65       | $I_\gamma$ : from 1972Hu10. Other: 2 (2001Vo24).               |
|                     |                  |                    |               |         |                  |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 1967Ko23.           |
|                     |                  |                    |               |         |                  |                    |            | $A_2=-0.83$ 22, $A_4=+0.72$ 33                                 |
|                     |                  |                    |               |         |                  |                    |            | (1967Ko20,1967Ko23).                                           |
| 8402.9              |                  | 3632.8             | 4.8 24        | 4769.86 |                  |                    |            |                                                                |
|                     |                  | 4224.7             | 9 5           | 4177.9  | 3/2              |                    |            |                                                                |
|                     |                  | 4343.4             | 23 6          | 4059.2  |                  |                    |            |                                                                |
|                     |                  | 5708.5             | 10 5          | 2693.9  | (3/2,5/2)        |                    |            |                                                                |
|                     |                  | 5756.7             | 7 4           | 2645.70 | (5/2,7/2)        |                    |            |                                                                |
|                     |                  | 6639.1             | 15 4          | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                |
|                     |                  | 8401.8             | 31 8          | 0       | 3/2 <sup>+</sup> |                    |            |                                                                |
| 8405.7              |                  | 3524.6             | 15 4          | 4880.96 |                  |                    |            |                                                                |
|                     |                  | 4232.0             | 1.6 8         | 4173.45 |                  |                    |            |                                                                |
|                     |                  | 4291.7             | 1.4 7         | 4113.7  | 7/2              |                    |            |                                                                |
|                     |                  | 5402.5             | 15 4          | 3002.75 | 5/2              |                    |            |                                                                |
|                     |                  | 6641.9             | 43 11         | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                |
|                     |                  | 8404.6             | 24 6          | 0       | 3/2 <sup>+</sup> |                    |            |                                                                |
| 8416.7              |                  | 3406.4             | 5&            | 5010.1  |                  |                    |            |                                                                |
|                     |                  | 4243.0             | 11&           | 4173.45 |                  |                    |            |                                                                |
|                     |                  | 4473.5             | 3&            | 3942.9  |                  |                    |            |                                                                |
|                     |                  | 5413.5             | 7&            | 3002.75 | 5/2              |                    |            |                                                                |
|                     |                  | 5770.5             | 8&            | 2645.70 | (5/2,7/2)        |                    |            |                                                                |
|                     |                  | 6652.9             | 22&           | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                |

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$^{34}\text{S}(\text{p},\gamma)$  [1976Me12](#),[1976Sp08](#),[1972Hu10](#) (continued) $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma$ † | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Comments                                                                                                                     |
|---------------------|------------------|--------------|---------------|---------|------------------|------------------------------------------------------------------------------------------------------------------------------|
| 8416.7              |                  | 7196.5       | 9&            | 1219.39 | 1/2 <sup>+</sup> |                                                                                                                              |
|                     |                  | 8415.6       | 35&           | 0       | 3/2 <sup>+</sup> |                                                                                                                              |
| 8464.3              |                  | 2809         | 8 4           | 5655    |                  |                                                                                                                              |
|                     |                  | 3248.3       | 5.8 29        | 5215.8  |                  |                                                                                                                              |
|                     |                  | 3454.0       | 8 4           | 5010.1  |                  |                                                                                                                              |
|                     |                  | 3609.7       | 3.3 17        | 4854.4  |                  |                                                                                                                              |
|                     |                  | 3839.7       | 4.3 22        | 4624.4  |                  |                                                                                                                              |
|                     |                  | 4286.1       | 5.2 26        | 4177.9  | 3/2              |                                                                                                                              |
|                     |                  | 4290.6       | 5.5 28        | 4173.45 |                  |                                                                                                                              |
|                     |                  | 4496.7       | 5.2 26        | 3967.3  |                  |                                                                                                                              |
|                     |                  | 4545.5       | 2.6 13        | 3918.48 | 3/2 <sup>+</sup> |                                                                                                                              |
|                     |                  | 5461.1       | 17 4          | 3002.75 | 5/2              |                                                                                                                              |
|                     |                  | 5769.9       | 16 4          | 2693.9  | (3/2,5/2)        |                                                                                                                              |
|                     |                  | 7244.1       | 9 4           | 1219.39 | 1/2 <sup>+</sup> |                                                                                                                              |
|                     |                  | 8463.2       | 10 3          | 0       | 3/2 <sup>+</sup> |                                                                                                                              |
| 8484.4              | 3/2 <sup>+</sup> | 2829         | 5.6 28        | 5655    |                  | $I_\gamma$ : other: 6 <a href="#">2001Vo24</a> .                                                                             |
|                     |                  | 3603.2       | 7.0 35        | 4880.96 |                  | $I_\gamma$ : other: 7 <a href="#">2001Vo24</a> .                                                                             |
|                     |                  | 3859.8       | 1.0 5         | 4624.4  |                  | $I_\gamma$ : other: 1 <a href="#">2001Vo24</a> .                                                                             |
|                     |                  | 4516.8       | 1.2 6         | 3967.3  |                  | $I_\gamma$ : other: 1 <a href="#">2001Vo24</a> .                                                                             |
|                     |                  | 4565.6       | 5.2 26        | 3918.48 | 3/2 <sup>+</sup> | $I_\gamma$ : other: 5 <a href="#">2001Vo24</a> .                                                                             |
|                     |                  | 5481.2       | 7 4           | 3002.75 | 5/2              | $I_\gamma$ : other: 7 <a href="#">2001Vo24</a> .                                                                             |
|                     |                  | 5790.0       | 20 5          | 2693.9  | (3/2,5/2)        | $I_\gamma$ : other: 20 <a href="#">2001Vo24</a> .                                                                            |
|                     |                  | 5838.2       | 2.8 14        | 2645.70 | (5/2,7/2)        | $I_\gamma$ : other: 3 <a href="#">2001Vo24</a> .                                                                             |
|                     |                  | 6720.6       | 46 12         | 1763.14 | 5/2 <sup>+</sup> | $I_\gamma$ : other: 46 <a href="#">2001Vo24</a> .                                                                            |
|                     |                  | 8483.3       | 4 2           | 0       | 3/2 <sup>+</sup> | $I_\gamma$ : other: 4 <a href="#">2001Vo24</a> .                                                                             |
| 8486.1              |                  | 2728         | 1.1 6         | 5758    |                  |                                                                                                                              |
|                     |                  | 3861.5       | 1.8 9         | 4624.4  |                  |                                                                                                                              |
|                     |                  | 4307.9       | <7.2          | 4177.9  | 3/2              | $I_\gamma$ : total $I_\gamma=4.8$ 24 for $\gamma$ decays to 4173 and 4178 doublet final levels ( <a href="#">1976Sp08</a> ). |
|                     |                  | 4312.4       | <7.2          | 4173.45 |                  | $I_\gamma$ : total $I_\gamma=4.8$ 24 for $\gamma$ decays to 4173 and 4178 doublet final levels ( <a href="#">1976Sp08</a> ). |
|                     |                  | 4426.6       | 1.9 10        | 4059.2  |                  |                                                                                                                              |
|                     |                  | 4518.5       | 7 4           | 3967.3  |                  |                                                                                                                              |
|                     |                  | 5482.9       | 8 4           | 3002.75 | 5/2              |                                                                                                                              |
|                     |                  | 5791.7       | 19 5          | 2693.9  | (3/2,5/2)        |                                                                                                                              |
|                     |                  | 6722.3       | 3.6 18        | 1763.14 | 5/2 <sup>+</sup> |                                                                                                                              |
|                     |                  | 7265.9       | 6.6 33        | 1219.39 | 1/2 <sup>+</sup> |                                                                                                                              |
|                     |                  | 8485.0       | 46 12         | 0       | 3/2 <sup>+</sup> |                                                                                                                              |
| 8514.3              |                  | 3659.7       | 4.4 22        | 4854.4  |                  |                                                                                                                              |
|                     |                  | 4336.1       | 2 1           | 4177.9  | 3/2              |                                                                                                                              |
|                     |                  | 4454.8       | 27 7          | 4059.2  |                  |                                                                                                                              |
|                     |                  | 4546.7       | 23 6          | 3967.3  |                  |                                                                                                                              |
|                     |                  | 4595.5       | 6.6 33        | 3918.48 | 3/2 <sup>+</sup> |                                                                                                                              |
|                     |                  | 7294.1       | 10 3          | 1219.39 | 1/2 <sup>+</sup> |                                                                                                                              |
|                     |                  | 8513.2       | 27 7          | 0       | 3/2 <sup>+</sup> |                                                                                                                              |
| 8571.9              |                  | 2848.3       | 2.2 11        | 5723.52 |                  |                                                                                                                              |
|                     |                  | 2917         | 1.3 7         | 5655    |                  |                                                                                                                              |
|                     |                  | 2972.1       | 0.7 4         | 5599.7  |                  |                                                                                                                              |
|                     |                  | 3408.4       | 1.8 9         | 5163.34 | 7/2              |                                                                                                                              |
|                     |                  | 3690.7       | 0.9 5         | 4880.96 |                  |                                                                                                                              |
|                     |                  | 4398.2       | 0.7 4         | 4173.45 |                  |                                                                                                                              |
|                     |                  | 4457.9       | 1.1 6         | 4113.7  | 7/2              |                                                                                                                              |
|                     |                  | 4653.1       | 1.1 6         | 3918.48 | 3/2 <sup>+</sup> |                                                                                                                              |
|                     |                  | 5408.5       | 1.6 8         | 3162.99 | 7/2 <sup>-</sup> |                                                                                                                              |
|                     |                  | 5568.7       | 4.5 23        | 3002.75 | 5/2              |                                                                                                                              |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10** (continued) $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$                             | $E_\gamma$ <sup><math>\dagger</math></sup> | $I_\gamma$ <sup><math>\#</math></sup> | $E_f$   | $J_f^\pi$        | Mult. <sup><math>b</math></sup> | $\delta$ <sup><math>b</math></sup> | Comments                                                                                                           |
|---------------------|---------------------------------------|--------------------------------------------|---------------------------------------|---------|------------------|---------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 8571.9              |                                       | 5877.5                                     | 0.6 3                                 | 2693.9  | (3/2,5/2)        |                                 |                                    |                                                                                                                    |
|                     |                                       | 5925.7                                     | 8 4                                   | 2645.70 | (5/2,7/2)        |                                 |                                    |                                                                                                                    |
|                     |                                       | 7351.7                                     | 5.4 27                                | 1219.39 | 1/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 8570.8                                     | 70 7                                  | 0       | 3/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
| 8580.5              |                                       | 2088                                       | 4.4 22                                | 6492    |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 3741.2                                     | 1.3 7                                 | 4839.10 |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 4402.3                                     | 4.4 22                                | 4177.9  | 3/2              |                                 |                                    |                                                                                                                    |
|                     |                                       | 4521.0                                     | 2 1                                   | 4059.2  |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 4661.7                                     | 2.3 12                                | 3918.48 | 3/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 5886.1                                     | 5.6 28                                | 2693.9  | (3/2,5/2)        |                                 |                                    |                                                                                                                    |
|                     |                                       | 7360.3                                     | 54 6                                  | 1219.39 | 1/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 8579.4                                     | 26 7                                  | 0       | 3/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
| 8590.0              |                                       | 2832                                       | 4.8 24                                | 5758    |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 2935                                       | 3.7 19                                | 5655    |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 3819.9                                     | 16 4                                  | 4769.86 |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 4416.3                                     | 13 3                                  | 4173.45 |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 4476.0                                     | 3.6 18                                | 4113.7  | 7/2              |                                 |                                    |                                                                                                                    |
|                     |                                       | 5426.6                                     | 8 4                                   | 3162.99 | 7/2 <sup>-</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 5586.8                                     | 22 6                                  | 3002.75 | 5/2              |                                 |                                    |                                                                                                                    |
|                     |                                       | 5943.8                                     | 5.5 28                                | 2645.70 | (5/2,7/2)        |                                 |                                    |                                                                                                                    |
|                     |                                       | 6826.2                                     | 14 4                                  | 1763.14 | 5/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 8588.9                                     | 10 3                                  | 0       | 3/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
| 8612.0              |                                       | 6848.1                                     | 50 5                                  | 1763.14 | 5/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 7391.8                                     | 17 4                                  | 1219.39 | 1/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 8610.9                                     | 33 8                                  | 0       | 3/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
| 8613.7              |                                       | 3732.5                                     | 1.6 8                                 | 4880.96 |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 4554.2                                     | 5.4 27                                | 4059.2  |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 4670.5 <sup>c</sup>                        | 4 &                                   | 3942.9  |                  |                                 |                                    | $I_\gamma$ : from <b>1967Da12</b> only; not observed by others.                                                    |
|                     |                                       | 4694.9                                     | 7 4                                   | 3918.48 | 3/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 5610.5                                     | 2.5 13                                | 3002.75 | 5/2              |                                 |                                    |                                                                                                                    |
|                     |                                       | 5919.3                                     | 4.3 22                                | 2693.9  | (3/2,5/2)        |                                 |                                    |                                                                                                                    |
|                     |                                       | 5967.5                                     | 23 6                                  | 2645.70 | (5/2,7/2)        |                                 |                                    |                                                                                                                    |
|                     |                                       | 6849.8 <sup>c</sup>                        | 9 &                                   | 1763.14 | 5/2 <sup>+</sup> |                                 |                                    | $I_\gamma$ : from <b>1967Da12</b> only; not observed by others.                                                    |
|                     |                                       | 7393.5 <sup>c</sup>                        | 2 &                                   | 1219.39 | 1/2 <sup>+</sup> |                                 |                                    | $I_\gamma$ : from <b>1967Da12</b> only; not observed by others.                                                    |
| 8618.1              | (3/2 <sup>+</sup> ,5/2 <sup>+</sup> ) | 8612.6                                     | 56 6                                  | 0       | 3/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 2894.5                                     | 2.1 11                                | 5723.52 |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 2963                                       | 1.7 9                                 | 5655    |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 3763.5                                     | 1.4 7                                 | 4854.4  |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 3993.5                                     | 1.5 8                                 | 4624.4  |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 4650.5                                     | 3.0 15                                | 3967.3  |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 4699.3                                     | 2.3 12                                | 3918.48 | 3/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 5614.9                                     | 16 4                                  | 3002.75 | 5/2              |                                 |                                    |                                                                                                                    |
|                     |                                       | 5923.7                                     | 14 4                                  | 2693.9  | (3/2,5/2)        |                                 |                                    | $A_2=+0.147$ 33, $A_4=+0.114$ 52 ( <b>1970Ko33</b> ).                                                              |
|                     |                                       | 5971.9                                     | 1.7 9                                 | 2645.70 | (5/2,7/2)        |                                 |                                    |                                                                                                                    |
|                     |                                       | 6854.2                                     | 23 6                                  | 1763.14 | 5/2 <sup>+</sup> |                                 |                                    | $A_2=+0.662$ 23, $A_4=+0.319$ 39 ( <b>1970Ko33</b> reported for 7855 $\gamma$ , 7855 is likely a misprint).        |
|                     |                                       | 8617.0                                     | 33 8                                  | 0       | 3/2 <sup>+</sup> |                                 |                                    | $A_2=-0.293$ 29, $A_4=-0.028$ 47 ( <b>1970Ko33</b> ).                                                              |
| 8630.1              | 7/2                                   | 2906.5                                     | 0.3 2                                 | 5723.52 |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 3466.6                                     | 8 4                                   | 5163.34 | 7/2              | D+Q                             | +0.60 10                           | Mult., $\delta$ : D+Q from $\gamma(\theta)$ in <b>1976Sp09</b> . $A_2=+0.60$ 3, $A_4=-0.08$ 4 ( <b>1976Sp09</b> ). |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$ | $E_\gamma^\dagger$ | $I_\gamma^\#$   | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                                                                                                                                                                    |
|---------------------|-----------|--------------------|-----------------|---------|------------------|--------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8630.1              | 7/2       | 3860.0             | 1.4 7           | 4769.86 |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 4282.1             | 12 3            | 4347.76 | 5/2,9/2          | D+Q                | -0.184 14  | Mult., $\delta$ : D+Q from $\gamma(\theta)$ in <a href="#">1976Sp09</a> .<br>$A_2=+0.11$ 2, $A_4=+0.02$ 2 ( <a href="#">1976Sp09</a> ).                                                                                                                                     |
|                     |           | 4456.4             | 3.6 18          | 4173.45 |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 4516.1             | 10 5            | 4113.7  | 7/2              | D+Q                | -0.06 2    | Mult., $\delta$ : D+Q from $\gamma(\theta)$ in <a href="#">1976Sp09</a> .<br>$A_2=+0.43$ 2, $A_4=-0.01$ 2 ( <a href="#">1976Sp09</a> ).                                                                                                                                     |
|                     |           | 4686.9             | 10 5            | 3942.9  |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 5466.7             | 3.5 18          | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 5626.9             | 1.4 7           | 3002.75 | 5/2              |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 5983.9             | 2.6 13          | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 6866.2             | 47 12           | 1763.14 | 5/2 <sup>+</sup> | D+Q                | -0.011 3   | Mult., $\delta$ : D+Q from $\gamma(\theta)$ in <a href="#">1976Sp09</a> .<br>$A_2=+0.10$ 33, $A_4=-0.65$ 46 ( <a href="#">1970Ko33</a> ).<br>$A_2=-0.387$ 7, $A_4=+0.012$ 8 ( <a href="#">1976Sp09</a> ).<br>$A_2=+0.398$ 46, $A_4=+0.019$ 78 ( <a href="#">1970Ko33</a> ). |
|                     |           | 8629.0             | 0.5 3           | 0       | 3/2 <sup>+</sup> |                    |            | $E_\gamma$ : other: 4689 7 ( <a href="#">1970Ko33</a> ).                                                                                                                                                                                                                    |
| 8640                | 5/2       | 4697               | 23 <sup>@</sup> | 3942.9  |                  |                    |            | $E_\gamma$ : other: 5473 7 ( <a href="#">1970Ko33</a> ).                                                                                                                                                                                                                    |
|                     |           | 5477               | 20 <sup>@</sup> | 3162.99 | 7/2 <sup>-</sup> |                    |            | $E_\gamma$ : other: 5630 7 ( <a href="#">1970Ko33</a> ).                                                                                                                                                                                                                    |
|                     |           | 5637               | 6 <sup>@</sup>  | 3002.75 | 5/2              |                    |            | $E_\gamma$ : other: 5941 7 ( <a href="#">1970Ko33</a> ).                                                                                                                                                                                                                    |
|                     |           | 5946               | 13 <sup>@</sup> | 2693.9  | (3/2,5/2)        |                    |            | $E_\gamma$ : other: 6874 7 ( <a href="#">1970Ko33</a> ).                                                                                                                                                                                                                    |
|                     |           | 6876               | 38 <sup>@</sup> | 1763.14 | 5/2 <sup>+</sup> |                    |            | $A_2=-0.52$ 12, $A_4=-0.29$ 19 ( <a href="#">1970Ko33</a> ).                                                                                                                                                                                                                |
|                     |           |                    |                 |         |                  |                    |            |                                                                                                                                                                                                                                                                             |
| 8641.4              | 3/2,5/2   | 2986               | 2.9 15          | 5655    |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 3055               | 2.5 13          | 5586    |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 3237.7             | 4.7 24          | 5403.5  |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 3802.1             | 2.6 13          | 4839.10 |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 3871.3             | 24 6            | 4769.86 |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 4016.8             | 1.9 10          | 4624.4  |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 4463.2             | <18             | 4177.9  | 3/2              |                    |            | $I_\gamma$ : total $I_\gamma=14$ 4 for $\gamma$ decays to 4173 and 4178 doublet final levels ( <a href="#">1976Sp08</a> ).                                                                                                                                                  |
|                     |           | 4467.6             | <18             | 4173.45 |                  |                    |            | $I_\gamma$ : total $I_\gamma=14$ 4 for $\gamma$ decays to 4173 and 4178 doublet final levels ( <a href="#">1976Sp08</a> ).                                                                                                                                                  |
|                     |           | 4581.9             | 9 5             | 4059.2  |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 4722.6             | 5.9 30          | 3918.48 | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                             |
| 8686.2              |           | 5947.0             | 5.2 26          | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 8640.3             | 28 7            | 0       | 3/2 <sup>+</sup> |                    |            | $A_2=-0.12$ 2, $A_4=-0.01$ 2 ( <a href="#">1976Sp09</a> ).                                                                                                                                                                                                                  |
|                     |           | 3100               | 2.1 11          | 5586    |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 3282.5             | 3.1 16          | 5403.5  |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 4508.0             | 2 1             | 4177.9  | 3/2              |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 4626.7             | 3.5 18          | 4059.2  |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 5522.7             | 3.7 19          | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 5991.8             | 2.8 14          | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 6922.3             | 4.8 24          | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 7466.0             | 1.9 10          | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                             |
| 8696.9              |           | 8685.0             | 76 8            | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 2891               | 2.9 15          | 5806    |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 3097.1             | 4.8 24          | 5599.7  |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 7476.7             | 12 3            | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 8695.7             | 80 8            | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                             |
| 8750.9              |           | 2945               | 2.2 11          | 5806    |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 3347.2             | 4.9 25          | 5403.5  |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 3534.9             | 2.4 12          | 5215.8  |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 4572.7             | 2.1 11          | 4177.9  | 3/2              |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 4577.1             | 2.3 12          | 4173.45 |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 6056.4             | 3.0 15          | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 6987.0             | 30 8            | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                             |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $E_\gamma^\dagger$  | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Comments                                                                                                 |
|---------------------|---------------------|---------------|---------|------------------|----------------------------------------------------------------------------------------------------------|
| 8750.9              | 7530.6 <sup>c</sup> | 2             | 1219.39 | 1/2 <sup>+</sup> | $I_\gamma$ : from 1967Da12 only; not observed by others.                                                 |
|                     | 8749.7              | 53 5          | 0       | 3/2 <sup>+</sup> |                                                                                                          |
| 8779.8              | 3179.9              | 4.3 22        | 5599.7  |                  |                                                                                                          |
|                     | 3563.8              | 6.6 33        | 5215.8  |                  |                                                                                                          |
|                     | 4601.6              | <19           | 4177.9  | 3/2              | $I_\gamma$ : total $I_\gamma=15$ 4 for $\gamma$ decays to 4173 and 4178 doublet final levels (1976Sp08). |
|                     | 4606.0              | <19           | 4173.45 |                  | $I_\gamma$ : total $I_\gamma=15$ 4 for $\gamma$ decays to 4173 and 4178 doublet final levels (1976Sp08). |
|                     | 4861.0              | 4.2 21        | 3918.48 | 3/2 <sup>+</sup> |                                                                                                          |
|                     | 5776.5              | 27 7          | 3002.75 | 5/2              |                                                                                                          |
|                     | 6085.3              | 12 3          | 2693.9  | (3/2,5/2)        |                                                                                                          |
|                     | 6133.5              | 13 3          | 2645.70 | (5/2,7/2)        |                                                                                                          |
|                     | 7015.9              | 6.7 34        | 1763.14 | 5/2 <sup>+</sup> |                                                                                                          |
|                     | 8778.6              | 11 3          | 0       | 3/2 <sup>+</sup> |                                                                                                          |
| 8787.2              | 3063.5              | 6.8 34        | 5723.52 |                  |                                                                                                          |
|                     | 3623.7              | 19 5          | 5163.34 | 7/2              |                                                                                                          |
|                     | 4017.1              | 6.7 34        | 4769.86 |                  |                                                                                                          |
|                     | 4609.0              | 9 5           | 4177.9  | 3/2              |                                                                                                          |
|                     | 5623.7              | 5.8 29        | 3162.99 | 7/2 <sup>-</sup> |                                                                                                          |
|                     | 5783.9              | 20 5          | 3002.75 | 5/2              |                                                                                                          |
|                     | 7023.3              | 25 6          | 1763.14 | 5/2 <sup>+</sup> |                                                                                                          |
|                     | 8786.0              | 8 4           | 0       | 3/2 <sup>+</sup> |                                                                                                          |
| 8798.4              | 4738.9              | 8 4           | 4059.2  |                  |                                                                                                          |
|                     | 5795.1              | 16 4          | 3002.75 | 5/2              |                                                                                                          |
|                     | 7034.5              | 7.0 35        | 1763.14 | 5/2 <sup>+</sup> |                                                                                                          |
|                     | 7578.1              | 7.0 35        | 1219.39 | 1/2 <sup>+</sup> |                                                                                                          |
|                     | 8797.2              | 62 6          | 0       | 3/2 <sup>+</sup> |                                                                                                          |
| 8820.9              | 5817.6              | 9 5           | 3002.75 | 5/2              |                                                                                                          |
|                     | 6126.4              | 18 5          | 2693.9  | (3/2,5/2)        |                                                                                                          |
|                     | 7057.0              | 21 5          | 1763.14 | 5/2 <sup>+</sup> |                                                                                                          |
|                     | 8819.7              | 52 5          | 0       | 3/2 <sup>+</sup> |                                                                                                          |
| 8824.2              | 5820.9              | 37 9          | 3002.75 | 5/2              |                                                                                                          |
|                     | 6129.7              | 6.0 3         | 2693.9  | (3/2,5/2)        |                                                                                                          |
|                     | 7060.3              | 16 4          | 1763.14 | 5/2 <sup>+</sup> |                                                                                                          |
|                     | 7603.9              | 22 6          | 1219.39 | 1/2 <sup>+</sup> |                                                                                                          |
|                     | 8823.0              | 19 5          | 0       | 3/2 <sup>+</sup> |                                                                                                          |
| 8829.3              | 3174                | 10 5          | 5655    |                  |                                                                                                          |
|                     | 3990.0              | 1.3 7         | 4839.10 |                  |                                                                                                          |
|                     | 4059.2              | 1.5 8         | 4769.86 |                  |                                                                                                          |
|                     | 4910.5              | 4.5 23        | 3918.48 | 3/2 <sup>+</sup> |                                                                                                          |
|                     | 5826.0              | 22 6          | 3002.75 | 5/2              |                                                                                                          |
|                     | 6134.8              | 13 3          | 2693.9  | (3/2,5/2)        |                                                                                                          |
|                     | 6183.0              | 33 8          | 2645.70 | (5/2,7/2)        |                                                                                                          |
|                     | 7065.4              | 9 5           | 1763.14 | 5/2 <sup>+</sup> |                                                                                                          |
|                     | 7609.0              | 2.6 13        | 1219.39 | 1/2 <sup>+</sup> |                                                                                                          |
|                     | 8828.1              | 3.8 19        | 0       | 3/2 <sup>+</sup> |                                                                                                          |
| 8833.1              | 3617.1              | 6.9 35        | 5215.8  |                  |                                                                                                          |
|                     | 4063.0              | 2.6 13        | 4769.86 |                  |                                                                                                          |
|                     | 4659.3              | 14 4          | 4173.45 |                  |                                                                                                          |
|                     | 4719.1              | 4.7 24        | 4113.7  | 7/2              |                                                                                                          |
|                     | 4773.6              | 11 3          | 4059.2  |                  |                                                                                                          |
|                     | 4914.3              | 11 3          | 3918.48 | 3/2 <sup>+</sup> |                                                                                                          |
|                     | 5669.6              | 3.7 19        | 3162.99 | 7/2 <sup>-</sup> |                                                                                                          |
|                     | 5829.8              | 6.7 34        | 3002.75 | 5/2              |                                                                                                          |
|                     | 6138.6              | 4.5 23        | 2693.9  | (3/2,5/2)        |                                                                                                          |
|                     | 6186.8              | 23 6          | 2645.70 | (5/2,7/2)        |                                                                                                          |

Continued on next page (footnotes at end of table)



$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10** (continued) $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$              | $E_\gamma^\dagger$  | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Comments                                                                   |
|---------------------|------------------------|---------------------|---------------|---------|------------------|----------------------------------------------------------------------------|
| 8833.1              |                        | 7069.2              | 9 5           | 1763.14 | 5/2 <sup>+</sup> |                                                                            |
|                     |                        | 8831.9              | 2.9 15        | 0       | 3/2 <sup>+</sup> |                                                                            |
| 8838.1              | 7/2 <sup>-</sup>       | 3674.6 <sup>‡</sup> |               | 5163.34 | 7/2              |                                                                            |
|                     |                        | 3956.9 <sup>‡</sup> |               | 4880.96 |                  |                                                                            |
|                     |                        | 4068.0              | 8 4           | 4769.86 |                  |                                                                            |
|                     |                        | 4490.0 <sup>‡</sup> |               | 4347.76 | 5/2,9/2          |                                                                            |
|                     |                        | 4778.6 <sup>‡</sup> |               | 4059.2  |                  |                                                                            |
|                     |                        | 5674.6              | 29 7          | 3162.99 | 7/2 <sup>-</sup> |                                                                            |
|                     |                        | 7074.2              | 63 6          | 1763.14 | 5/2 <sup>+</sup> | $A_2=-0.20$ 2, $A_4=-0.00$ 2 ( <a href="#">1981Bi05</a> ).                 |
| 8856.1              |                        | 3974.9              | 3.4 17        | 4880.96 |                  |                                                                            |
|                     |                        | 4231.4              | 5.1 26        | 4624.4  |                  |                                                                            |
|                     |                        | 4677.9              | 3.5 18        | 4177.9  | 3/2              |                                                                            |
|                     |                        | 4937.3              | 11 3          | 3918.48 | 3/2 <sup>+</sup> |                                                                            |
|                     |                        | 5692.6              | 51 5          | 3162.99 | 7/2 <sup>-</sup> |                                                                            |
|                     |                        | 6161.6              | 2.8 14        | 2693.9  | (3/2,5/2)        |                                                                            |
|                     |                        | 6209.8              |               | 2645.70 | (5/2,7/2)        |                                                                            |
|                     |                        | 7092.2              | 7 4           | 1763.14 | 5/2 <sup>+</sup> |                                                                            |
|                     |                        | 8854.9              | 16 4          | 0       | 3/2 <sup>+</sup> |                                                                            |
| 8868.6              |                        | 4690.4              | 2.6 13        | 4177.9  | 3/2              |                                                                            |
|                     |                        | 5865.3              | 16 4          | 3002.75 | 5/2              |                                                                            |
|                     |                        | 6174.1              | 12 3          | 2693.9  | (3/2,5/2)        |                                                                            |
|                     |                        | 7104.7              | 4.4 22        | 1763.14 | 5/2 <sup>+</sup> |                                                                            |
|                     |                        | 7648.3              | 16 4          | 1219.39 | 1/2 <sup>+</sup> |                                                                            |
|                     |                        | 8867.4              | 49 12         | 0       | 3/2 <sup>+</sup> |                                                                            |
| 8886.1              |                        | 3231                | 6 3           | 5655    |                  |                                                                            |
|                     |                        | 3286.2              | 6.6 33        | 5599.7  |                  |                                                                            |
|                     |                        | 3670.1              | 2.2 11        | 5215.8  |                  |                                                                            |
|                     |                        | 3722.6              | 9 4           | 5163.34 | 7/2              |                                                                            |
|                     |                        | 4116.0              | 4.5 23        | 4769.86 |                  |                                                                            |
|                     |                        | 4707.9              | 8 4           | 4177.9  | 3/2              |                                                                            |
|                     |                        | 4967.2              | 4.1 21        | 3918.48 | 3/2 <sup>+</sup> |                                                                            |
|                     |                        | 5882.8              | 8 4           | 3002.75 | 5/2              |                                                                            |
|                     |                        | 6239.8              | 26 7          | 2645.70 | (5/2,7/2)        |                                                                            |
|                     |                        | 7122.2              | 12 3          | 1763.14 | 5/2 <sup>+</sup> |                                                                            |
|                     |                        | 8884.9              | 14 4          | 0       | 3/2 <sup>+</sup> |                                                                            |
| 8893.3              | (5/2,7/2) <sup>+</sup> | 3293.4 <sup>C</sup> |               | 5599.7  |                  | $I_\gamma$ : 1 from <a href="#">2001Vo24</a> only; not observed by others. |
|                     |                        | 3307                | 1.6 8         | 5586    |                  |                                                                            |
|                     |                        | 4779.3              | 9 4           | 4113.7  | 7/2              | $I_\gamma$ : other: 9 <a href="#">2001Vo24</a> .                           |
|                     |                        | 4950.0              | 3.9 20        | 3942.9  |                  | $I_\gamma$ : other: 4 <a href="#">2001Vo24</a> .                           |
|                     |                        | 5729.8              | 29 7          | 3162.99 | 7/2 <sup>-</sup> | $I_\gamma$ : other: 29 <a href="#">2001Vo24</a> .                          |
|                     |                        | 6198.8              | 37 9          | 2693.9  | (3/2,5/2)        | $I_\gamma$ : other: 37 <a href="#">2001Vo24</a> .                          |
|                     |                        | 7129.4              | 19 5          | 1763.14 | 5/2 <sup>+</sup> | $I_\gamma$ : other: 19 <a href="#">2001Vo24</a> .                          |
|                     |                        | 8892.1              | 1.0 5         | 0       | 3/2 <sup>+</sup> | $I_\gamma$ : other: 1 <a href="#">2001Vo24</a> .                           |
| 8904.8              |                        | 2724                | 4.3 22        | 6181    |                  |                                                                            |
|                     |                        | 3250                | 3.4 17        | 5655    |                  |                                                                            |
|                     |                        | 3894.5              | 3.9 20        | 5010.1  |                  |                                                                            |
|                     |                        | 4050.2              | 6.2 31        | 4854.4  |                  |                                                                            |
|                     |                        | 4065.5              | 4.5 23        | 4839.10 |                  |                                                                            |
|                     |                        | 4726.6              | 18 5          | 4177.9  | 3/2              |                                                                            |
|                     |                        | 4845.2              | 5.7 29        | 4059.2  |                  |                                                                            |
|                     |                        | 6210.3              | 15 4          | 2693.9  | (3/2,5/2)        |                                                                            |
|                     |                        | 7684.5              | 10 3          | 1219.39 | 1/2 <sup>+</sup> |                                                                            |
|                     |                        | 8903.6              | 29 7          | 0       | 3/2 <sup>+</sup> |                                                                            |
| 8906.8              | 5/2 <sup>+</sup>       | 3252                |               | 5655    |                  | $I_\gamma$ : 4 from <a href="#">2001Vo24</a> only; not observed by others. |
|                     |                        | 3261                | 3.7 19        | 5646    |                  | $I_\gamma$ : other: 2 <a href="#">2001Vo24</a> .                           |

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$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$  | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                              |
|---------------------|------------------|---------------------|---------------|---------|------------------|--------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 8906.8              | 5/2 <sup>+</sup> | 3306.9              | 2.2 11        | 5599.7  |                  |                    |            |                                                                                                                                       |
|                     |                  | 3321                | 4             | 5586    |                  |                    |            | $I_\gamma$ : other: 4 2001Vo24.                                                                                                       |
|                     |                  | 3387                | 2.5 13        | 5520    |                  |                    |            |                                                                                                                                       |
|                     |                  | 3690.8              | 4.4 22        | 5215.8  |                  |                    |            |                                                                                                                                       |
|                     |                  | 4136.7              | 5.6 28        | 4769.86 |                  |                    |            | $I_\gamma$ : other: 6 2001Vo24.                                                                                                       |
|                     |                  | 4963.5              | 13 3          | 3942.9  |                  |                    |            | $I_\gamma$ : other: 15 2001Vo24.                                                                                                      |
|                     |                  | 7142.9              | 69 7          | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 69 2001Vo24.                                                                                                      |
| 8919.8              |                  | 3274                | 5.6 28        | 5646    |                  |                    |            |                                                                                                                                       |
|                     |                  | 4571.7              | 8 4           | 4347.76 | 5/2,9/2          |                    |            |                                                                                                                                       |
|                     |                  | 5756.3              | 57 6          | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                       |
|                     |                  | 6273.5              | 26 7          | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                                       |
| 8953.0              | 3/2              | 7155.9              | 4 2           | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                                                                       |
|                     |                  | 7732.7              |               | 1219.39 | 1/2 <sup>+</sup> | D+Q                |            | Mult.: from $\gamma(\theta)$ in 1976Sp08; $\delta=-0.34$ 2 or $\delta=-5.0$ 4 (1976Sp09).                                             |
|                     |                  | 8951.8              |               | 0       | 3/2 <sup>+</sup> | D+Q                |            | $A_2=+0.12$ 3, $A_4=-0.02$ 4 (1976Sp09).<br>Mult.: from $\gamma(\theta)$ in 1976Sp08; $\delta=-0.39$ 5 or $\delta=-8.2$ 7 (1976Sp09). |
| 8984.2              |                  | 3260.5              | 0.8 4         | 5723.52 |                  |                    |            | $A_2=-0.19$ 2, $A_4=+0.02$ 2 (1976Sp09).                                                                                              |
|                     |                  | 4144.8              | 1.2 6         | 4839.10 |                  |                    |            |                                                                                                                                       |
|                     |                  | 4214.1              | 0.5 3         | 4769.86 |                  |                    |            |                                                                                                                                       |
|                     |                  | 4359.5              | 0.5 3         | 4624.4  |                  |                    |            |                                                                                                                                       |
|                     |                  | 4806.0              | 1.5 8         | 4177.9  | 3/2              |                    |            |                                                                                                                                       |
|                     |                  | 4810.4              | 1.5 8         | 4173.45 |                  |                    |            |                                                                                                                                       |
|                     |                  | 5065.3              | 9 5           | 3918.48 | 3/2 <sup>+</sup> |                    |            |                                                                                                                                       |
|                     |                  | 5820.7              | 6 3           | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                       |
|                     |                  | 6289.7              | 1.8 9         | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                                       |
|                     |                  | 6337.9              | 1.2 6         | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                                       |
|                     |                  | 7220.3              | 16 4          | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                                                                       |
|                     |                  | 8983.0              | 60 6          | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                                                                       |
| 9081.4              | 5/2 <sup>+</sup> | 3357.7              | 1             | 5723.52 |                  |                    |            | $I_\gamma$ : from 2001Vo24.                                                                                                           |
|                     |                  | 4200.2              | 1             | 4880.96 |                  |                    |            | $I_\gamma$ : from 2001Vo24.                                                                                                           |
|                     |                  | 4242.0 <sup>‡</sup> |               | 4839.10 |                  |                    |            |                                                                                                                                       |
|                     |                  | 4311.3 <sup>‡</sup> |               | 4769.86 |                  |                    |            |                                                                                                                                       |
|                     |                  | 4456.7 <sup>‡</sup> |               | 4624.4  |                  |                    |            |                                                                                                                                       |
|                     |                  | 4903.1              | 2             | 4177.9  | 3/2              |                    |            | $I_\gamma$ : from 2001Vo24.                                                                                                           |
|                     |                  | 4907.6              | 2             | 4173.45 |                  |                    |            | $I_\gamma$ : from 2001Vo24.                                                                                                           |
|                     |                  | 5162.5              | 9             | 3918.48 | 3/2 <sup>+</sup> | D(+Q)              | -0.002 9   | $I_\gamma$ : from 2001Vo24.<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1976Sp09.<br>$A_2=-0.43$ 3, $A_4=+0.05$ 3 (1976Sp09).       |
|                     |                  | 5917.9              | 6             | 3162.99 | 7/2 <sup>-</sup> |                    |            | $I_\gamma$ : from 2001Vo24.                                                                                                           |
|                     |                  | 6386.9              | 2             | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : from 2001Vo24.                                                                                                           |
|                     |                  | 6435.1              | 1             | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : from 2001Vo24.                                                                                                           |
|                     |                  | 7317.4              | 16            | 1763.14 | 5/2 <sup>+</sup> | D+Q                | +0.11 2    | $I_\gamma$ : from 2001Vo24.<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1976Sp09.<br>$A_2=+0.555$ 14, $A_4=+0.03$ 2 (1976Sp09).     |
|                     |                  | 9080.1              | 60            | 0       | 3/2 <sup>+</sup> | D+Q                | +0.089 9   | $I_\gamma$ : from 2001Vo24.<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1976Sp09.<br>$A_2=-0.243$ 11, $A_4=+0.05$ 2 (1976Sp09).     |
| 9157.1              | 5/2              | 3941.1              | 7.4 19        | 5215.8  |                  |                    |            |                                                                                                                                       |
|                     |                  | 4386.9              | 6.9 17        | 4769.86 |                  |                    |            |                                                                                                                                       |
|                     |                  | 4978.8              | <16           | 4177.9  | 3/2              |                    |            | $I_\gamma$ : total $I_\gamma=13$ 3 for $\gamma$ decays to 4173 and 4178 doublet final levels (1976Sp08).                              |
|                     |                  | 4983.3              | <16           | 4173.45 |                  |                    |            | $I_\gamma$ : total $I_\gamma=13$ 3 for $\gamma$ decays to 4173 and 4178 doublet final levels (1976Sp08).                              |

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$^{34}\text{S}(\text{p},\gamma)$  [1976Me12](#),[1976Sp08](#),[1972Hu10](#) (continued) $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                                                                                                                                              |
|---------------------|--------------------|---------------|---------|------------------|--------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9157.1              | 5993.6             | 9.6 24        | 3162.99 | 7/2 <sup>-</sup> | D+Q                | +0.10 2    | Mult., $\delta$ : from $\gamma(\theta)$ in <a href="#">2001Vo24</a> .<br>A <sub>2</sub> =-0.15 8, A <sub>4</sub> =-0.31 9, A <sub>6</sub> =-0.05 8 ( <a href="#">1996Ka21</a> ).                                                                      |
|                     | 6153.8             | 8.4 21        | 3002.75 | 5/2              |                    |            |                                                                                                                                                                                                                                                       |
|                     | 6462.6             | 1.3 7         | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                                                                                                                                                       |
|                     | 6510.8             | 2.9 15        | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                                                                                                                                                       |
|                     | 7393.1             | 7.5 19        | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                       |
|                     | 9155.8             | 43 4          | 0       | 3/2 <sup>+</sup> | D+Q                | +0.09 1    | Mult., $\delta$ : from $\gamma(\theta)$ in <a href="#">2001Vo24</a> . Other: $\delta(\text{Q/D})=0.47$<br>50 ( <a href="#">1996Ka21</a> ).<br>A <sub>2</sub> =-0.29 6, A <sub>4</sub> =-0.38 8, A <sub>6</sub> =-0.10 6 ( <a href="#">1996Ka21</a> ). |

<sup>†</sup> From level-energy differences, unless otherwise noted.

<sup>‡</sup>  $\gamma$  branches observed in [1981Bi05](#). Branching ratios are not given.

<sup>#</sup> For levels below 8.27 MeV ( $E_p < 1.95$  MeV),  $I_\gamma$  is mainly from [1976Me12](#) and for levels above 8.27 MeV ( $E_p > 1.95$  MeV),  $I_\gamma$  is mainly from [1976Sp08](#), unless otherwise noted.

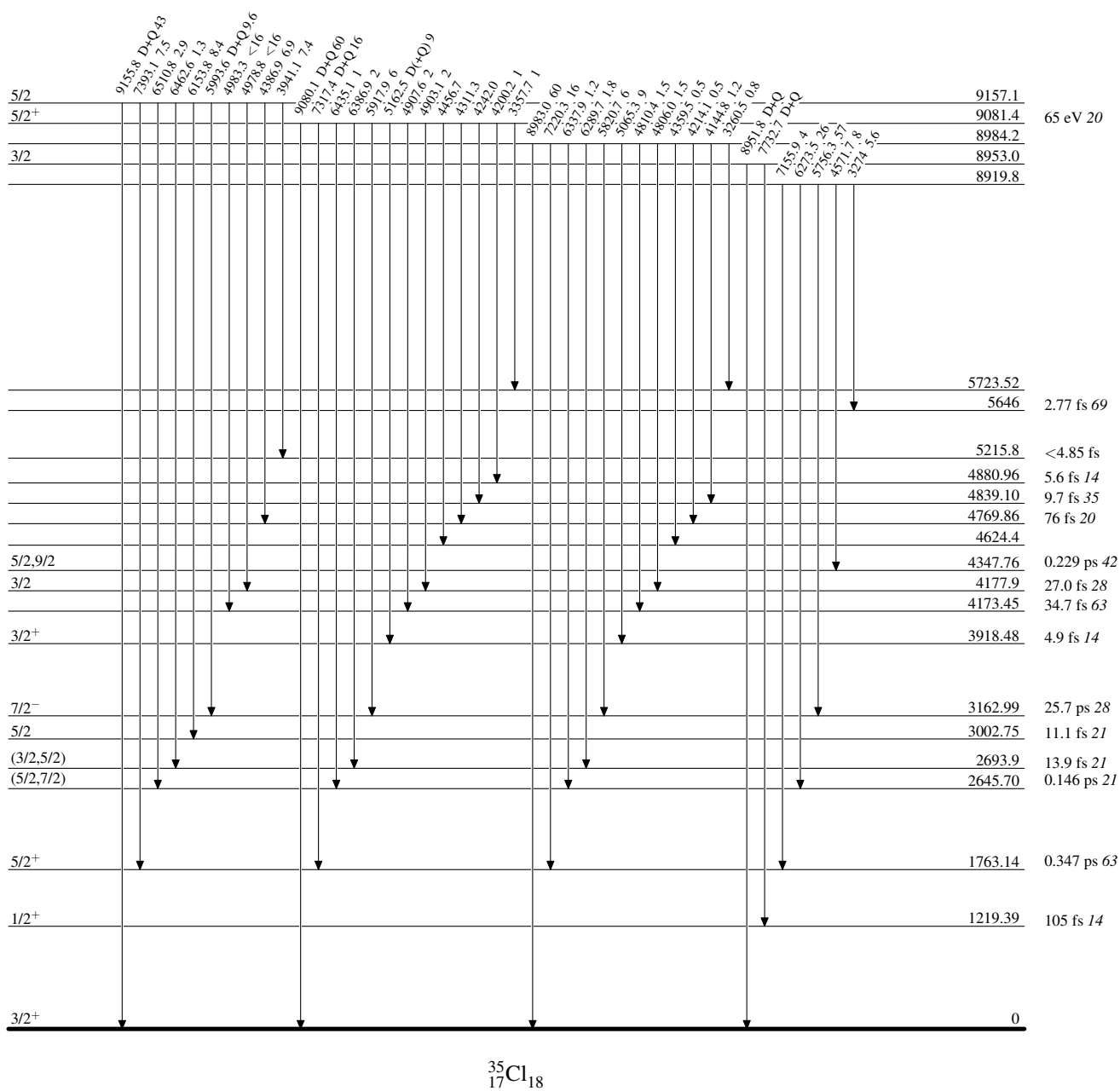
@ From [1970Ko33](#).

& From [1967Da12](#).

<sup>a</sup> From [1963Ha32](#).

<sup>b</sup> From  $\gamma(\theta)$  and/or  $\gamma\gamma(\theta)$  data as given under comments.

<sup>c</sup> Placement of transition in the level scheme is uncertain.

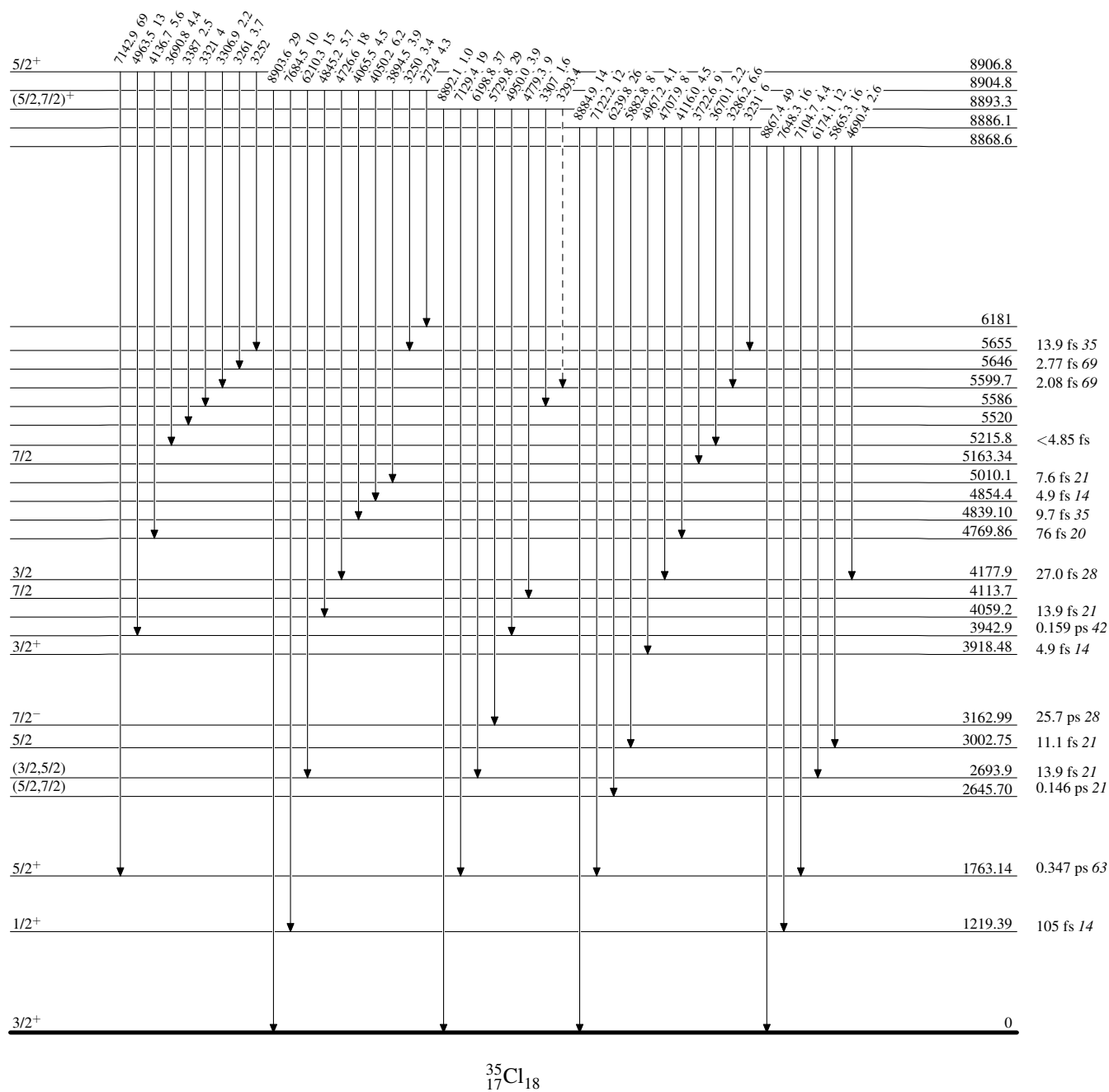


$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

Legend

## Level Scheme (continued)

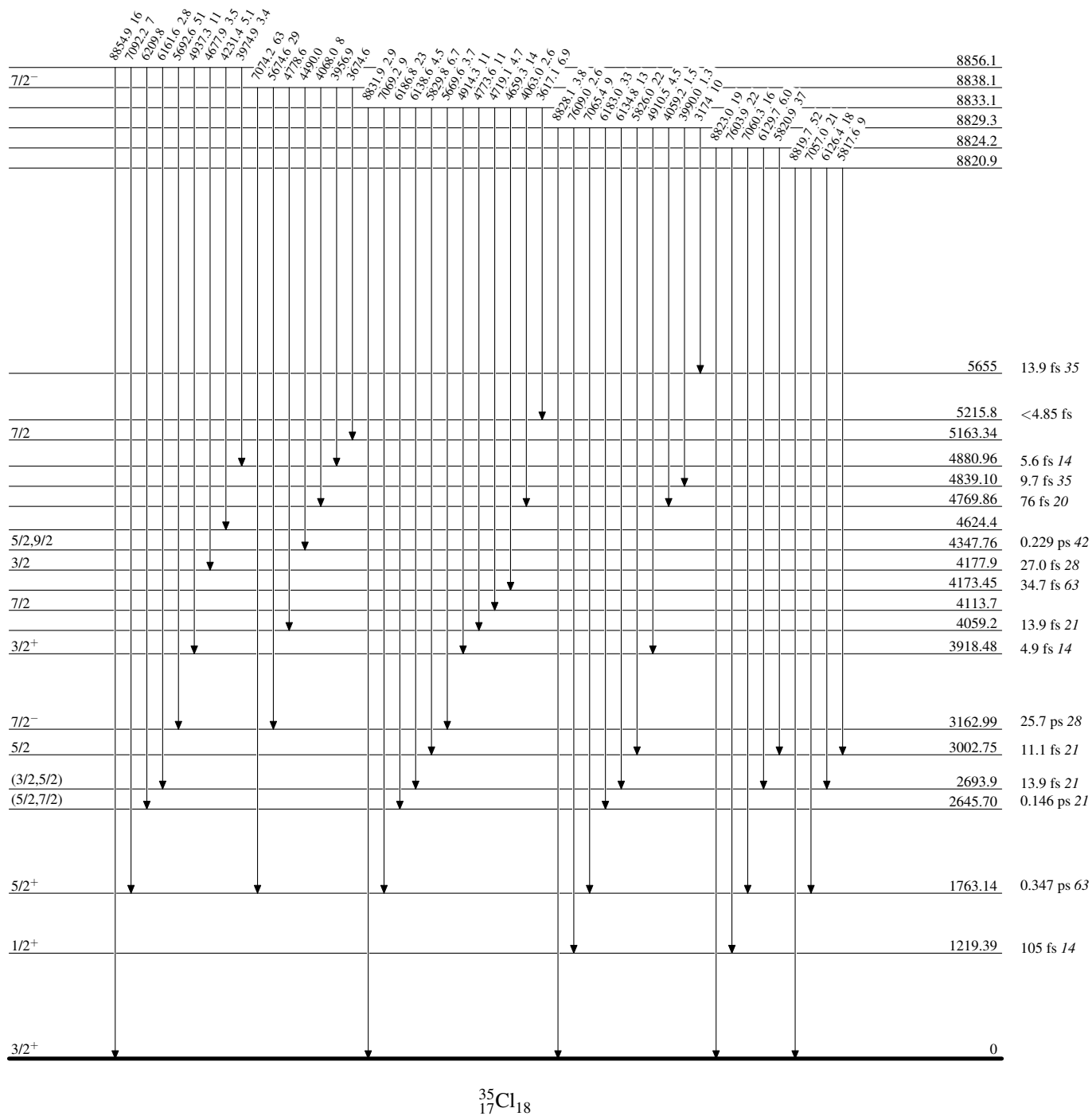
Intensities: % photon branching from each level

-----►  $\gamma$  Decay (Uncertain)

$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

Intensities: % photon branching from each level

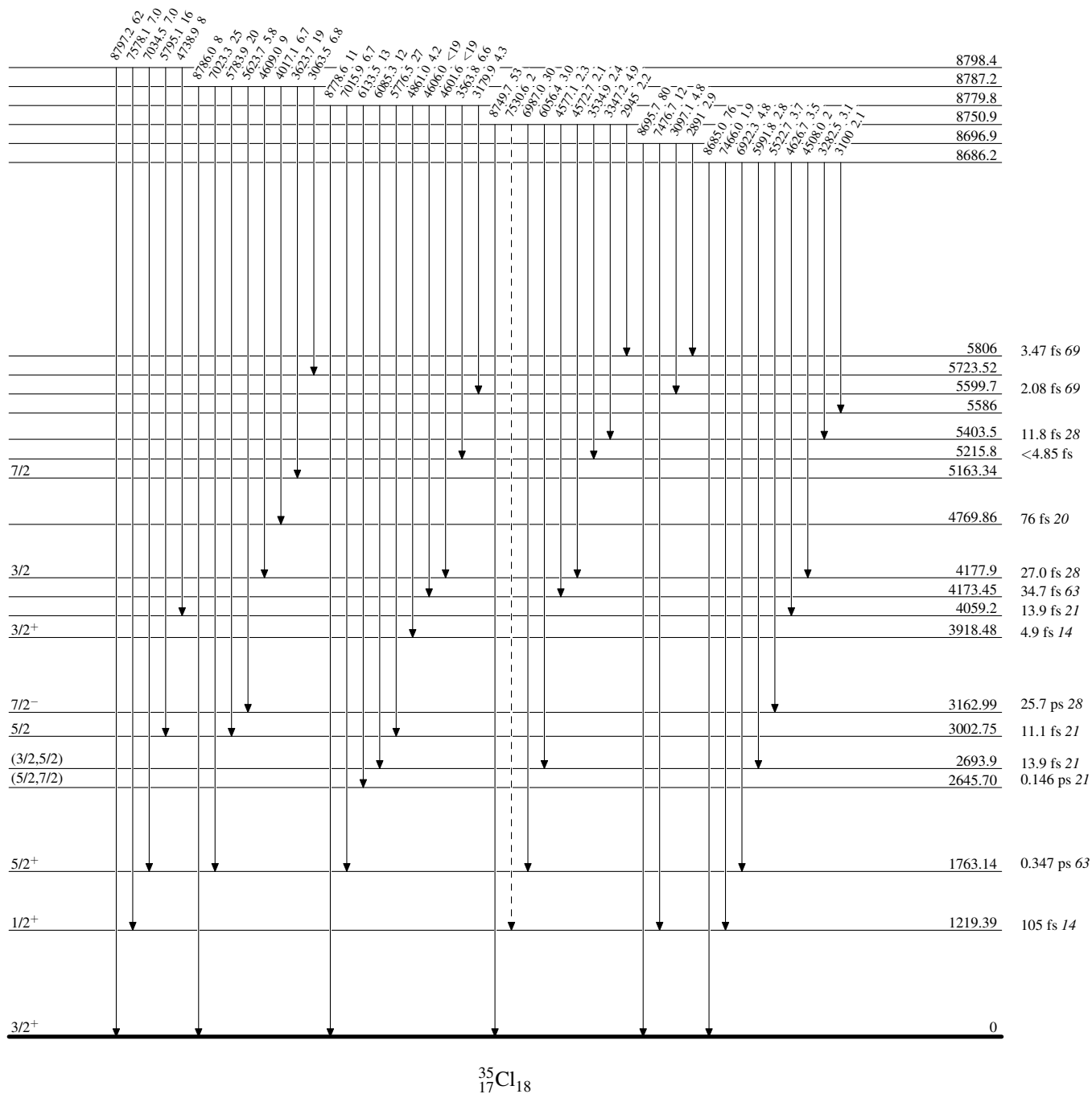


$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

Legend

## Level Scheme (continued)

Intensities: % photon branching from each level

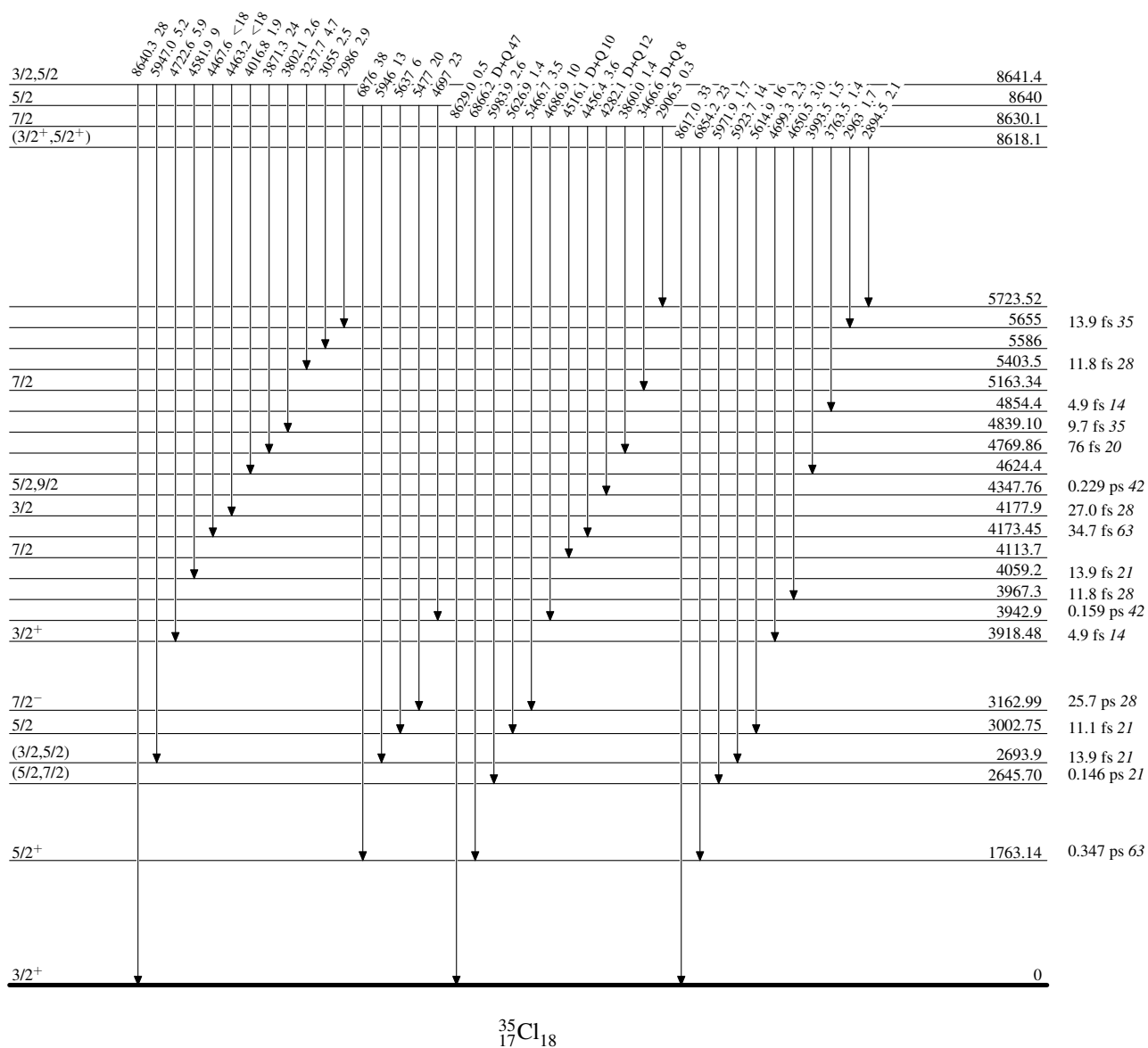
-----►  $\gamma$  Decay (Uncertain)



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

Intensities: % photon branching from each level

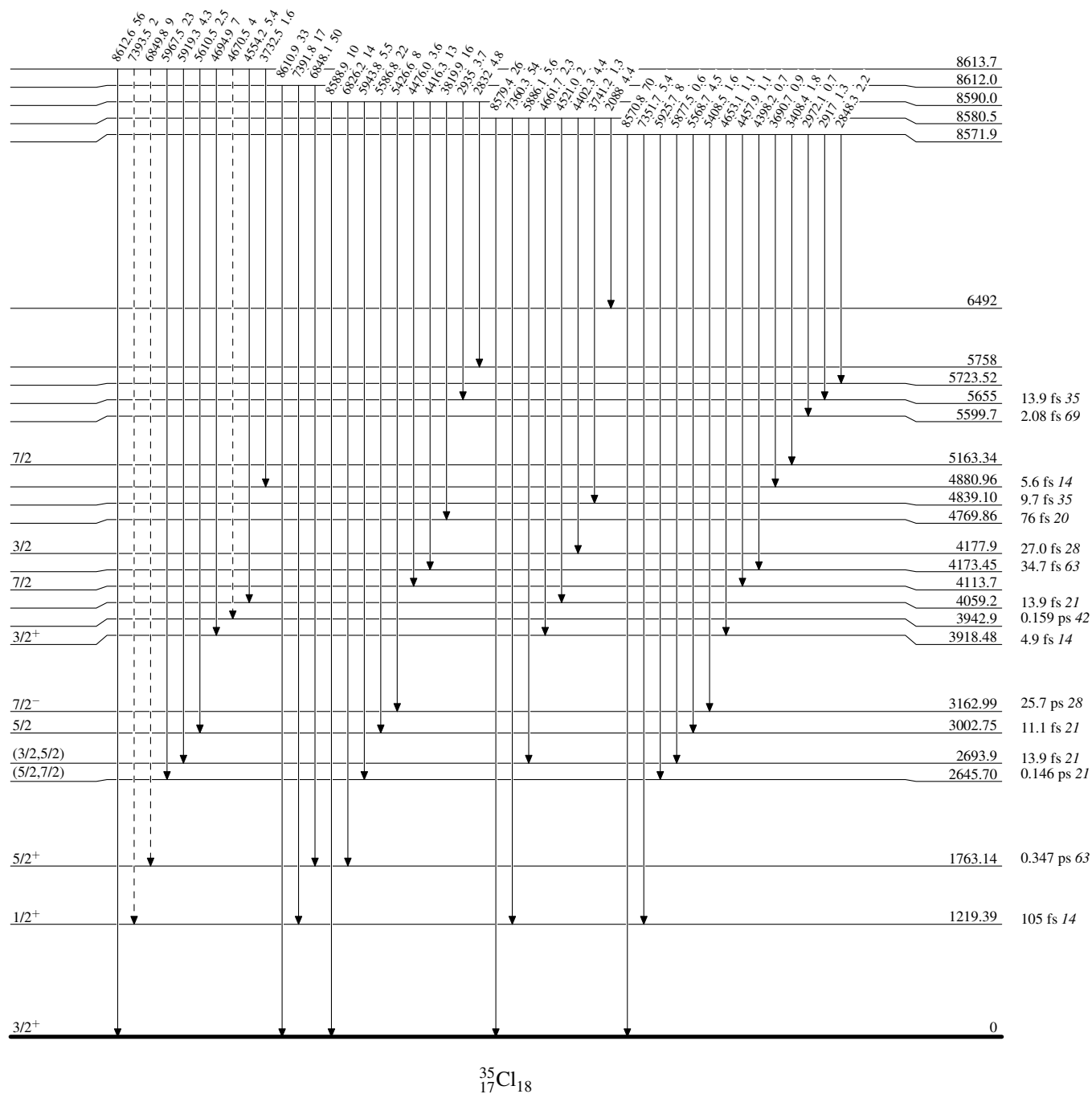
 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

Legend

## Level Scheme (continued)

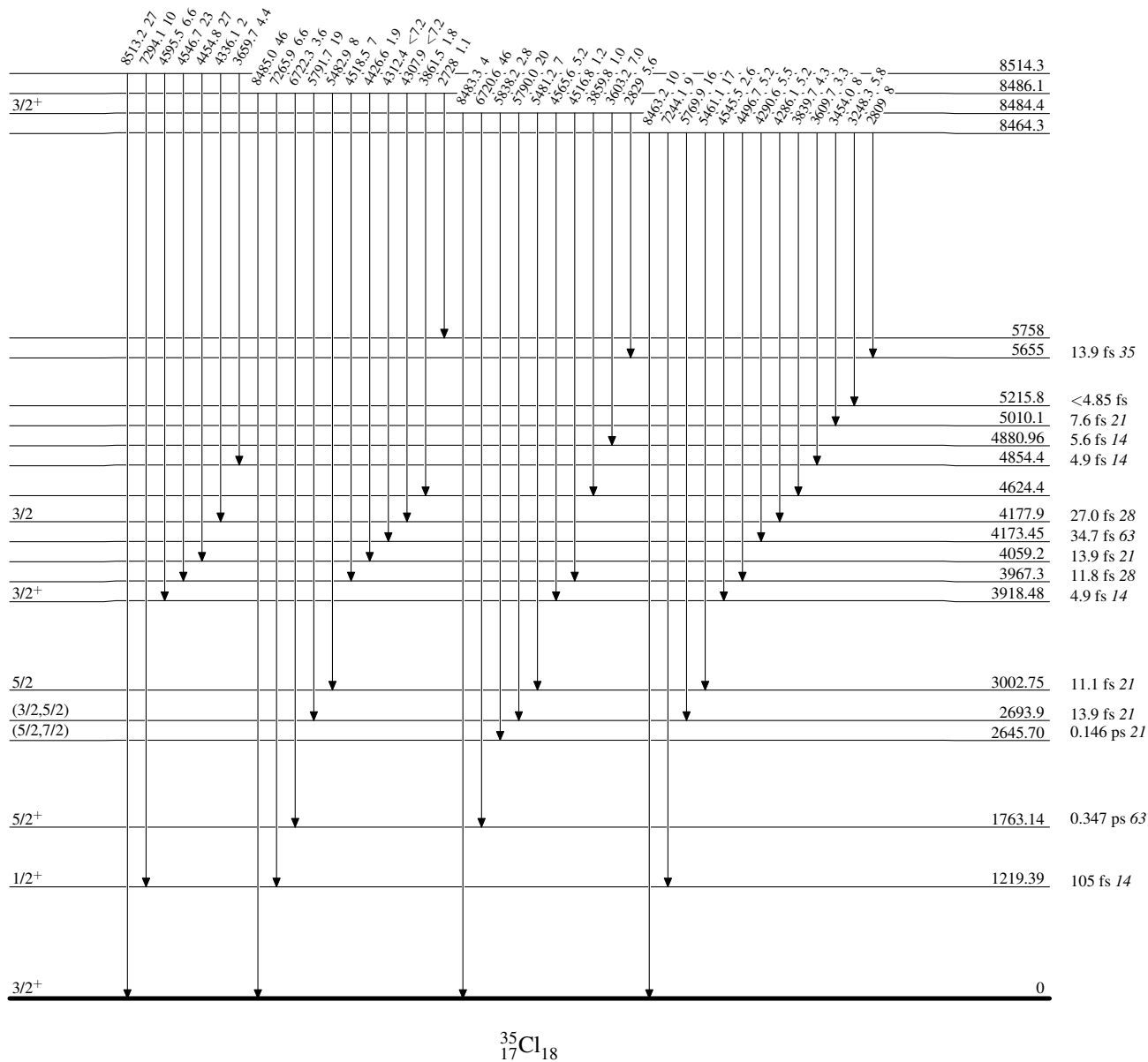
Intensities: % photon branching from each level

-----►  $\gamma$  Decay (Uncertain)

$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

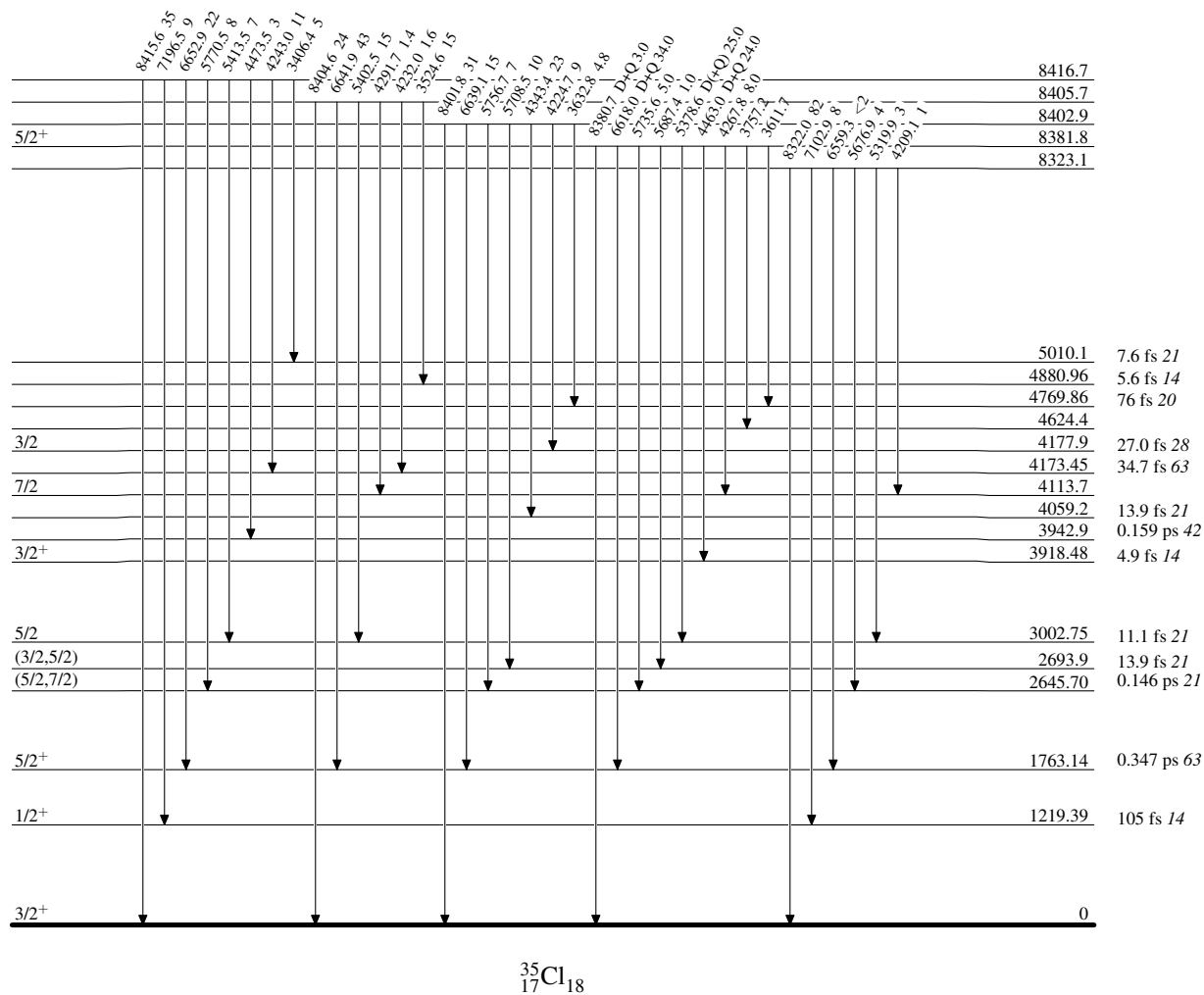
Intensities: % photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

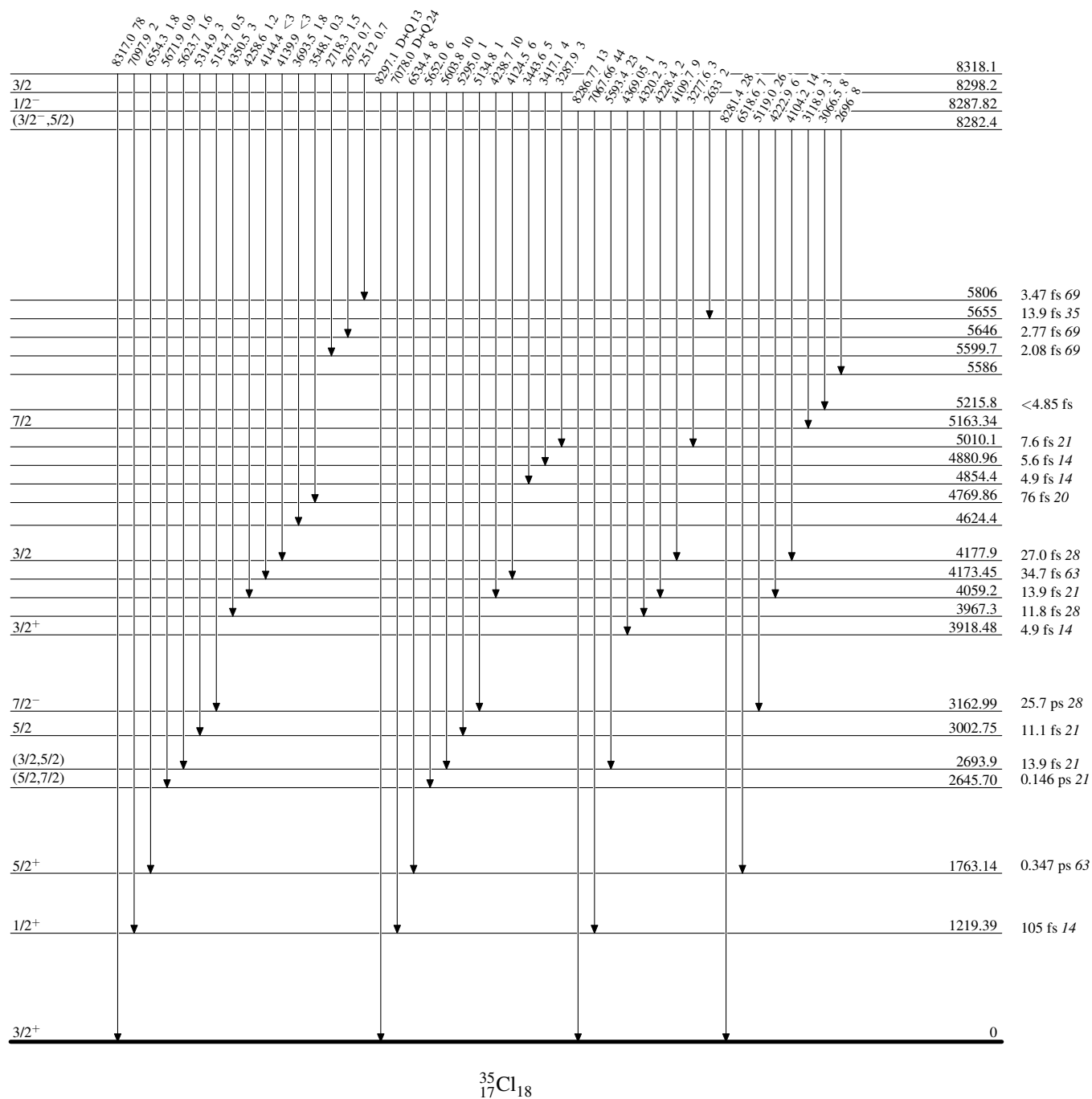
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

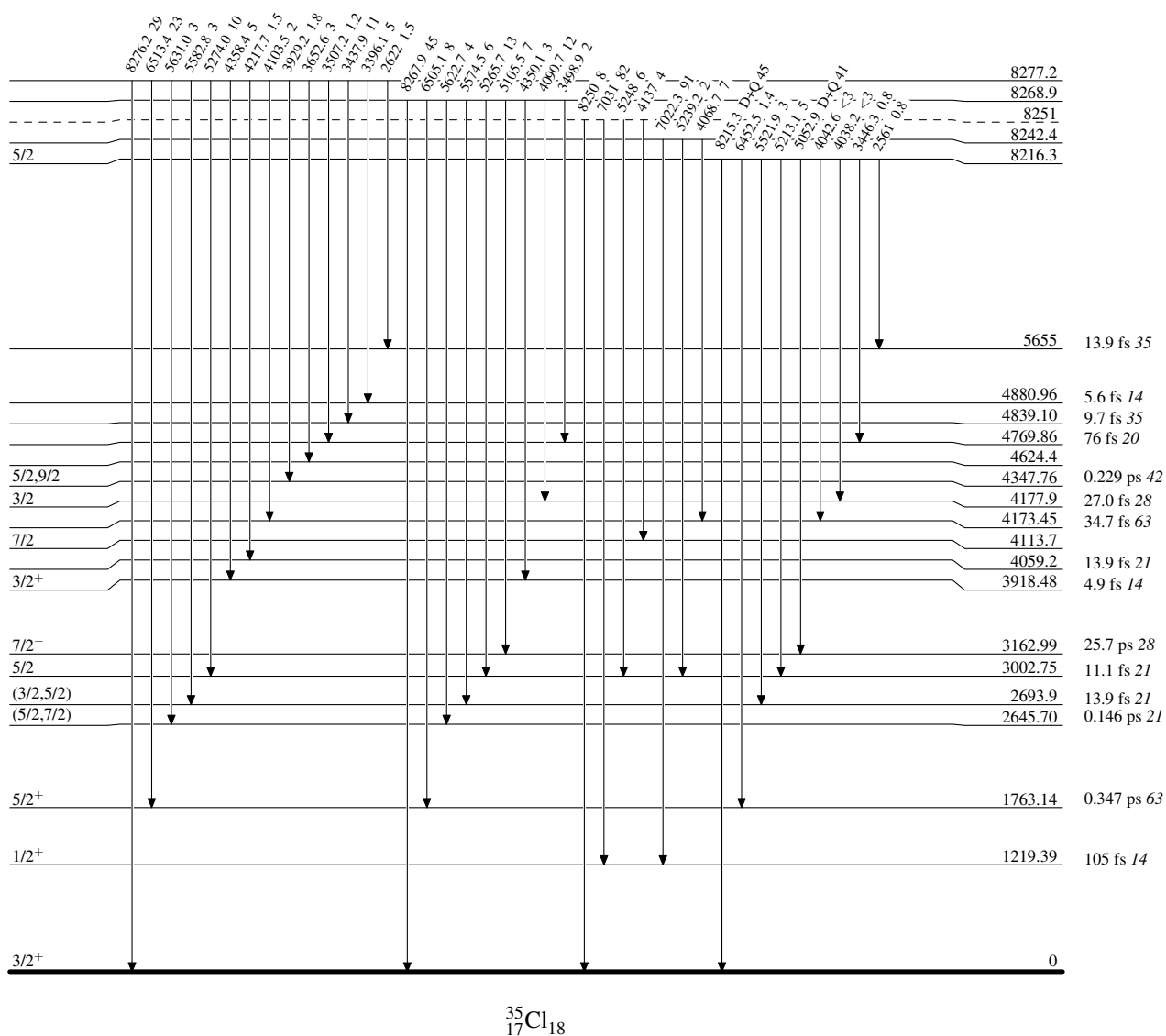
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

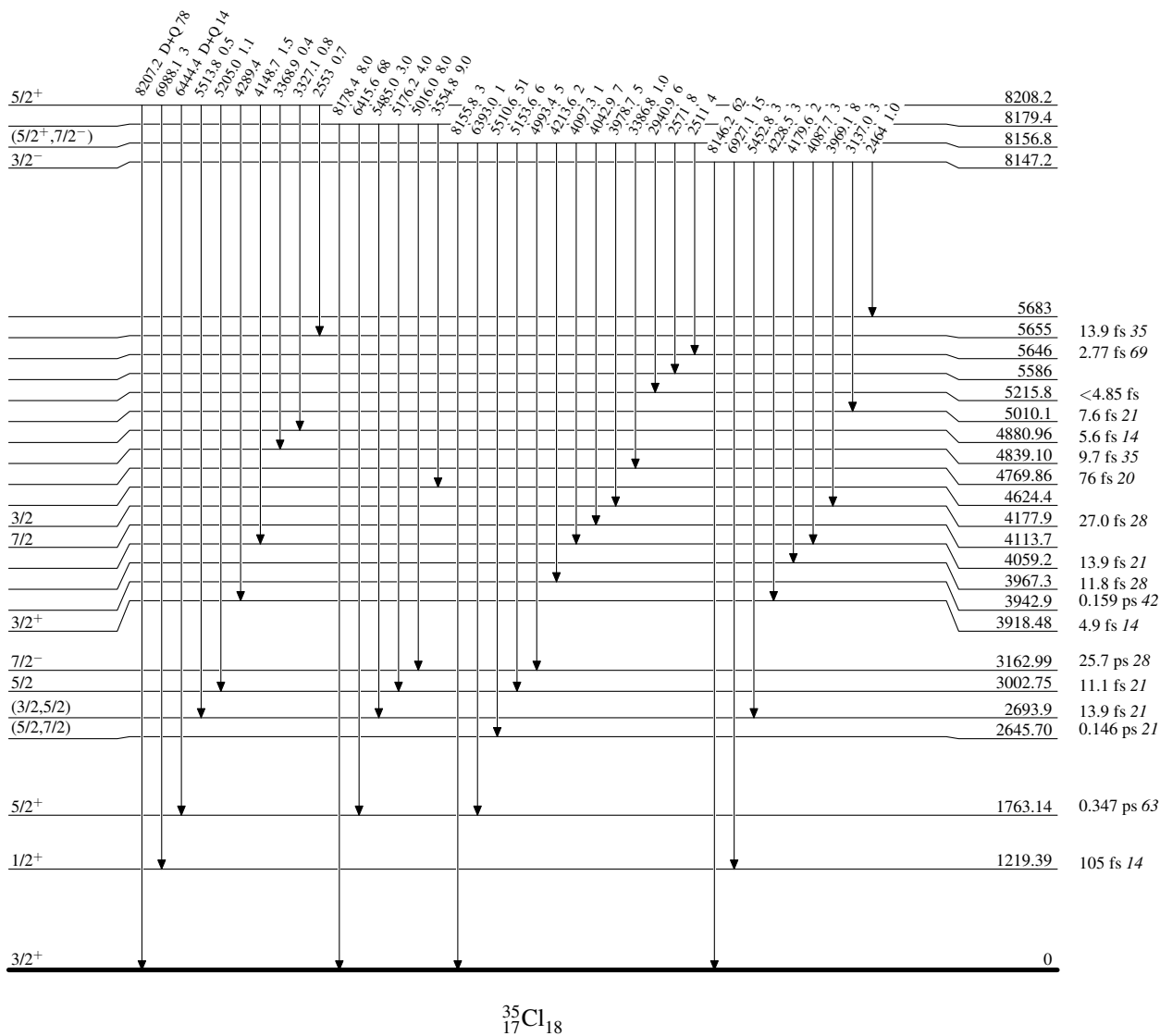
Intensities: % photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

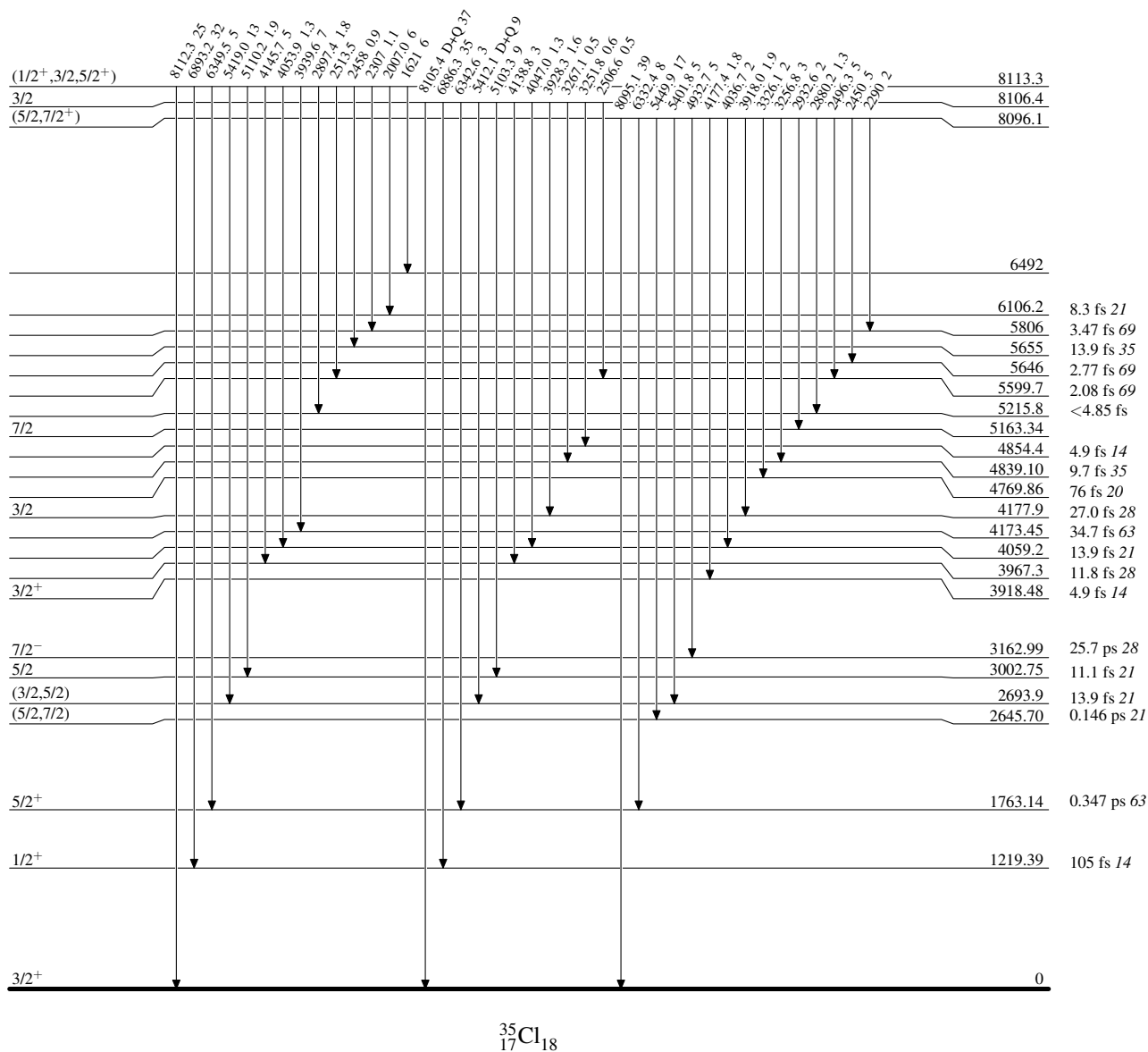
Intensities: % photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

Intensities: % photon branching from each level

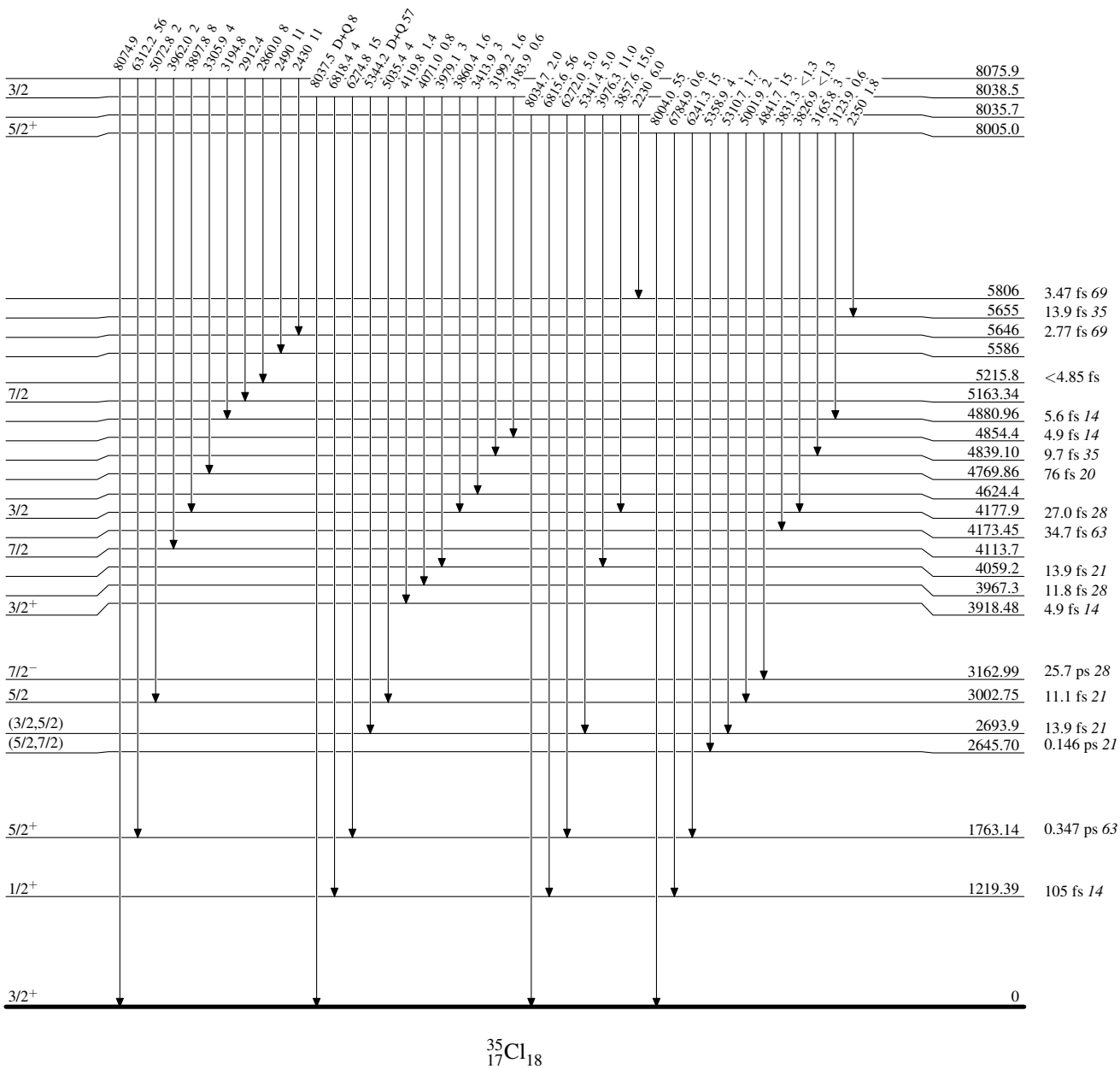
 $^{35}_{17}\text{Cl}_{18}$



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

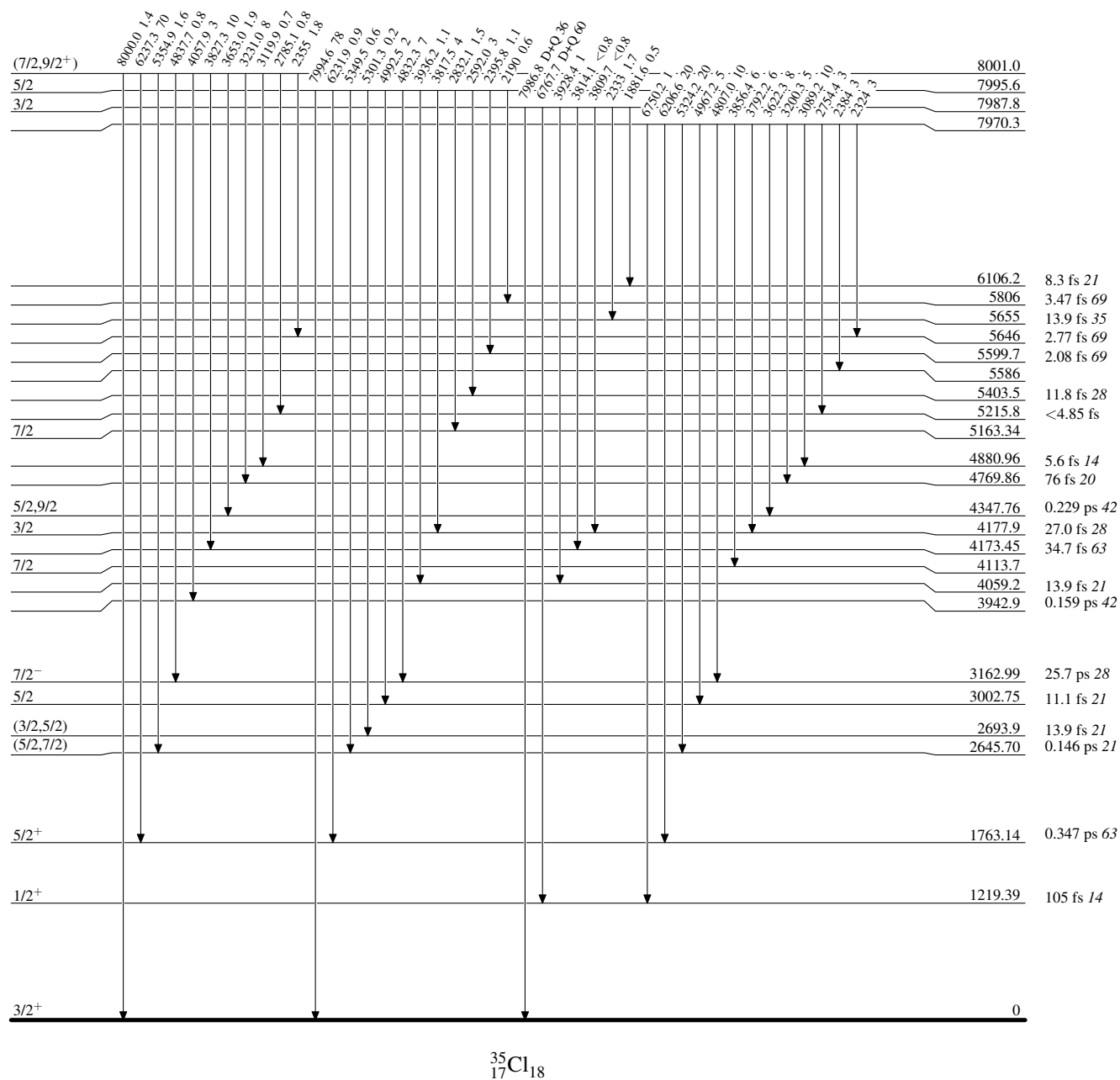
Intensities: % photon branching from each level

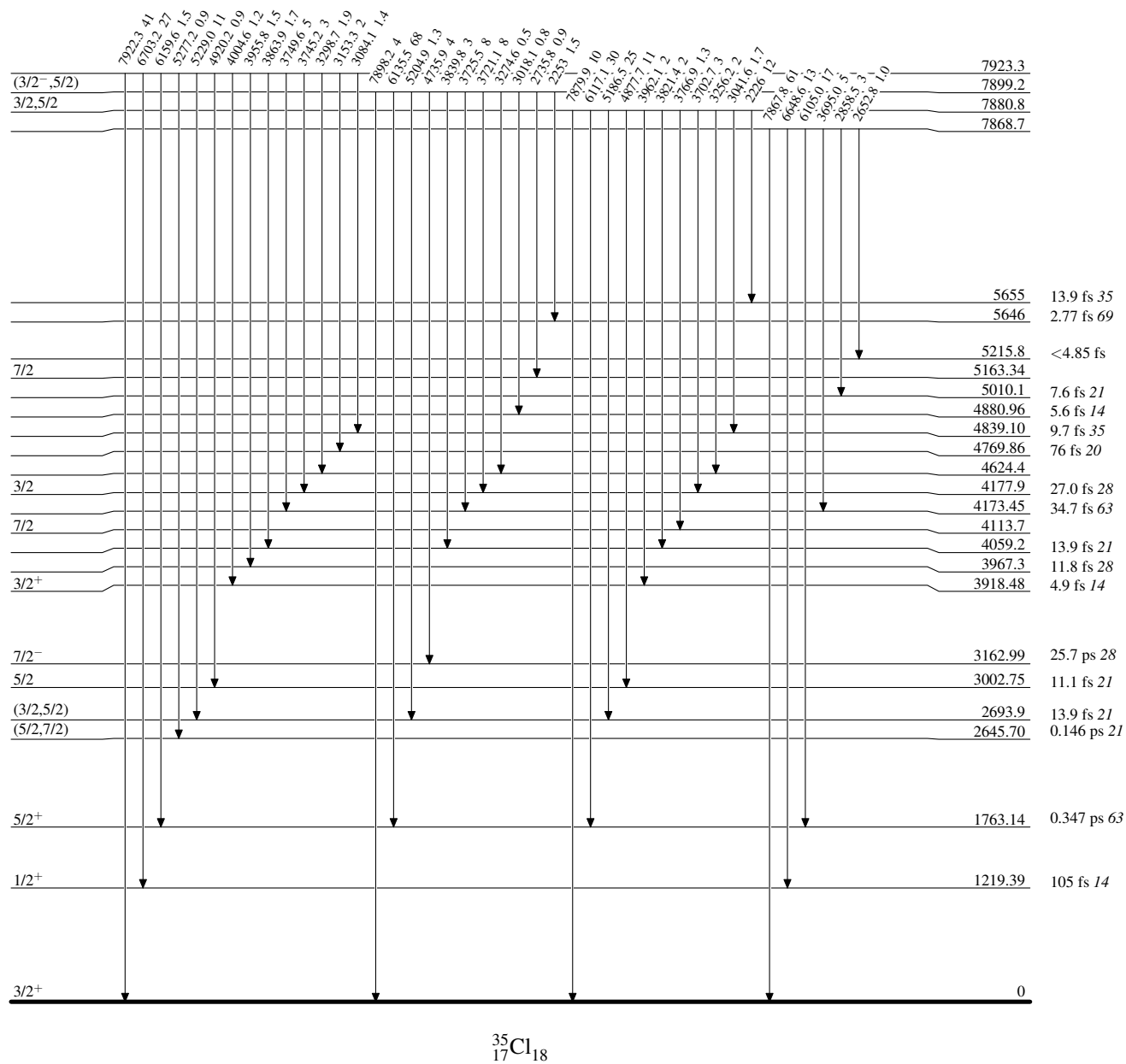


$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

Intensities: % photon branching from each level

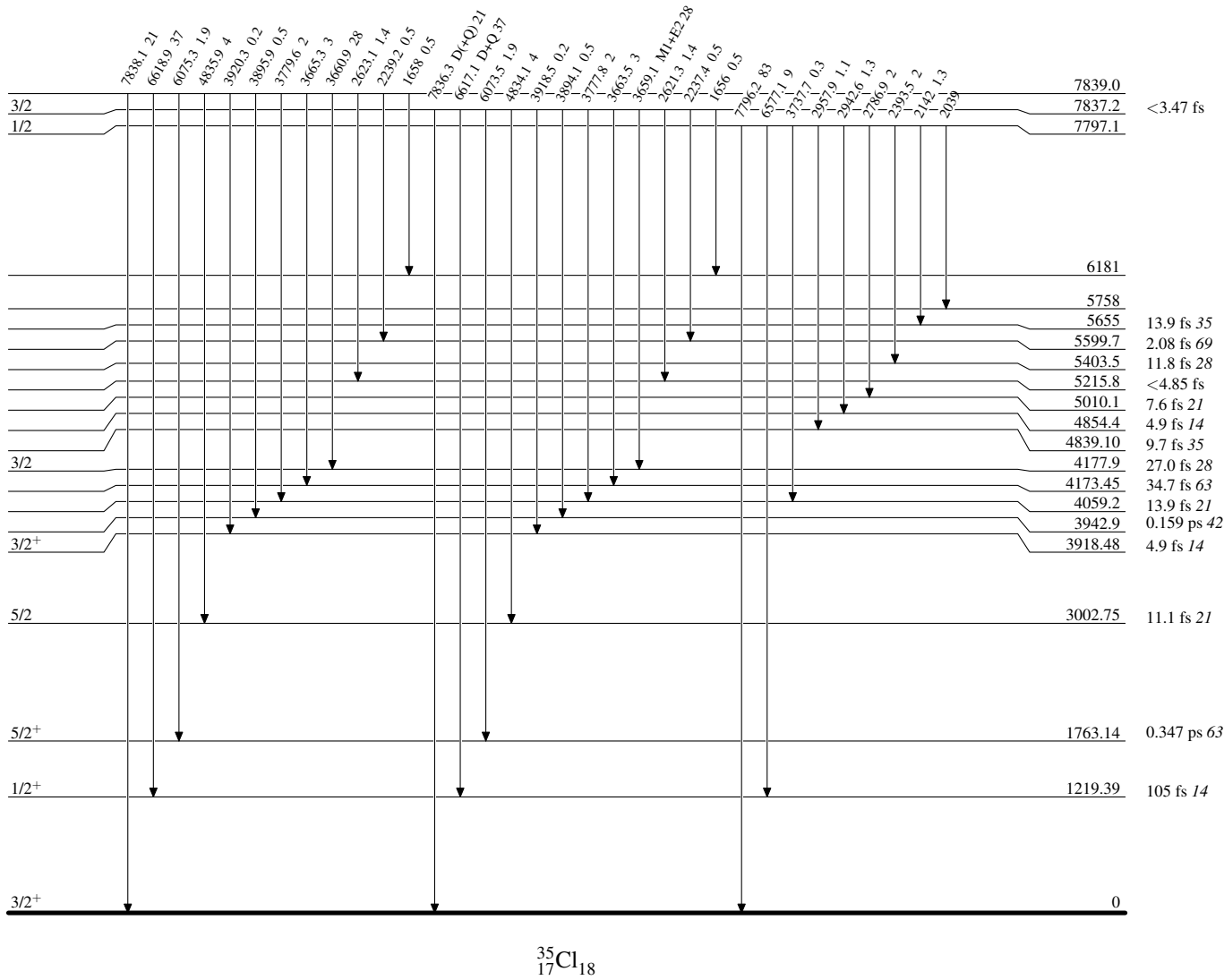




<sup>34</sup>S(p,γ) 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

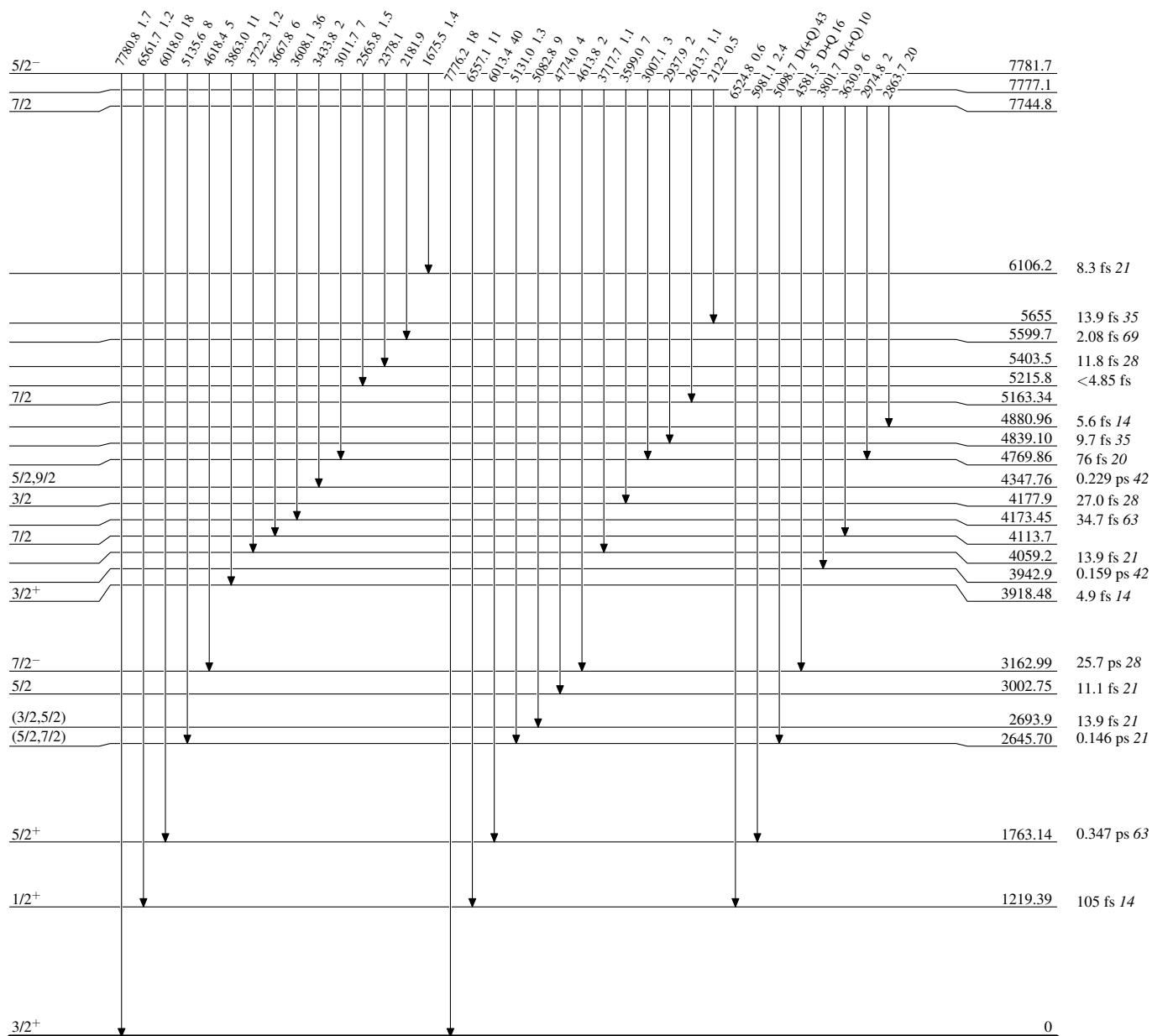


<sup>35</sup><sub>17</sub>Cl<sub>18</sub>

$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

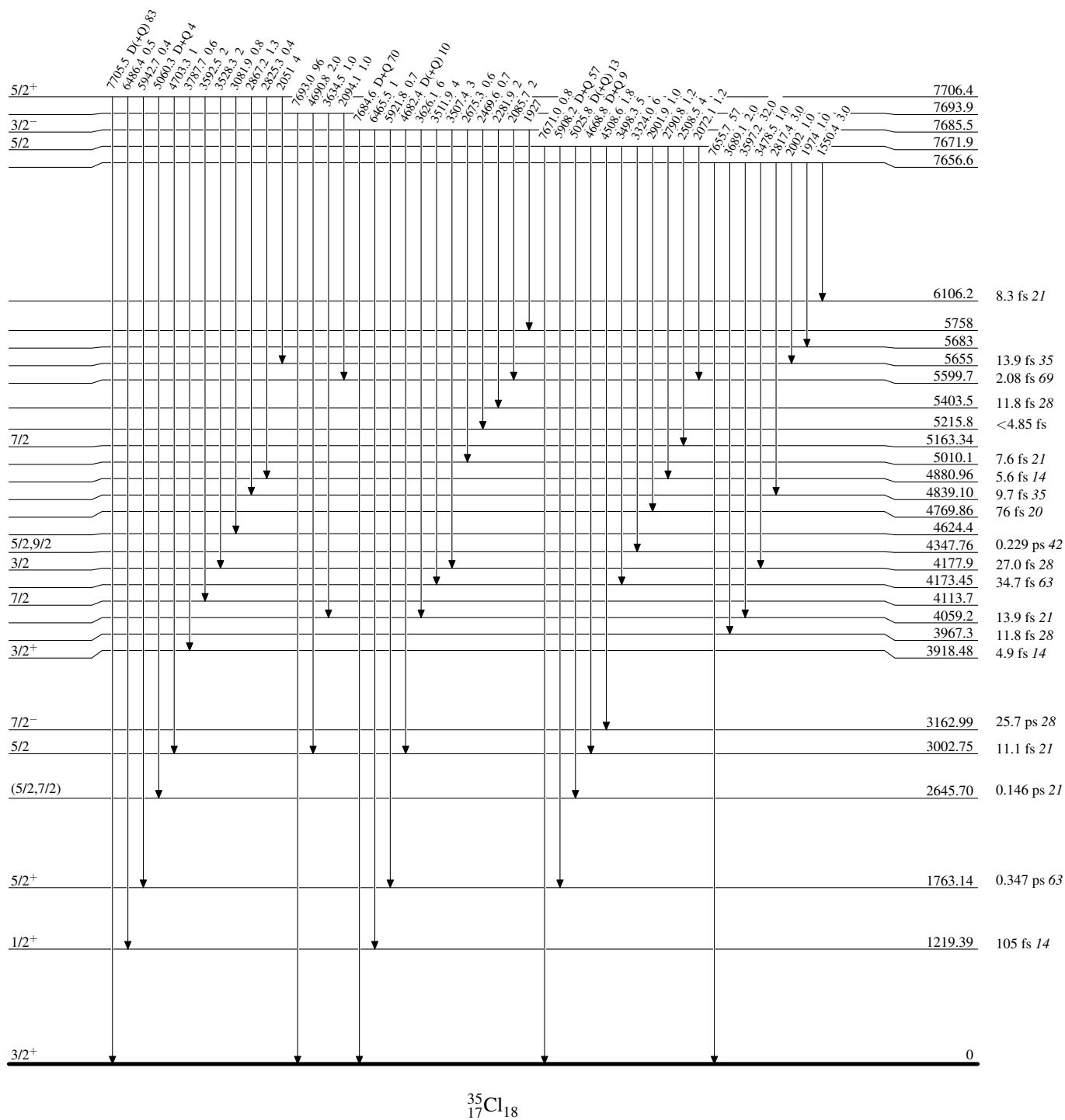
Intensities: % photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

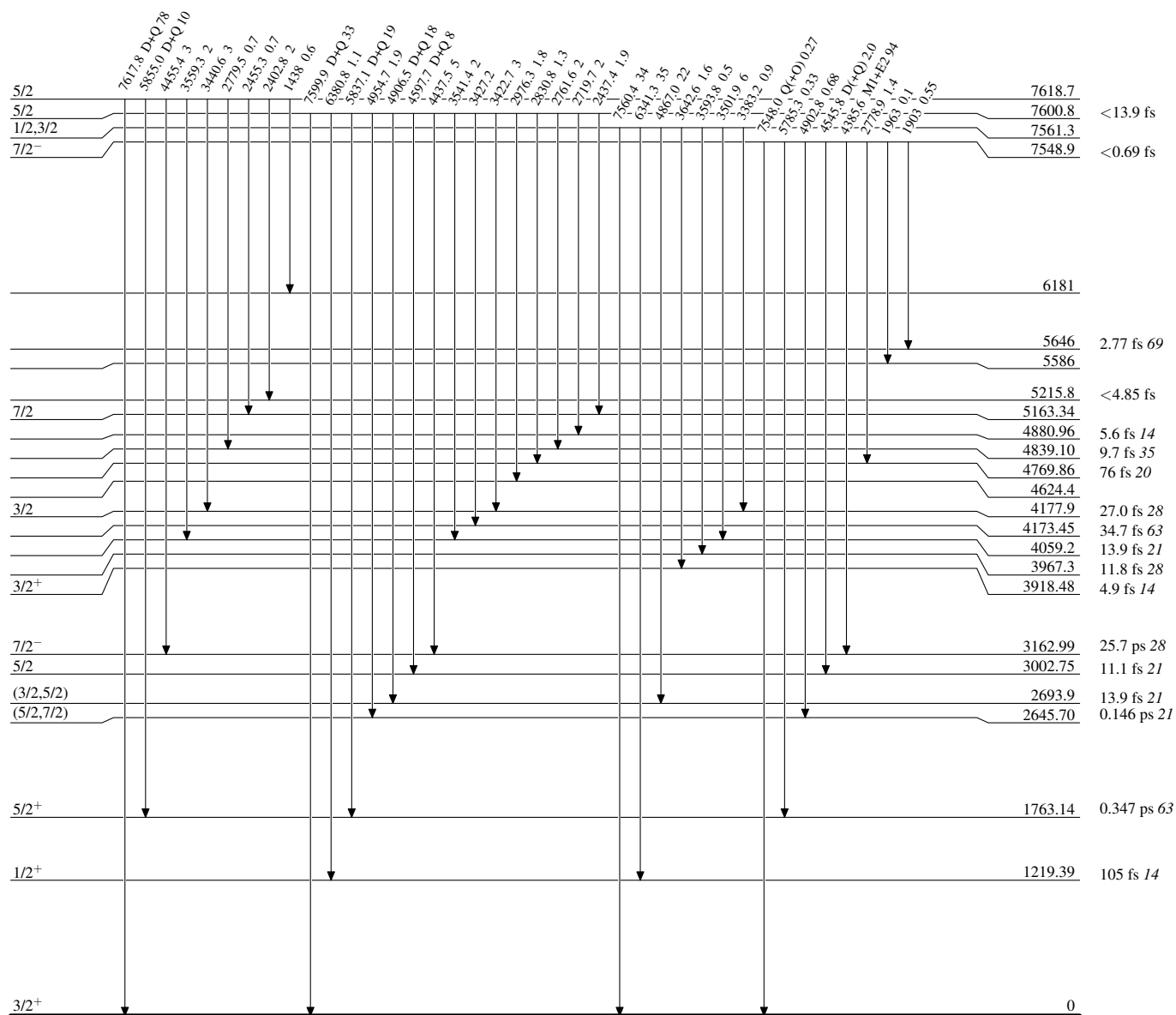
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

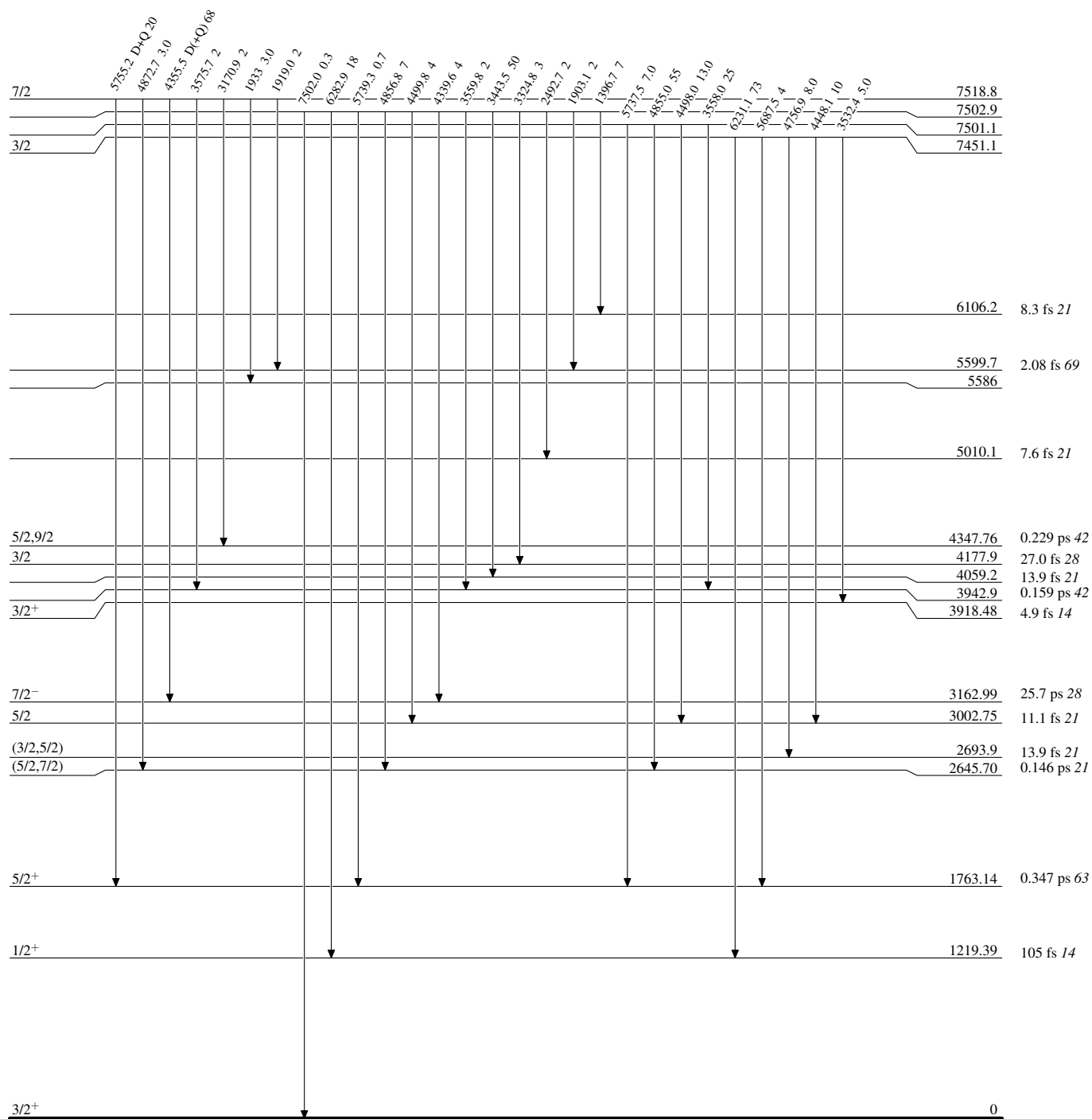
Intensities: % photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

Intensities: % photon branching from each level

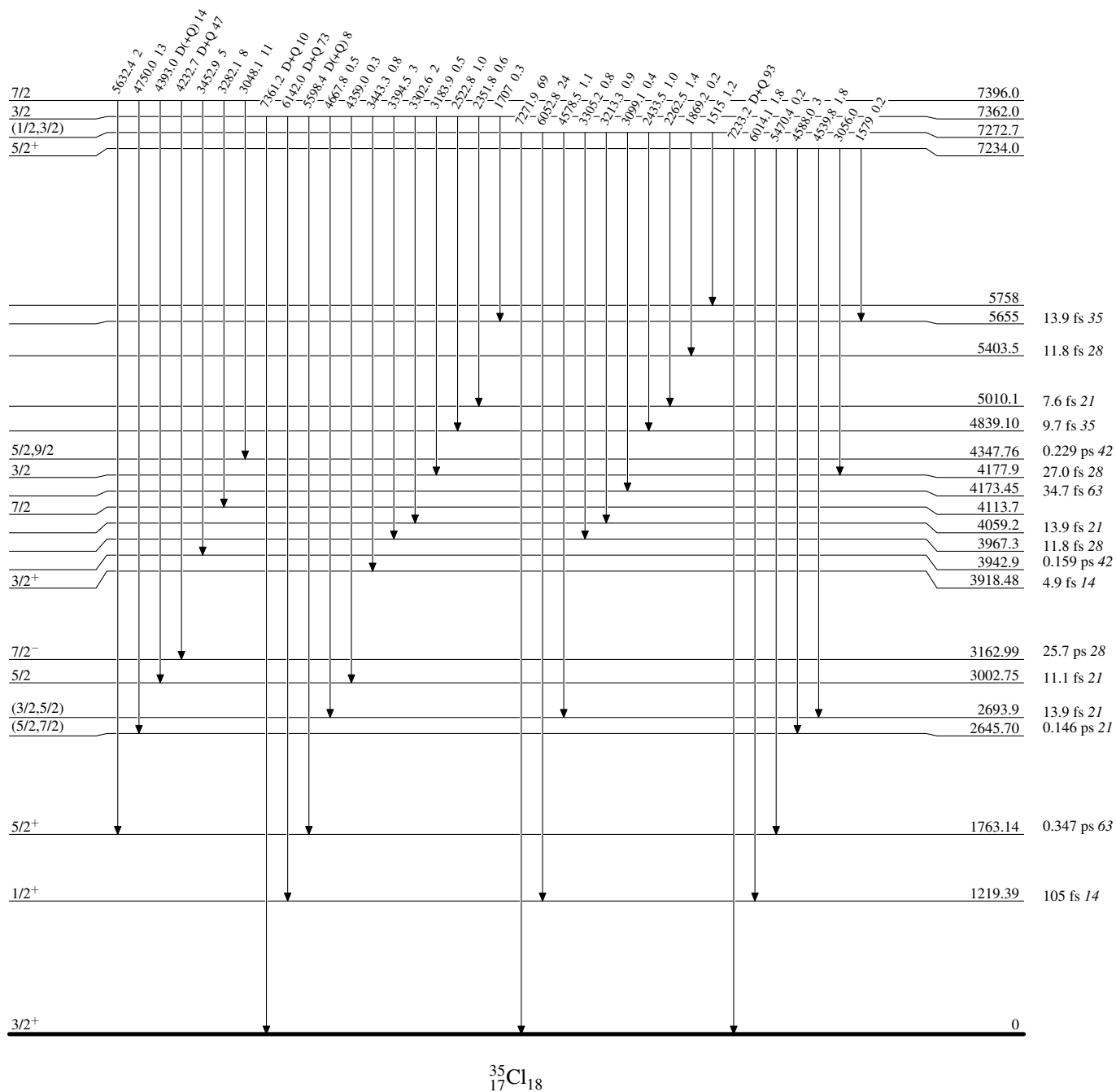
 $^{35}_{17}\text{Cl}_{18}$



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

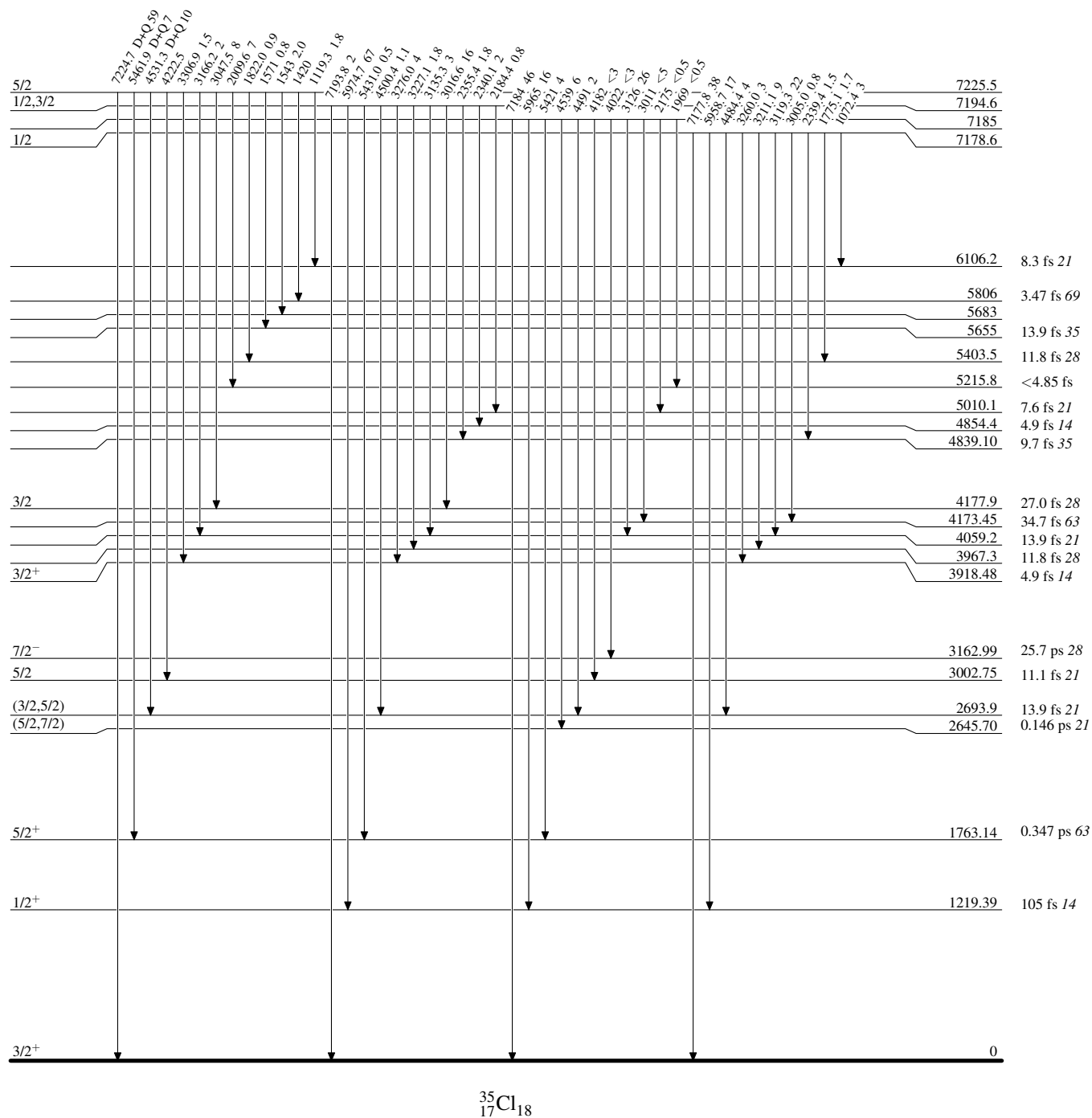
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

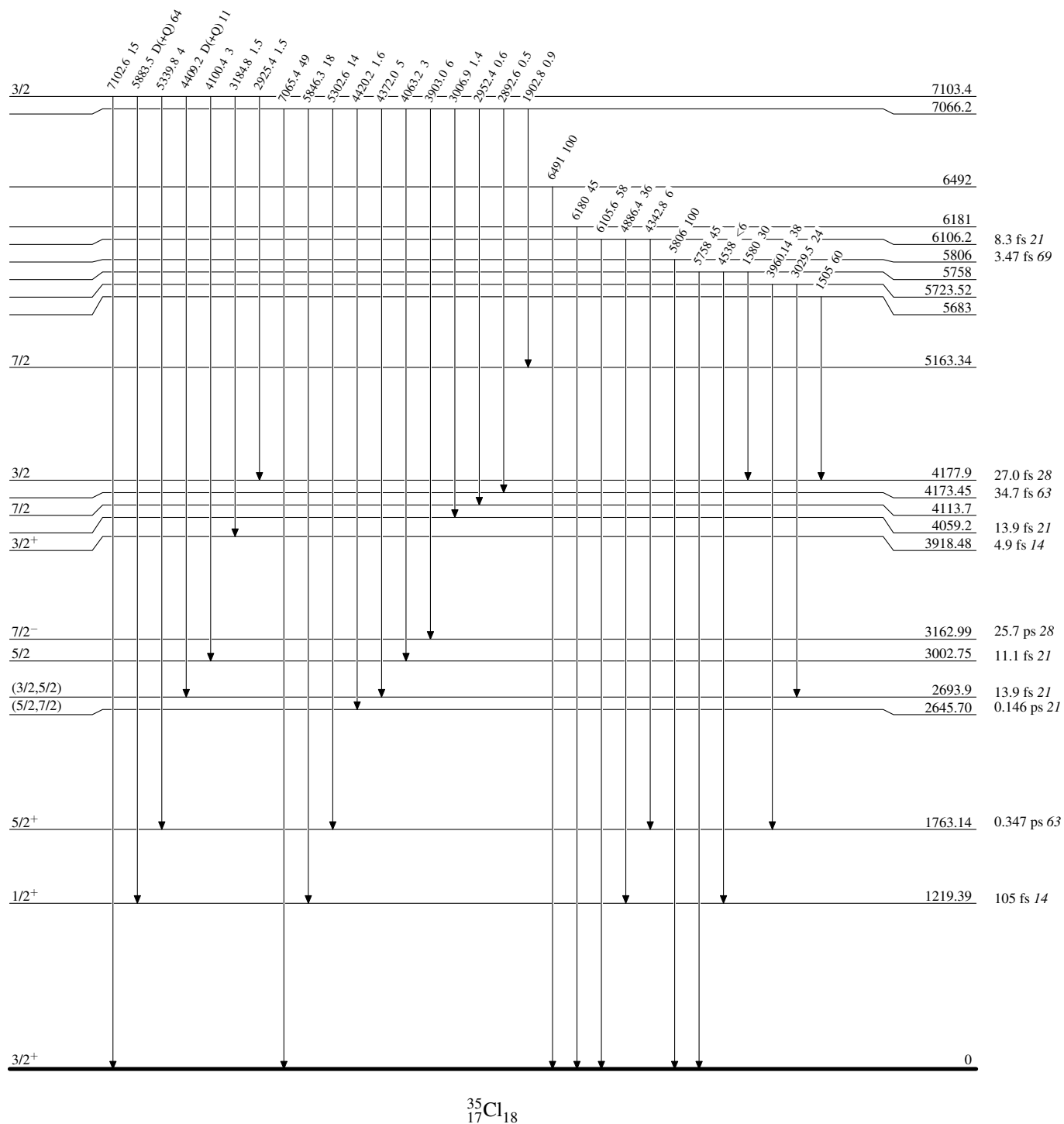
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

Intensities: % photon branching from each level

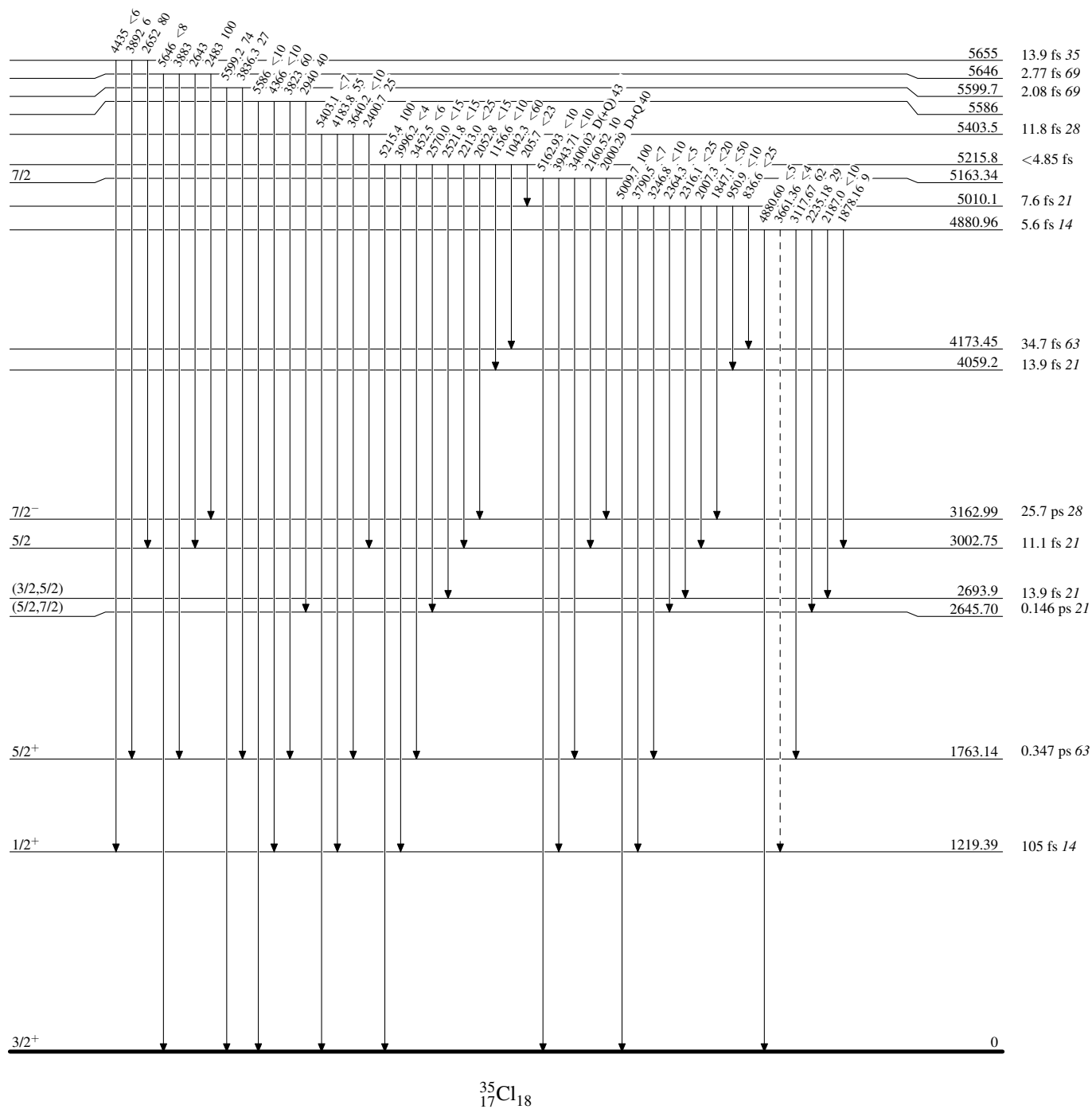


$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

Legend

## Level Scheme (continued)

Intensities: % photon branching from each level

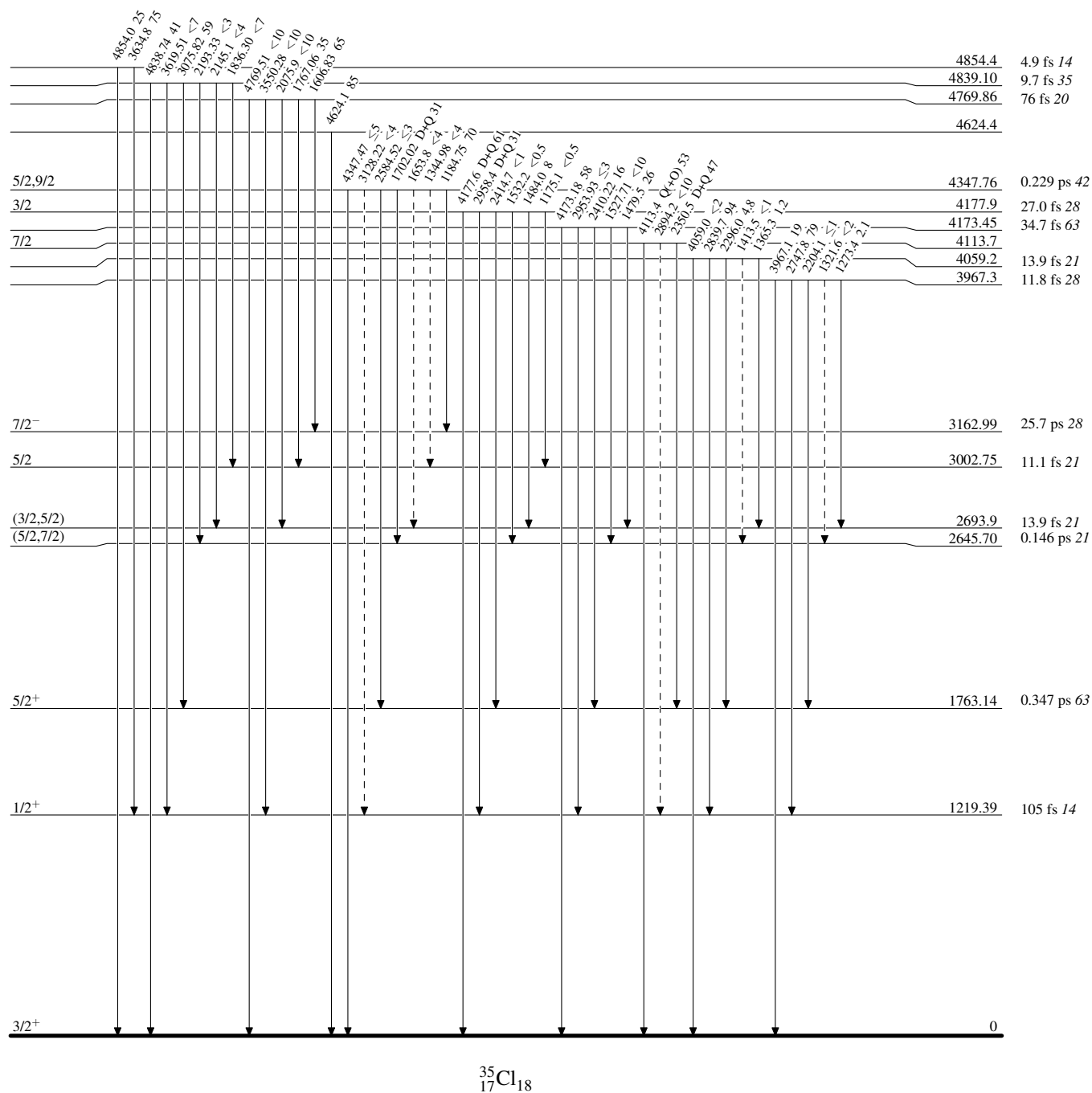
-----►  $\gamma$  Decay (Uncertain)

$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

Legend

## Level Scheme (continued)

Intensities: % photon branching from each level

-----►  $\gamma$  Decay (Uncertain) $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

Legend

## Level Scheme (continued)

Intensities: % photon branching from each level

-----►  $\gamma$  Decay (Uncertain)